

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Exterior Lighting - Edge & MKX

2008 ACCESSORIES & BODY, CAB

Exterior Lighting - Edge & MKX

SPECIFICATIONS

TORQUE SPECIFICATIONS

Description	Nm	lb-in
Headlamp assembly bolts	4	35
Liftgate lamp assembly nuts	6	53
Rear lamp screws	3	27

DESCRIPTION AND OPERATION

EXTERIOR LIGHTING

NOTE: If equipped, the perimeter lighting feature can be enabled or disabled. When enabled, the exterior lamps are illuminated when the UNLOCK button is pressed on the remote keyless entry (RKE) transmitter. Refer to HANDLES, LOCKS, LATCHES AND ENTRY SYSTEMS article.

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

Battery Saver

NOTE: Time-out is 1 minute if the vehicle has less than 80 km (50 miles).

The battery saver is internal to the SJB and provides automatic shut-off of the headlamps and parking lamps after 10 minutes in order to save battery voltage. A timer is started when the ignition switch is turned to the OFF position with the headlamp switch in the HEADLAMPS ON position, or the headlamp switch is placed in the HEADLAMPS ON position with the ignition switch off. When the time-out period has elapsed, the voltage is automatically shut off to the lamps by the battery saver.

The battery saver does **not** control the parking lamps if the headlamp switch is in the PARKING LAMPS ON position.

Headlamps

The headlamp system is a quad-beam pattern system. It consists of replaceable low and high beam bulbs. The headlamp assembly has the park/turn indicator lamp integrated into the headlamp assembly. The SJB supplies voltage to the headlamps when demanded.

NOTE: Some vehicles are equipped with the adaptive headlighting option. Refer to

Adaptive Headlighting .

If the headlamps are on when the ignition switch is turned off, the battery saver feature turns off the headlamps after 10 minutes have elapsed.

Autolamps

The autolamp system provides light sensitive automatic on/off control of the exterior lamps. The autolamp system keeps the exterior lamps on for a pre-selected period of time after the ignition switch is turned to the OFF position (20 seconds is the factory default setting). The pre-selected time lapse is adjustable up to approximately 3 minutes.

Headlamps On With Wipers On Feature

On vehicles equipped with autolamps, there is also a feature called the headlamps on with wipers on feature. If the headlamp switch is in the AUTOLAMPS ON position and the front wipers are on for more than 10 seconds (except during a mist wipe or while the wipers are on to clear washer fluid during a wash condition), the exterior lamps are turned on. The exterior lamps are turned off when the ignition switch is changed to the OFF or ACC position, the headlamp switch is placed in the OFF position, or the front wipers are off for more than 30 seconds. The exception to this is when the exterior lights are on because of darkness determined by the light sensor.

For the headlamps on with wipers on feature, refer to **WIPERS AND WASHERS** article.

Stoplamps

The stoplamp switch is located on the brake pedal assembly. The stoplamps are supplied voltage when the brake pedal is applied.

Turn Signal/Hazard Lamps

The front turn signal lamps are located within the headlamp assemblies. The rear lamps share functionality with the stoplamps. When the multifunction switch is placed in the LH or RH TURN positions, the SJB routes voltage to the LH or RH turn signal lamps. The SJB then cycles the voltage on and off approximately 80 times per minute. If a front or rear turn bulb is inoperative, the SJB cycles the voltage on and off approximately 160 times per minute.

The hazard switch is located on the center instrument panel finish panel. When the hazard switch is pressed, the SJB supplies voltage to all the turn lamps. The SJB cycles the voltage on and off approximately 80 times per minute.

Parking Lamps

The SJB supplies voltage to the parking lamps when demanded. The parking lamp system includes the following:

- Front parking lamps

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- Front side marker lamps
- Rear parking lamps
- Rear side marker lamps
- License plate lamps
- Liftgate lamp assembly (MKX only)

Fog Lamps

The fog lamp switch is integrated into the headlamp switch. When the fog lamp switch is engaged, the SJB provides ground for the fog lamp relay coil (integrated into the SJB), which then energizes and routes voltage to the fog lamps. The fog lamps can be turned on when the ignition switch is in the RUN or ACC position, the parking lamps are on, and the high beam headlamps are off.

Reversing Lamps

When the transmission is placed in REVERSE (R), the PCM provides ground for the reversing lamp relay coil. The reversing lamps receive voltage when the reversing lamp relay is energized. The reversing lamps are located within the rear lamp assemblies.

For MKX, the reversing lamps are LEDs integrated into the rear lamp assemblies.

Trailer Tow

When the vehicle's rear stop/turn or parking lamps are illuminated, voltage is also supplied to a corresponding trailer tow relay (coil side) located in the battery junction box (BJB).

Adaptive Headlighting

Vehicles equipped with adaptive front lighting have a module (located on the bottom of the LH headlamp assembly) to control the left/right aiming of the headlamp bulbs. Depending on the inputs received (steering wheel angle and vehicle speed for example), the adaptive front lighting module can change the angle at which the headlamps are aimed (left or right) to improve nighttime visibility around curves.

There are sensors and motors within each headlamp that the module uses to control the angle of the headlamp beams and are not repaired separately.

Daytime Running Lamps (DRL)

The SJB can be configured to turn the DRL on for this vehicle. Refer to **MODULE CONFIGURATION** article for additional information on this programmable parameter.

The DRL system operates the low beam headlamps at a reduced intensity.

The SJB supplies pulse-width modulated (PWM) voltage to illuminate the low beam headlamps at a reduced intensity when the following conditions are met:

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- The ignition switch is in the RUN position.
- The headlamps have not been turned on by the autolamp system or the headlamp switch.
- The transmission is not in PARK (P).

DIAGNOSTIC TESTS

HEADLAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

Exterior Lighting

The SJB monitors the headlamp switch position by sending voltage reference signals on multiple circuits to the headlamp switch. There is one circuit for each headlamp switch position. At any given time, one of the signal circuits is routed to ground.

If the SJB does not detect any of the inputs to the headlamp switch are active (routed to ground) for 5 seconds, the SJB turns on the exterior lights and keeps them on for 10 minutes after the ignition switch is turned off (or 10 minutes from the time the SJB does not detect any headlamp switch input if the ignition switch was already off). Additionally, if the SJB detects multiple circuits short to ground, the SJB implements a planned strategy depending on the multiple inputs received.

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If either of these situations occur, the SJB should **NOT** be ruled immediately as being at fault. This is normal behavior of the SJB design as it has detected a fault with the inputs from the headlamp switch.

High Beam and Flash-to-Pass

The SJB also monitors the multifunction switch for a flash-to-pass or high beam request. There are 2 voltage reference circuits which monitor this. When the multifunction switch is in the FLASH-TO-PASS or HIGH BEAM position, the voltage signal is routed to ground.

NOTE: **The flash-to-pass feature does not require any input from the headlamp switch.**

When the SJB receives an input requesting the headlamps on, the SJB supplies voltage to the low beams. If the low beams are on and the SJB receives a request for high beams (or just a flash-to-pass request), the SJB energizes an internal relay which routes voltage to the headlamps.

Field-Effect Transistor (FET) Protection

The SJB utilizes a FET protective circuit strategy for many of its outputs (for example, the headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault is detected. A continuous DTC is stored at that time of the fault. The circuit then resets after a customer demand of the function (switching the component on, battery saver being energized). When an excessive circuit load occurs several times, the module shuts down the output until a repair procedure is carried out. At the same time, the continuous DTC that was stored on the first failure, does not clear by a command to clear the continuous DTCs. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful on-demand self-test proves that the fault has been repaired. After the on-demand self-test has successfully completed (no on-demand DTCs present), the continuous DTC clears and the circuit function returns.

Each circuit has 3 predefined levels of short circuit tolerance established in the module based on each circuits' capability. When the first or second level is reached, the continuous DTC associated with the circuit sets along with DTC B106E. These DTCs may be cleared using the Clear DTC operation on the scan tool as long as the fault itself has been corrected. If any of the circuits are shorted past the third level, then DTCs B106F and B1342 set along with the associated continuous DTC. This DTC cannot be cleared and the module must be replaced.

The SJB FET protected output circuits for the headlamp system are for the LH low beam output and the RH low beam output circuits.

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
	• Smart junction box

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• Headlamp switch • Multifunction switch • Headlamp assembly	(SJB) fuse(s): ○ 7 (10A) (LH low beam) ○ 8 (10A) (RH low beam) ○ 23 (15A) (high beams) • Bulb(s) • Bulb holder(s) • Wiring, terminals or connectors • SJB
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3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
- Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
- Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.

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8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B1472	Lamp Headlamp Input Circuit Short to Ground	Go to Pinpoint Test D .
B1510	Flash To Pass Switch Circuit Short to Ground	Go to Pinpoint Test D .
B1696	Autolamp On Circuit Short To Ground	Go to Pinpoint Test D .
B2008	Wipers On Signal Circuit Short to Ground	Go to Pinpoint Test D .
B2A22	Headlamp Off Circuit Open	ADDRESS any other DTCs first. Go to Pinpoint Test D .
B2A23	High Beam Input Circuit Short to Ground	Go to Pinpoint Test D .
B2A2E	Right Front Lamp Low Beam Circuit Open	If the low beam is inoperative, go to Pinpoint Test B . If the low beam is always on, go to Pinpoint Test D .
B2A2F	Right Front Lamp Low Beam Circuit Short to Ground	Go to Pinpoint Test B .
B2A30	Left Front Lamp Low Beam Circuit Open	If the low beam is inoperative, go to Pinpoint Test B . If the low beam is always on, go to Pinpoint Test D .
B2A31	Left Front Lamp Low Beam Circuit Short to Ground	Go to Pinpoint Test B .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
No communication with the smart junction box (SJB)	- Fuse - Wiring, terminals or connectors - SJB	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
	NOTE:	

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Both low beams are inoperative	Due to varying wattage ratings and resulting current draw, certain aftermarket headlamp bulbs may cause the SJB to shut down the short circuit protection, resulting in the low beams becoming inoperative. - Incorrect bulbs - SJB - Fuse - Wiring, terminals or connectors - Joint connector - Multifunction switch - SJB	- VERIFY that the bulbs meet Ford specifications. If the bulbs do not meet Ford specifications, INSTALL the correct bulbs. RUN the on-demand self-test (required to clear certain DTCs and reset the SJB). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation. - CHECK for recorded SJB DTCs. - If DTCs are present, REFER to DTC Charts in this service information. - If no DTCs are present, INSTALL a new SJB. REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article. TEST the system for normal operation. - Go to Pinpoint Test A.
Both high beams are inoperative		
One low beam headlamp is inoperative	- Fuse - Wiring, terminals or connectors - Joint connector - Headlamp assembly (MKX) - SJB	Go to Pinpoint Test B.
One high beam headlamp is inoperative	- Wiring, terminals or connectors - Joint connector - Headlamp assembly (MKX)	Go to Pinpoint Test C.

• The headlamp(s) are on continuously	• Wiring, terminals or connectors • Joint connector • Headlamp switch • Multifunction switch • Windshield wiper motor • SJB	• Go to Pinpoint Test D .
• The flash-to-pass feature is inoperative	• Wiring, terminals or connectors • Multifunction switch • SJB	• Go to Pinpoint Test E .

Pinpoint Tests

Pinpoint Test A: Both High Beams Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

When the headlamp switch is placed in the HEADLAMPS ON position, the smart junction box (SJB) monitors the high beam request circuit from the multifunction switch through circuit CLF17 (WH/OG). When the multifunction switch is placed in the HIGH BEAM position, the signal is routed to ground through circuit GD116 (BK/VT) (which routes through a joint connector). The SJB then energizes the high beam relay (integrated into the SJB) which supplies voltage to the headlamps through circuit CLF08 (YE/VT).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Joint connector
- Multifunction switch
- SJB

PINPOINT TEST A: BOTH HIGH BEAMS ARE INOPERATIVE

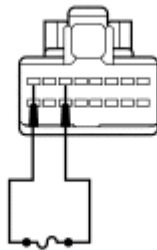
CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK THE HIGH BEAM INDICATOR

- Key in ON position.
- Place the headlamp switch in the HEADLAMPS ON position.
- Place the multifunction switch in the HIGH BEAM position.
- **Does the high beam indicator illuminate?**
YES : Go to A6.
NO : Go to A2.

A2 CHECK THE MULTIFUNCTION SWITCH

- Place the headlamp switch in the OFF position.
- Key in OFF position.
- Disconnect: Multifunction Switch C202



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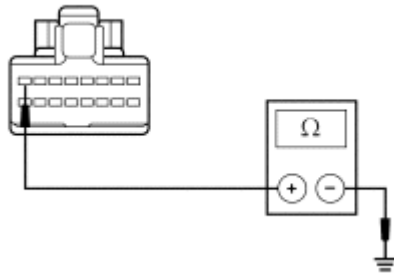
Fig. 1: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (5A) jumper wire between the multifunction switch C202-6, circuit CLF17 (WH/OG), harness side and the multifunction switch C202-8, circuit GD116 (BK/VT), harness side.
- Place the headlamp switch in the HEADLAMPS ON position.
- **Do the high beams illuminate?**
YES : REMOVE the jumper wire. INSTALL a new multifunction switch. REFER to **STEERING COLUMN SWITCHES** article. TEST the system for normal operation.
NO : REMOVE the jumper wire. Go to A3.

A3 CHECK CIRCUIT GD116 (BK/VT) FOR CONTINUITY TO GROUND

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

- Place the headlamp switch in the OFF position.



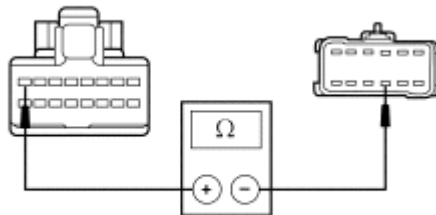
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Fig. 2: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-8, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to A5.
NO : Go to A4.

A4 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN

- Disconnect: Joint Connector 6 C2365



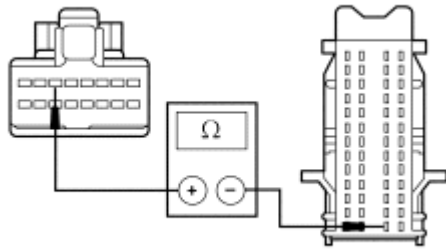
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Fig. 3: Measuring Resistance Between Multifunction Switch C202-8, Circuit GD116 (BK/VT) & Joint Connector 6 C2365-D
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-8, circuit GD116 (BK/VT), harness side and the joint connector 6 C2365-D, circuit GD116 (BK/VT), harness side.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

A5 CHECK CIRCUIT CLF17 (WH/OG) FOR AN OPEN

- Disconnect: SJB C2280b



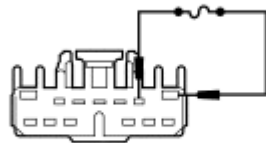
N0057366

Fig. 4: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-6, circuit CLF17 (WH/OG), harness side and the SJB C2280b-27, circuit CLF17 (WH/OG), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to A7.
NO : REPAIR the circuit. TEST the system for normal operation.

A6 CHECK CIRCUIT CLF08 (YE/VT) FOR AN OPEN

- Disconnect: SJB C2280e



N0057367

Fig. 5: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (20A) jumper wire between the SJB C2280e-1, circuit CBB53 (GY/BN), harness side and the SJB C2280e-2, circuit CLF08 (YE/VT), harness side.
- Key in ON position.
- **Do the high beams illuminate?**
YES : VERIFY the SJB fuse 23 (15A) is OK. If OK, go to A7.
NO : REPAIR the circuit. TEST the system for normal operation.

A7 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion

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- damaged pins
- pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: One Low Beam Headlamp Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

When the smart junction box (SJB) detects the headlamp switch in the HEADLAMPS ON position and the multifunction switch in the LOW BEAM position, the SJB provides voltage through circuits CLF04 (BN/BU) and CLF05 (BU/GN) to the LH and RH low beam headlamps, respectively. Ground is supplied on circuit GD123 (BK/GY) for RH headlamp. Ground for the LH headlamp is routed through a joint connector through circuit GD120 (BK/GN).

DTC Description	Fault Trigger Conditions
B2A2E - Right Front Lamp Low Beam Circuit Open	An on-demand DTC that sets when the SJB detects an open in the RH low beam output circuit.
B2A2F - Right Front Lamp Low Beam Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground in the RH low beam output circuit.
B2A30 - Left Front Lamp Low Beam Circuit Open	An on-demand DTC that sets when the SJB detects an open in the LH low beam output circuit.
B2A31 - Left Front Lamp Low Beam Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground in the LH low beam output circuit.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Joint connector
- Headlamp assembly (MKX)
- SJB

PINPOINT TEST B: ONE LOW BEAM HEADLAMP IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Due to varying wattage ratings and resulting current draw, certain aftermarket headlamp bulbs may cause the SJB to shut down the short circuit protection, resulting in the low beams becoming inoperative. Verify that the bulbs meet Ford specifications. If the bulbs do not meet Ford specifications, install the correct bulbs. Run the on-demand self-test (required to clear certain DTCs and reset the SJB). Correct any unresolved DTCs. Clear all DTCs. Test the system for normal operation.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

B1 USE THE RECORDED DTCs FROM THE SJB SELF-TEST

- Check the recorded results from the SJB self-test.
- **Is DTCs B2A2F or B2A31 recorded?**

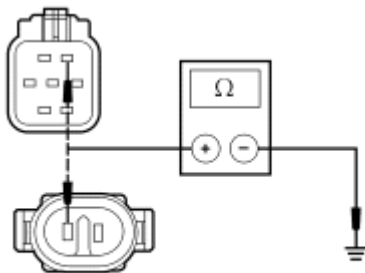
YES : For MKX, go to B7.

For Edge, go to B8.

NO : Go to B2.

B2 CHECK THE HEADLAMP GROUND CIRCUIT FOR CONTINUITY TO GROUND

- Key in OFF position.
- Disconnect: Inoperative Headlamp



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Fig. 6: Measuring Resistance Between Inoperative Headlamp, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the inoperative headlamp, harness side and ground as follows:

Connector-	
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Pin	Circuit
Edge	
LH low beam C1111-A	GD120 (BK/GN)
RH low beam C1112-A	GD123 (BK/GY)
MKX	
LH headlamp C1021a-2	GD120 (BK/GN)
RH headlamp C1041a-2	GD123 (BK/GY)

- Is the resistance less than 5 ohms?

YES :

For MKX, go to B5.

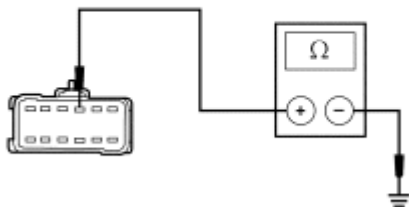
For Edge, VERIFY the SJB fuse 7 (10A) (LH low beam) or fuse 8 (10A) (RH low beam) is OK. If OK Go to B6.

NO : For RH headlamp, REPAIR circuit GD123 (BK/GY) for an open. CLEAR the DTCs. REPEAT the self-test.

For LH headlamp, go to B3.

B3 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN (JOINT CONNECTOR TO GROUND)

- Disconnect: Joint Connector 7 C2200



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Fig. 7: Measuring Resistance Between Joint Connector 7 C2200-J, Circuit GD120 (BK/GN) & Ground

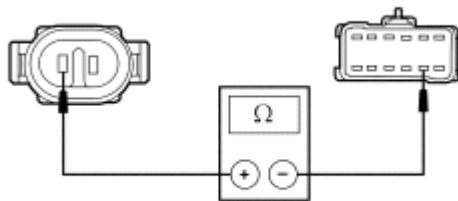
Courtesy of FORD MOTOR CO.

- Measure the resistance between the joint connector 7 C2200-J, circuit GD120 (BK/GN), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : Go to B4.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

B4 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN (HEADLAMP TO JOINT CONNECTOR)

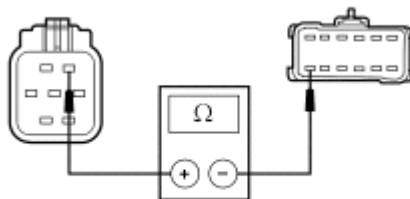


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Fig. 8: Measuring Resistance Between LH Low Beam C1111-A, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-E

Courtesy of FORD MOTOR CO.

- For Edge, measure the resistance between the LH low beam C1111-A, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-E, circuit GD120 (BK/GN), harness side.



N0062211

Fig. 9: Measuring Resistance Between LH Headlamp C1021A-2, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-A

Courtesy of FORD MOTOR CO.

- For MKX, measure the resistance between the LH headlamp C1021a-2, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-A, circuit GD120 (BK/GN), harness side.

- **Is the resistance less than 5 ohms?**

YES : INSTALL a new joint connector. CLEAR the DTCs. REPEAT the self-test.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

B5 CHECK THE SJB HEADLAMP VOLTAGE SUPPLY CIRCUIT FOR VOLTAGE

- Place the headlamp switch in the HEADLAMPS ON position.

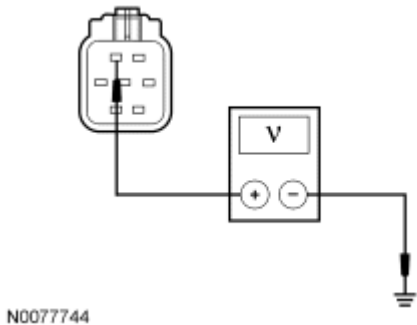


Fig. 10: Measuring Voltage Between Inoperative Headlamp, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between the inoperative headlamp, harness side and ground as follows:

Connector-Pin	Circuit
LH headlamp C1021a-1	CLF04 (BN/BU)
RH headlamp C1041a-1	CLF05 (BU/GN)

- Is the voltage greater than 10 volts?

YES : REPAIR or INSTALL a new headlamp assembly. REFER to Headlamp Assembly. CLEAR the DTCs. REPEAT the self-test.

NO : VERIFY the SJB fuse 7 (10A) (LH low beam) or fuse 8 (10A) (RH low beam) is OK. If OK, go to B6.

B6 CHECK THE SJB HEADLAMP VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Place the headlamp switch in the OFF position.
- Disconnect: SJB C2280e
- Measure the resistance between the inoperative headlamp, harness side and the SJB as follows:

Headlamp Connector-Pin	SJB Connector-Pin	Circuit
Edge		
LH low beam C1111-B	SJB C2280e-13	CLF04 (BN/BU)
RH low		

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beam C1112-B	SJB C2280e-4	CLF05 (BU/GN)
MKX		
LH headlamp C1021a-1	SJB C2280e-13	CLF04 (BN/BU)
RH headlamp C1041a-1	SJB C2280e-4	CLF05 (BU/GN)

- Is the resistance less than 5 ohms?

YES : Go to B9.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

B7 CHECK THE HEADLAMP ASSEMBLY

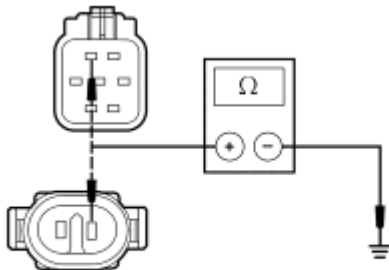
- Key in OFF position.
- Disconnect: Inoperative Headlamp
- Key in ON position.
- Repeat the SJB on-demand self-test.
- Is DTC B2A2F or B2A31 retrieved again?

YES : Go to B8.

NO : INSTALL a new headlamp assembly. REFER to **Headlamp Assembly**. TEST the system for normal operation.

B8 CHECK CIRCUIT CLF04 (BN/BU) (LH HEADLAMP) OR CIRCUIT CLF05 (BU/GN) (RH HEADLAMP) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280e
- Disconnect: Inoperative Headlamp



N0062213

Fig. 11: Measuring Resistance Between Inoperative Headlamp, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the inoperative headlamp, harness side and ground as follows:

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Connector-Pin	Circuit
Edge	
LH low beam C1111-B	CLF04 (BN/BU)
RH low beam C1112-B	CLF05 (BU/GN)
MKX	
LH headlamp C1021a-1	CLF04 (BN/BU)
RH headlamp C1041a-1	CLF05 (BU/GN)

- **Is the resistance greater than 10,000 ohms?**

YES : Go to B9.

NO : REPAIR the circuit in question. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

B9 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test C: One High Beam Headlamp Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

When the smart junction box (SJB) detects the headlamp switch in the HEADLAMPS ON position and the multifunction switch in the HIGH BEAM position, the SJB energizes an internal relay. When the relay is energized, the relay routes voltage through circuit CLF08 (YE/VT) to the high beams. Ground is supplied on circuit GD123 (BK/GY) for the RH headlamp. Ground for the LH headlamp is routed through a joint connector through circuit GD120 (BK/GN).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector
- Headlamp assembly (MKX)

PINPOINT TEST C: ONE HIGH BEAM HEADLAMP IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

C1 CHECK THE LOW BEAM OPERATION

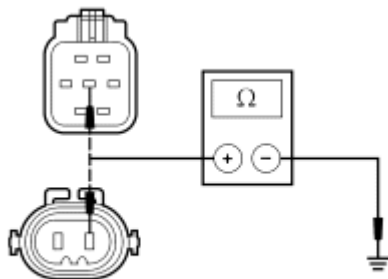
- Check the operation of the low beams.
- **Do the low beams operate correctly?**

YES : Go to C2.

NO : Go to **Pinpoint Test B** .

C2 CHECK FOR CONTINUITY TO GROUND

- Key in OFF position.
- Disconnect: Inoperative Headlamp



N0062214

Fig. 12: Measuring Resistance Between Inoperative Lamp, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the inoperative lamp, harness side and ground as follows:

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Inoperative Lamp	Connector-Pin	Circuit
Edge		
LH high beam	C1020-A	GD120 (BK/GN)
RH high beam	C1040-A	GD123 (BK/GY)
MKX		
LH headlamp	C1021a-4	GD120 (BK/GN)
RH headlamp	C1041a-4	GD123 (BK/GY)

- **Is the resistance less than 5 ohms?**

YES : For Edge, REPAIR circuit CLF08 (YE/VT) for an open. TEST the system for normal operation.

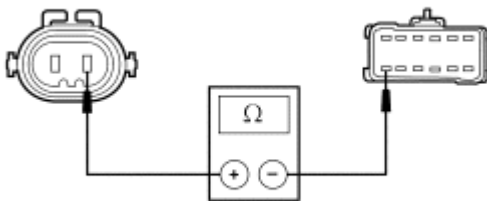
For MKX, go to C4.

NO : For the RH headlamp, REPAIR circuit GD123 (BK/GY) for an open. TEST the system for normal operation.

For the LH headlamp, go to C3.

C3 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN (HEADLAMP TO JOINT CONNECTOR)

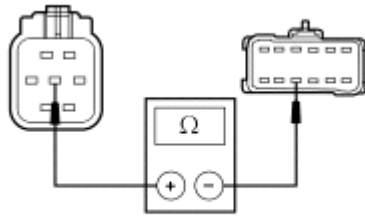
- Disconnect: Joint Connector 7 C2200



N0062215

Fig. 13: Measuring Resistance Between LH High Beam C1020-A, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-A
Courtesy of FORD MOTOR CO.

- For Edge, measure the resistance between the LH high beam C1020-A, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-A, circuit GD120 (BK/GN), harness side.



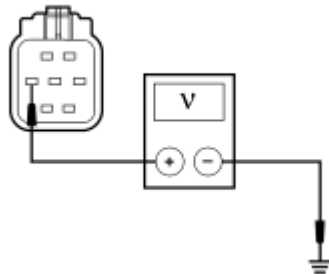
N0062216

Fig. 14: Measuring Resistance Between LH High headlamp C1021A-4, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-C
 Courtesy of FORD MOTOR CO.

- For MKX, measure the resistance between the LH headlamp C1021a-4, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-C, circuit GD120 (BK/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

C4 CHECK CIRCUIT CLF08 (YE/VT) FOR AN OPEN

- Place the headlamp switch in the HEADLAMPS ON position.
- Place the multifunction switch in the HIGH BEAM position.



N0077746

Fig. 15: Measuring Voltage Between Inoperative Lamp, Harness Side & Ground
 Courtesy of FORD MOTOR CO.

- Measure the voltage between the inoperative lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
LH headlamp	C1021a-3	CLF08 (YE/VT)
RH headlamp	C1041a-3	CLF08 (YE/VT)

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- **Is the voltage greater than 10 volts?**

YES : REPAIR or INSTALL a new headlamp assembly. REFER to **Headlamp Assembly**. TEST the system for normal operation.

NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test D: The Headlamp(s) Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation - Low Beams

The smart junction box (SJB) sends reference signals to the headlamp switch through circuits CLF19 (VT/GN) (AUTOLAMPS), CLF23 (WH/VT) (OFF), CLS34 (GY) (PARKING LAMPS) and CLF18 (BU/WH) (HEADLAMPS). At any given time, the headlamp switch routes one of the input circuits to ground through joint connector 6 through circuit GD116 (BK/VT).

When the SJB detects the headlamp switch in the HEADLAMPS ON position and the multifunction switch in the LOW BEAM position, the SJB provides voltage through circuits CLF04 (BN/BU) and CLF05 (BU/GN) to the LH and RH low beam headlamps, respectively.

On vehicles equipped with autolamps, the SJB also monitors input from the windshield wiper motor through circuit CRW01 (WH) for the headlamps on with wipers on feature.

High Beams

The SJB sends a reference signal to the multifunction switch through circuits CLF27 (GN/BN) and CLF17 (WH/OG). When the multifunction switch is placed in the FLASH-TO-PASS or HIGH BEAM position, the voltage signal is routed to ground. When the SJB detects a request for flash-to-pass or high beams (when the low beams are on), the SJB energizes the high beam relay (integrated into the SJB), which routes voltage through circuit CLF08 (YE/VT) to the headlamps.

DTC Description	Fault Trigger Conditions
B1472 - Lamp Headlamp Input Circuit Short to Ground	An on-demand DTC that sets when the SJB detects the headlamps on input circuit is short to ground.
B1510 - Flash To Pass Switch Circuit Short to Ground	An on-demand DTC that sets when the SJB detects the flash-to-pass request input circuit is short to ground.
B1696 - Autolamp On Circuit Short To Ground	An on-demand DTC that sets when the SJB detects the autolamps on request input circuit is short to ground.
B2008 - Wipers On Signal Circuit Short to Ground	An on-demand DTC that sets when the SJB detects the wipers on input circuit is short to ground.
B2A22 - Headlamp Off Circuit Open	An on-demand DTC that sets when the SJB detects the headlamp switch off input circuit open.

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B2A23 - High Beam Input Circuit Short to Ground	An on-demand DTC that sets when the SJB detects a short to ground on the high beam request input circuit.
B2A2E - Right Front Lamp Low Beam Circuit Open	An on-demand DTC that sets when the SJB detects the RH low beam output circuit shorted to voltage.
B2A30 - Left Front Lamp Low Beam Circuit Open	An on-demand DTC that sets when the SJB detects the LH low beam output circuit shorted to voltage.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector
- Headlamp switch
- Multifunction switch
- Windshield wiper motor
- SJB

PINPOINT TEST D: THE HEADLAMP(S) ARE ON CONTINUOUSLY

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

D1 VERIFY THE HEADLAMP OPERATION

- Key in ON position.
- **Are the high beam headlamps illuminated?**
YES : Go to D13.
NO : Go to D2.

D2 CHECK THE SJB HEADLAMP SWITCH PIDs

- Enter the following diagnostic mode on the diagnostic tool: SJB Data Logger

NOTE: Make sure the headlamp switch is lined up in the correct position when monitoring the PIDs.

- Monitor the SJB headlamp switch PIDs (HD_LMP_S2 and HEAD_L) while moving the headlamp switch through all positions. Refer to the following chart.

Headlamp Switch Position	HD_LMP_S2 Expected Value	HEAD_L Expected Value
--------------------------------	--------------------------------	-----------------------------

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Autolamps	AUTOLAMP	OFF
OFF	-	OFF
Parklamps	-	-
Headlamp	-	HEADLAMP

- **Do the headlamp switch positions agree with the PIDs?**

YES : Go to D9.

NO : Go to D3.

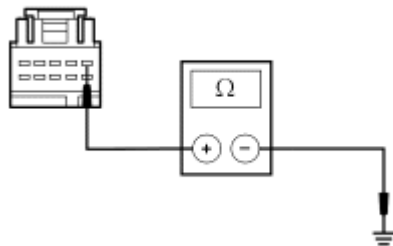
D3 CHECK THE HEADLAMP SWITCH

- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Carry out the headlamp switch component test. Refer to appropriate COMPONENT TESTING article.
- **Is the headlamp switch OK?**

YES : Go to D4.

NO : INSTALL a new headlamp switch. REFER to **Headlamp Switch**. CLEAR the DTCs. REPEAT the self-test.

D4 CHECK FOR CONTINUITY TO GROUND FROM THE HEADLAMP SWITCH



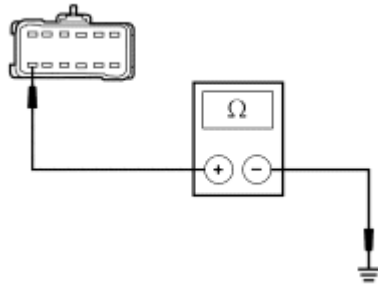
N0038799

Fig. 16: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-1, circuit GD116 (BK/VT), harness side and ground.
 - **Is the resistance less than 5 ohms?**
- YES** : Go to D7.
- NO** : Go to D5.

D5 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN (JOINT CONNECTOR TO GROUND)

- Disconnect: Joint Connector 6 C2365



N0062217

Fig. 17: Measuring Resistance Between Joint Connector 6 C2365-A, Circuit GD116 (BK/VT) & Ground

Courtesy of FORD MOTOR CO.

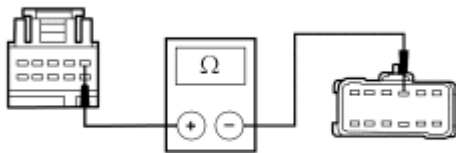
- Measure the resistance between the joint connector 6 C2365-A, circuit GD116 (BK/VT), harness side and ground.

- **Is the resistance less than 5 ohms?**

YES : Go to D6.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

D6 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN (HEADLAMP SWITCH TO JOINT CONNECTOR)



N0062218

Fig. 18: Measuring Resistance Between Headlamp Switch C205-1, Circuit GD116 (BK/VT) & Joint Connector 6 C2365-J

Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-1, circuit GD116 (BK/VT), harness side and joint connector 6 C2365-J, circuit GD116 (BK/VT), harness side.

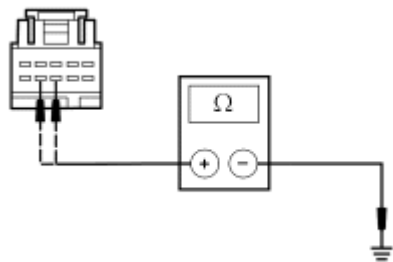
- **Is the resistance less than 5 ohms?**

YES : INSTALL a new joint connector. CLEAR the DTCs. REPEAT the self-test.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

D7 CHECK THE HEADLAMP SWITCH INPUT CIRCUITS FOR A SHORT TO GROUND

- Disconnect: SJB C2280b



N0077747

Fig. 19: Measuring Resistance Between Headlamp Switch, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch, harness side and ground as follows:

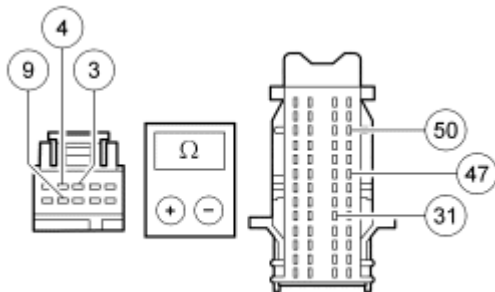
Connector-Pin	Circuit
C205-9 (if equipped with autolamps)	CLF19 (VT/GN)
C205-8	CLF18 (BU/WH)

- Are the resistances greater than 10,000 ohms?

YES : Go to D8.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

D8 CHECK THE HEADLAMP SWITCH INPUT CIRCUITS FOR AN OPEN



N0077748

Fig. 20: Measuring Resistance Between Headlamp Switch, Harness Side & SJB
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch, harness side and the SJB, harness side as follows:

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Headlamp Switch Connector-Pin	SJB Connector-Pin	Circuit
C205-9 (if equipped with autolamps)	C2280b-50	CLF19 (VT/GN)
C205-4	C2280b-47	CLF23 (WH/VT)
C205-3	C2280b-31	CLS34 (GY)

- Are the resistances less than 5 ohms?

YES : Go to D16.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

D9 CHECK FOR SJB DTC B2008

- Check the recorded results from the SJB self-test.

- Is DTC B2008 present?

YES : Go to D10.

NO : Go to D12.

D10 CHECK THE WINDSHIELD WIPER MOTOR

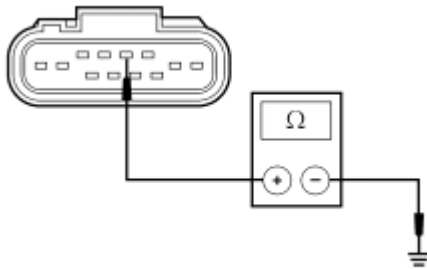
- Key in OFF position.
- Disconnect: Windshield Wiper Motor C1552
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB Self-Test
- Repeat the SJB self-test.
- Is DTC B2008 still present?

YES : Go to D11.

NO : INSTALL a new windshield wiper motor. REFER to **WIPERS AND WASHERS** article. TEST the system for normal operation.

D11 CHECK CIRCUIT CRW01 (WH) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280f



N0046955

Fig. 21: Checking Circuit For Short To Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the windshield wiper motor C1552-2, circuit CRW01 (WH), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to D16.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

D12 CHECK THE SJB OUTPUT

- Disconnect: SJB C2280e
- **Does the LH or RH headlamp continue to illuminate?**
YES : REPAIR circuit CLF04 (BN/BU) (LH headlamp) or circuit CLF05 (BU/GN) (RH headlamp) for a short to voltage as necessary. CLEAR the DTCs. REPEAT the self-test.
NO : Go to D16.

D13 CHECK CIRCUIT CLF08 (YE/VT) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: SJB C2280e
- Key in ON position.
- **Do the high beams continue to illuminate?**
YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
NO : Go to D14.

D14 CHECK THE MULTIFUNCTION SWITCH

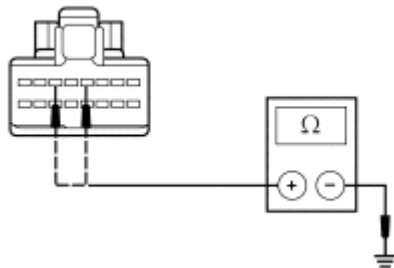
- Key in OFF position.
- Connect: SJB C2280e
- Disconnect: Multifunction Switch C202
- Key in ON position.
- **Do the high beams continue to illuminate?**
YES : Go to D15.

NO : INSTALL a new multifunction switch. REFER to **STEERING COLUMN SWITCHES** article. CLEAR the DTCs. REPEAT the self-test.

D15 CHECK CIRCUIT CLF17 (WH/OG) (HIGH BEAM) OR CIRCUIT CLF27 (GN/BN)

(FLASH-TO-PASS) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280b



N0037272

Fig. 22: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-6, circuit CLF17 (WH/OG), harness side and ground; and between the multifunction switch C202-4, circuit CLF27 (GN/BN), harness side and ground.
- **Are the resistances greater than 10,000 ohms?**
YES : Go to D16.
NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

D16 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test E: The Flash-To-Pass Feature Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a reference signal to the multifunction switch through circuit CLF27 (GN/BN). When the multifunction switch is placed in the FLASH-TO-PASS position, the voltage signal is routed to ground. When the SJB detects a request for the flash-to-pass, the SJB provides voltage to the high beam headlamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Multifunction switch
- SJB

PINPOINT TEST E: THE FLASH-TO-PASS FEATURE IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 VERIFY THE HIGH BEAM HEADLAMP OPERATION

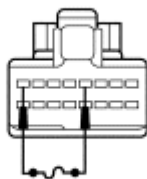
- Key in ON position.
- Place the headlamp switch in the HEADLAMPS ON position.
- Place the multifunction switch in the HIGH BEAM position while observing the headlamps.
- **Do the high beam headlamps illuminate?**

YES : Go to E2.

NO : Go to **Pinpoint Test A** .

E2 CHECK THE MULTIFUNCTION SWITCH

- Key in OFF position.
- Disconnect: Multifunction Switch C202



N0037273

Fig. 23: Checking Multifunction Switch
Courtesy of FORD MOTOR CO.

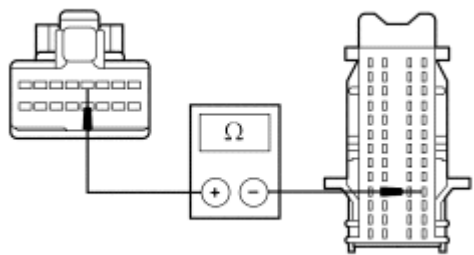
- Connect a fused (5A) jumper wire between the multifunction switch C202-4, circuit CLF27 (GN/BN), harness side and the multifunction switch C202-8, circuit GD116 (BK/VT), harness side.
- **Do the high beams illuminate?**

YES : REMOVE the jumper wire. INSTALL a new multifunction switch. REFER to **STEERING COLUMN SWITCHES** article. TEST the system for normal operation.

NO : REMOVE the jumper wire. Go to E3.

E3 CHECK CIRCUIT CLF27 (GN/BN) FOR AN OPEN

- Disconnect: SJB C2280b



N0057371

Fig. 24: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-4, circuit CLF27 (GN/BN), harness side and the SJB C2280b-43, circuit CLF27 (GN/BN), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to E4.

NO : REPAIR the circuit. TEST the system for normal operation.

E4 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

AUTOLAMPS

Special Tools

Illustration	Tool Name	Tool Number

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 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

NOTE: For the perimeter lighting feature, refer to HANDLES, LOCKS, LATCHES AND ENTRY SYSTEMS article.

NOTE: For the headlamps on with the wipers on feature, refer to WIPERS AND WASHERS article.

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

Exterior Lighting

The SJB monitors the headlamp switch position by sending voltage reference signals on multiple circuits to the headlamp switch. There is one circuit for each headlamp switch position. At any given time, one of the signal circuits is routed to ground.

If the SJB does not detect any of the inputs to the headlamp switch are active (routed to ground) for 5 seconds, the SJB turns on the exterior lights and keeps them on for 10 minutes after the ignition switch is turned off (or 10 minutes from the time the SJB does not detect any headlamp switch input if the ignition switch was already off). Additionally, if the SJB detects multiple circuits short to ground, the SJB implements a planned strategy depending on the multiple inputs received.

If either of these situations occur, the SJB should **NOT** be ruled immediately as being at fault. This is normal behavior of the SJB design as it has detected a fault with the inputs from the headlamp switch.

Autolamps Feature

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The SJB monitors the light sensor with a voltage reference signal. The light sensor input to the SJB varies with the ambient light conditions.

When the SJB receives an input from the headlamp switch indicating a request for the autolamps, the SJB monitors the light sensor for the ambient light condition. If the SJB determines the ambient light level is dark, the SJB supplies voltage to the exterior lamps.

If the ignition switch is already in the RUN position and the vehicle enters a dark/lighted area (such as when entering/exiting a non-lighted tunnel during the daytime), the transition from light to dark (or dark to light) needs to last 15 seconds before the SJB turns the exterior lamps on or off. This strategy is to prevent the exterior lamps from unnecessarily flashing on and off.

The SJB does react quicker under extreme light conditions. If the SJB has detected a very dark condition, the exterior lamps are turned on after 1.5 seconds. If the SJB has detected a very high ambient light level, the exterior lamps are turned off after 4 seconds.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Headlamp switch	• Wiring, terminals or connectors • Light sensor • Smart junction box (SJB)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:

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- Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
- Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B1472	Lamp Headlamp Input Circuit Short to Ground	Go to Pinpoint Test G .
B1696	Autolamp On Circuit Short To Ground	Go to Pinpoint Test G .
B1791	Autolamp Sensor Input Circuit Open	Go to Pinpoint Test G .
B1793	Autolamp Sensor Input Circuit Short to Ground	Go to Pinpoint Test G .
B2008	Wipers On Signal Circuit Short to Ground	Go to Pinpoint Test G .
B2A22	Headlamp Off Circuit Open	ADDRESS any other DTCs first. Go to Pinpoint Test G .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
	• Fuse	

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• No communication with the smart junction box (SJB)	• Wiring, terminals or connectors • SJB	• REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
• The autolamps are inoperative	• Light sensor • SJB	• Go to Pinpoint Test F .
• The autolamps are on continuously	• Wiring, terminals or connectors • Joint connector • Light sensor • Headlamp switch • Windshield wiper motor • SJB	• Go to Pinpoint Test G .

Pinpoint Tests

Pinpoint Test F: The Autolamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

When the smart junction box (SJB) detects the headlamp switch in the AUTOLAMPS ON position, the SJB monitors the light sensor input. The light sensor determines the amount of light based on the input received from the ambient lighting conditions. When the ambient light level has reached a point (determined by the internal programming of the SJB), the SJB provides voltage to the headlamps, parking and license plate lamps.

This pinpoint test is intended to diagnose the following:

- Light sensor
- SJB

PINPOINT TEST F: THE AUTOLAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 CHECK THE MANUAL HEADLAMP OPERATION

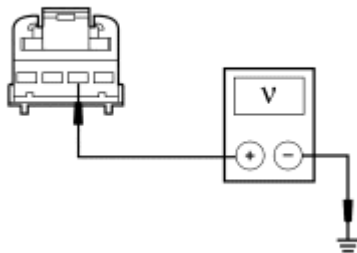
- Place the headlamp switch in the HEADLAMPS ON position.
- **Do the headlamps operate correctly?**

YES : Go to F2.

NO : REFER to Headlamps.

F2 CHECK FOR VOLTAGE TO THE LIGHT SENSOR

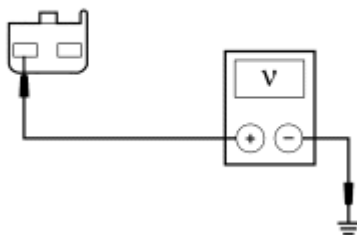
- Key in OFF position.
- Disconnect: Light Sensor C286 or C287
- Key in ON position.
- Place the headlamp switch in the AUTOLAMPS ON position.



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Fig. 25: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- For vehicles with EATC, measure the voltage between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



N0057373

Fig. 26: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- For vehicles with manual temperature control, measure the voltage between the light sensor C287a-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is the voltage approximately 5 volts?**

YES : INSTALL a new light sensor. REFER to Light Sensor. CLEAR the DTCs. REPEAT the self-test.

NO : Go to F3.

F3 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.

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- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test G: The Autolamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a reference voltage signal to the headlamp switch through circuit CLF19 (VT/GN) and a reference voltage signal to the light sensor through circuit VLF14 (BU/BN). Ground for the light sensor is provided by circuit GD115 (BK/GY).

When the headlamp switch is placed in the AUTOLAMPS ON position, the voltage signal is routed to ground. When the SJB detects the headlamp switch in the AUTOLAMPS ON position, the SJB monitors the light sensor input. The light sensor determines the amount of light based on the input received from the ambient lighting conditions. When the ambient light level has reached a point (determined by the internal programming of the SJB), the SJB provides voltage to the headlamps, parking and license plate lamps.

DTC Description	Fault Trigger Conditions
B1472 - Lamp Headlamp Input Circuit Short to Ground	An on-demand DTC that sets when the SJB detects the headlamps on input circuit is shorted to ground.
B1696 - Autolamp On Circuit Short To Ground	An on-demand DTC that sets when the SJB detects the autolamps on input circuit shorted to ground.
B1791 - Autolamp Sensor Input Circuit Open	An on-demand DTC that sets when the SJB detects an open on the light sensor input circuit.
B1793 - Autolamp Sensor Input Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the light sensor input circuit.
B2008 - Wipers On Signal Circuit Short to Ground	A n on-demand DTC that sets when the SJB detects the wipers on input circuit is short to ground.
	An on-demand DTC that sets when the SJB detects

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B2A22 - Headlamp Off Circuit Open | the headlamp switch off circuit open.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector
- Light sensor
- Headlamp switch
- Windshield wiper motor
- SJB

PINPOINT TEST G: THE AUTOLAMPS ARE ON CONTINUOUSLY

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

G1 USE THE RECORDED DTCs FROM THE SJB SELF-TEST

- Check the recorded results from the SJB self-test.
- **Was DTC B2008, B1791 or B1793 recorded?**

YES : For DTC B2008, go to G10.

For DTC B1791, go to G12.

For DTC B1793, go to G16.

NO : Go to G2.

G2 CHECK THE SJB HEADLAMP SWITCH PIDs

- Enter the following diagnostic mode on the diagnostic tool: SJB Data Logger

NOTE: Make sure the headlamp switch is lined up in the correct position when monitoring the PIDs.

- Monitor the SJB headlamp switch PIDs (HD_LMP_S2 and HEAD_L) while moving the headlamp switch through all positions. Refer to the following chart.

Headlamp Switch Position	HD_LMP_S2 Expected Value	HEAD_L Expected Value
Autolamps	AUTOLAMP	OFF

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OFF	-	OFF
PARKING LAMPS	-	-
HEADLAMPS	-	HEADLAMP

- Do the headlamp switch positions agree with the PIDs?

YES : Go to G9.

NO : Go to G3.

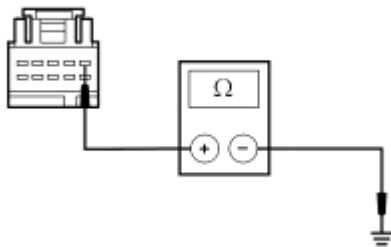
G3 CHECK THE HEADLAMP SWITCH

- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Carry out the headlamp switch component test. Refer to appropriate COMPONENT TESTING article.
- Is the headlamp switch OK?

YES : Go to G4.

NO : INSTALL a new headlamp switch. REFER to Headlamp Switch. CLEAR the DTCs. REPEAT the self-test.

G4 CHECK FOR CONTINUITY TO GROUND FROM THE HEADLAMP SWITCH



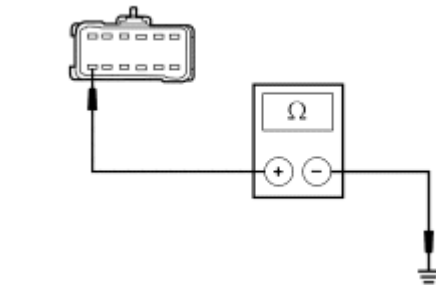
N0038799

Fig. 27: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-1, circuit GD116 (BK/VT), harness side and ground.
 - Is the resistance less than 5 ohms?
- YES** : Go to G7.
- NO** : Go to G5.

G5 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN (JOINT CONNECTOR TO GROUND)

- Disconnect: Joint Connector 6 C2365



N0062217

Fig. 28: Measuring Resistance Between Joint Connector 6 C2365-A, Circuit GD116 (BK/VT) & Ground

Courtesy of FORD MOTOR CO.

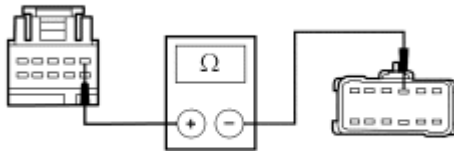
- Measure the resistance between the joint connector 6 C2365-A, circuit GD116 (BK/VT), harness side and ground.

- **Is the resistance less than 5 ohms?**

YES : Go to G6.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G6 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN (HEADLAMP SWITCH TO JOINT CONNECTOR)



N0062218

Fig. 29: Measuring Resistance Between Headlamp Switch C205-1, Circuit GD116 (BK/VT) & Joint Connector 6 C2365-J

Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-1, circuit GD116 (BK/VT), harness side and joint connector 6 C2365-J, circuit GD116 (BK/VT), harness side.

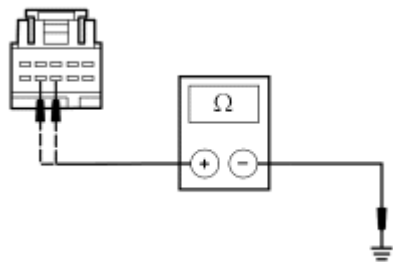
- **Is the resistance less than 5 ohms?**

YES : INSTALL a new joint connector. CLEAR the DTCs. REPEAT the self-test.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G7 CHECK THE HEADLAMP SWITCH INPUT CIRCUITS FOR A SHORT TO GROUND

- Disconnect: SJB C2280b



N0077747

Fig. 30: Measuring Resistance Between Headlamp Switch, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch, harness side and ground as follows:

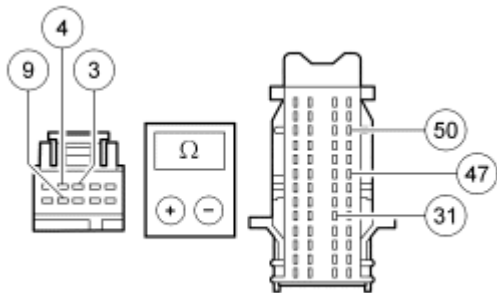
Connector-Pin	Circuit
C205-9	CLF19 (VT/GN)
C205-8	CLF18 (BU/WH)

- Are the resistances greater than 10,000 ohms?

YES : Go to G8.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

G8 CHECK THE HEADLAMP SWITCH INPUT CIRCUITS FOR AN OPEN



N0077748

Fig. 31: Measuring Resistance Between Headlamp Switch, Harness Side & SJB
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch, harness side and the SJB, harness side as follows:

Headlamp Switch	SJB Connector-	Circuit
-----------------	----------------	---------

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Connector-Pin	Pin	
C205-9	C2280b-50	CLF19 (VT/GN)
C205-4	C2280b-47	CLF23 (WH/VT)
C205-3	C2280b-31	CLS34 (GY)

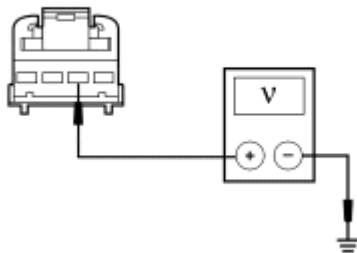
- Are the resistances less than 5 ohms?

YES : Go to G18.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

G9 CHECK FOR VOLTAGE TO THE LIGHT SENSOR

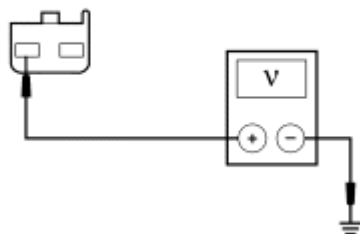
- Key in OFF position.
- Disconnect: Light Sensor C286 or C287
- Key in ON position.
- Place the headlamp switch in the AUTOLAMPS ON position.



N0057372

Fig. 32: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- For vehicles with EATC, measure the voltage between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



N0057373

Fig. 33: Checking Circuit For Voltage

Courtesy of FORD MOTOR CO.

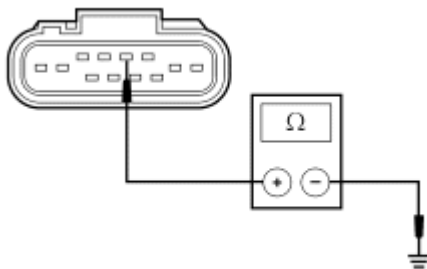
- For vehicles with manual temperature control, measure the voltage between the light sensor C287a-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is the voltage approximately 5 volts?**
YES : INSTALL a new light sensor. REFER to **Light Sensor**. CLEAR the DTCs. REPEAT the self-test.
NO : Go to G18.

G10 CHECK THE WINDSHIELD WIPER MOTOR

- Disconnect: Windshield Wiper Motor C1552
- Enter the following diagnostic mode on the diagnostic tool: SJB Self-Test
- Repeat the SJB self-test.
- **Is DTC B2008 still present?**
YES : Go to G11.
NO : INSTALL a new windshield wiper motor. REFER to **WIPERS AND WASHERS** article. TEST the system for normal operation.

G11 CHECK CIRCUIT CRW01 (WH) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280f



N0046955

Fig. 34: Checking Circuit For Short To Ground
Courtesy of FORD MOTOR CO.

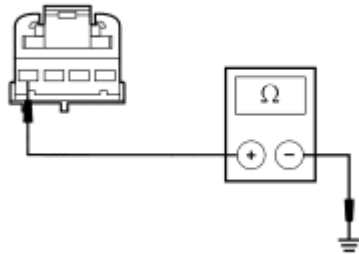
- Measure the resistance between the windshield wiper motor C1552-2, circuit CRW01 (WH), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to G18.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G12 CHECK CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.
- Disconnect: Light Sensor C286 or C287

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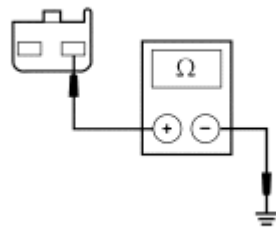
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N0038803

Fig. 35: Measuring Resistance
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the resistance between the light sensor C286-4, circuit GD115 (BK/GY), harness side and ground.



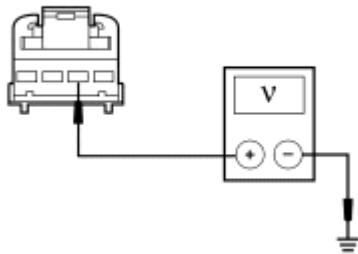
N0057377

Fig. 36: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- On vehicles with manual climate control, measure the resistance between the light sensor C287-B, circuit GD115 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to G13.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G13 CHECK CIRCUIT VLF14 (BU/BN) FOR A SHORT TO VOLTAGE

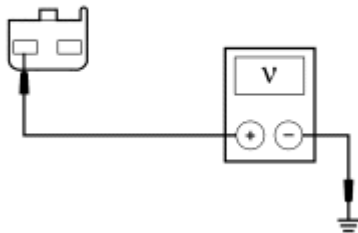
- Disconnect: SJB C2280b
- Key in ON position.



N0057372

Fig. 37: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the voltage between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



N0057373

Fig. 38: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

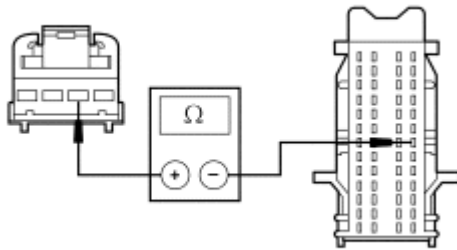
- On vehicles with manual climate control, measure the voltage between the light sensor C287-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
NO : Go to G14.

G14 CHECK CIRCUIT VLF14 (BU/BN) FOR AN OPEN

- Key in OFF position.

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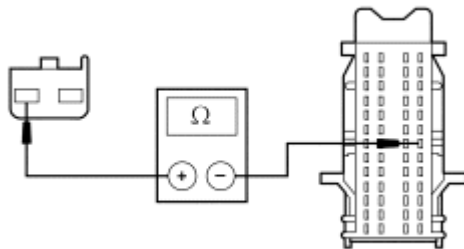
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N0057378

Fig. 39: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the resistance between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and the SJB C2280b-46, circuit VLF14 (BU/BN), harness side.



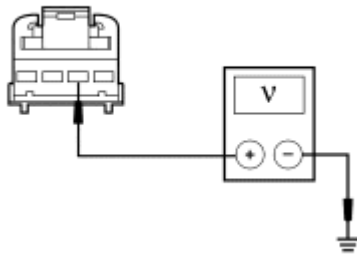
N0057379

Fig. 40: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- On vehicles with manual climate control, measure the resistance between the light sensor C287-A, circuit VLF14 (BU/BN), harness side and the SJB C2280b-46, circuit VLF14 (BU/BN), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to G15.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G15 CHECK FOR VOLTAGE TO THE LIGHT SENSOR (DTC B1791)

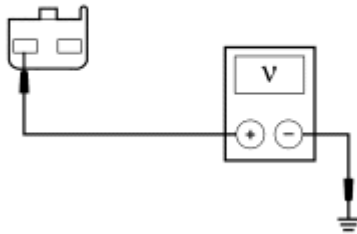
- Connect: SJB C2280b
- Key in ON position.
- Place the headlamp switch in the AUTOLAMPS ON position.



N0057372

Fig. 41: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the voltage between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



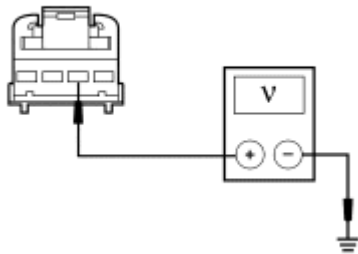
N0057373

Fig. 42: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- On vehicles with manual climate control, measure the voltage between the light sensor C287-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is the voltage approximately 5 volts?**
YES : INSTALL a new light sensor. REFER to Light Sensor. CLEAR the DTCs. REPEAT the self-test.
NO : Go to G18.

G16 CHECK FOR VOLTAGE TO THE LIGHT SENSOR (DTC B1793)

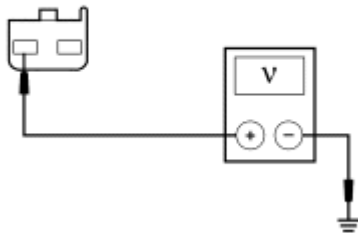
- Disconnect: Light Sensor C286 or C287
- Key in ON position.
- Place the headlamp switch in the AUTOLAMPS ON position.



N0057372

Fig. 43: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the voltage between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



N0057373

Fig. 44: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

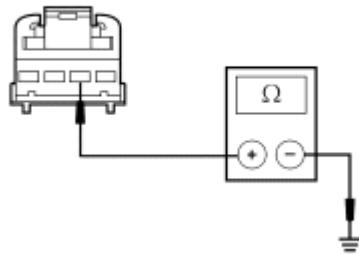
- On vehicles with manual temperature control, measure the voltage between the light sensor C287-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is the voltage approximately 5 volts?**

YES : INSTALL a new light sensor. REFER to Light Sensor. CLEAR the DTCs. REPEAT the self-test.

NO : Go to G17.

G17 CHECK CIRCUIT VLF14 (BU/BN) FOR A SHORT TO GROUND

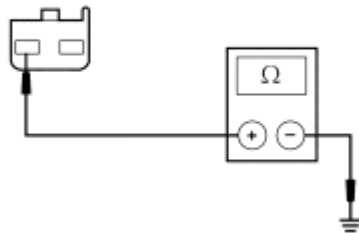
- Key in OFF position.
- Disconnect: SJB C2280b



N0057374

Fig. 45: Checking Circuit For Short To Ground
Courtesy of FORD MOTOR CO.

- On vehicles with EATC, measure the resistance between the light sensor C286-2, circuit VLF14 (BU/BN), harness side and ground.



N0057375

Fig. 46: Checking Circuit For Short To Ground
Courtesy of FORD MOTOR CO.

- On vehicles with manual temperature control, measure the resistance between the light sensor C287-A, circuit VLF14 (BU/BN), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to G18.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

G18 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES**

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article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

STOPLAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

The SJB monitors the input from the stoplamp switch. When the brake pedal is applied, voltage is routed to the SJB and the high mounted stoplamp. The SJB then supplies voltage to the stoplamps.

Field-Effect Transistor (FET) Protection

The SJB utilizes a FET protective circuit strategy for many of its outputs (for example, the headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault is detected. A continuous DTC is stored at that time of the fault. The circuit then resets after a customer demand of the function (switching the component on, battery saver being energized). When an excessive circuit load occurs several times, the module shuts down the output until a repair procedure is carried out. At the same time, the continuous DTC that was stored on the first failure, does not clear by a command to clear the continuous DTCs. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful on-demand self-test proves that the fault has been repaired. After the on-demand self-test has successfully completed (no on-demand DTCs present), the continuous DTC clears and the circuit function returns.

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Each circuit has 3 predefined levels of short circuit tolerance established in the module based on each circuits' capability. When the first or second level is reached, the continuous DTC associated with the circuit sets along with DTC B106E. These DTCs may be cleared using the Clear DTC operation on the scan tool as long as the fault itself has been corrected. If any of the circuits are shorted past the third level, then DTCs B106F and B1342 set along with the associated continuous DTC. This DTC cannot be cleared and the module must be replaced.

The SJB FET protected output circuits for the stoplamp system are for the LH rear stop/turn lamp output and the RH rear stop/turn lamp output circuits.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Stoplamp switch	• Battery junction box (BJB) fuse 71 (10A) (stoplamp switch) • Smart junction box (SJB) fuse 6 (20A) (rear stoplamps) • Bulb(s) • Bulb holder(s) • Wiring, terminals or connectors • SJB

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

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5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B1485	Brake Pedal Input Short to Battery	Go to Pinpoint Test J .
B2044	Left Rear Stop Lamp Circuit Short to Ground	Go to Pinpoint Test I .
B2045	Left Rear Stop Lamp Circuit Open	If the lamp is inoperative, go to Pinpoint Test I . If the lamp is always on, go to Pinpoint Test J .
B2046	Right Rear Stop Lamp Circuit Short to Ground	Go to Pinpoint Test I .
B2047	Right Rear Stop Lamp Circuit Open	If the lamp is inoperative, go to Pinpoint Test I . If the lamp is always on, go to Pinpoint Test J .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• No communication with	• Fuse	• REFER to MODULE

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the smart junction box (SJB)	• Wiring, terminals or connectors • SJB	<u>COMMUNICATIONS NETWORK</u> article.
• All the stoplamps are inoperative	• Fuse • Wiring, terminals or connectors • Stoplamp switch	• Go to Pinpoint Test H .
• One or more stoplamps are inoperative	• Wiring, terminals or connectors • Bulb holder • High mounted stoplamp • SJB	• Go to Pinpoint Test I .
• The stoplamps are on continuously	• Wiring, terminals or connectors • Stoplamp switch • PCM • SJB	• Go to Pinpoint Test J .

Pinpoint Tests

Pinpoint Test H: All The Stoplamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

The stoplamp switch is supplied voltage through circuit SBB71 (WH/RD) from the battery junction box (BJB). When the brake pedal is applied, the stoplamp switch routes voltage to the smart junction box (SJB) and the high mounted stoplamp through circuit CCB08 (VT/WH).

This pinpoint test is intended to diagnose the following:

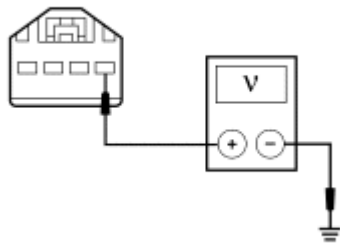
- Fuse
- Wiring, terminals or connectors
- Stoplamp switch

PINPOINT TEST H: ALL THE STOPLAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

H1 CHECK CIRCUIT SBB71 (WH/RD) FOR AN OPEN

- Key in OFF position.
- Disconnect: Stoplamp Switch C2064

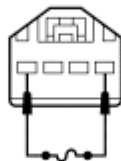


N0038805

Fig. 47: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the stoplamp switch C2064-1, circuit SBB71 (WH/RD), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : Go to H2.
NO : VERIFY the BJB fuse 71 (10A) is OK. If OK, REPAIR the circuit. TEST the system for normal operation.

H2 CHECK THE STOPLAMP SWITCH



N0038807

Fig. 48: Checking Stoplamp Switch
Courtesy of FORD MOTOR CO.

- Connect a fused (5A) jumper wire between the stoplamp switch C2064-1, circuit SBB71 (WH/RD), harness side and the stoplamp switch C2064-4, circuit CCB08 (VT/WH), harness side.
- **Are the stoplamps illuminated?**
YES : INSTALL a new stoplamp switch. REFER to **Stoplamp Switch**. TEST the system for normal operation.
NO : REPAIR circuit CCB08 (VT/WH) for an open. TEST the system for normal operation.

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Pinpoint Test I: One Or More Stoplamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

When the brake pedal is applied, the stoplamp switch routes voltage to the smart junction box (SJB) and the high mounted stoplamp through circuit CCB08 (VT/WH).

When the SJB detects the brake pedal is applied, the SJB provides voltage to the LH and RH rear stoplamps through circuits CLS18 (GY/BN) and CLS19 (VT/OG), respectively.

Ground for the LH and RH rear stoplamps is provided through circuits GD149 (BK/GY) and GD151 (BK/GN), respectively. Ground for the high mounted stoplamp is provided through circuit GD149 (BK/GY).

DTC Description	Fault Trigger Conditions
B2044 - Left Rear Stop Lamp Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the LH rear stop/turn lamp voltage supply circuit.
B2045 - Left Rear Stop Lamp Circuit Open	An on-demand DTC that sets when the SJB detects an open on the LH rear stop/turn lamp voltage supply circuit.
B2046 - Right Rear Stop Lamp Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the RH rear stop/turn lamp voltage supply circuit.
B2047 - Right Rear Stop Lamp Circuit Open	An on-demand DTC that sets when the SJB detects an open on the RH rear stop/turn lamp voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Bulb holder
- High mounted stoplamp
- SJB

PINPOINT TEST I: ONE OR MORE STOPLAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

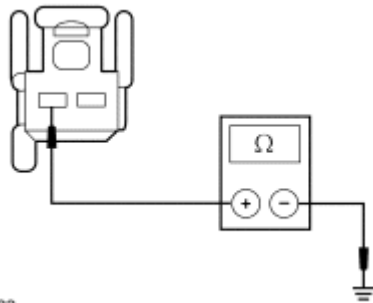
NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

I1 DETERMINE THE INOPERATIVE STOPLAMP

- Key in OFF position.
- Apply the brake pedal and observe the stoplamps.
- **Is the high mounted stoplamp illuminated?**
YES : Go to I4.
NO : Go to I2.

I2 CHECK CIRCUIT GD149 (BK/GY) FOR AN OPEN

- Disconnect: High Mounted Stoplamp C904

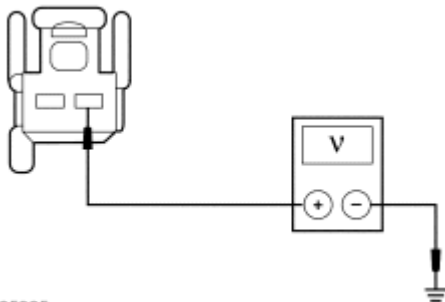


N0037283

Fig. 49: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the high mounted stoplamp C904-2, circuit GD149 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to I3.
NO : REPAIR the circuit. TEST the system for normal operation.

I3 CHECK CIRCUIT CCB08 (VT/WH) FOR AN OPEN (TO HIGH MOUNTED STOPLAMP)



A0025385

Fig. 50: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- While applying the brake pedal, measure the voltage between the high mounted stoplamp C904-1, circuit CCB08 (VT/WH), harness side and ground.

- **Is the voltage greater than 10 volts?**

YES : INSTALL a new high mounted stoplamp. TEST the system for normal operation.

NO : REPAIR the circuit. TEST the system for normal operation.

I4 USE THE RECORDED DTCs FROM THE SJB SELF-TEST

- Check the recorded results from the SJB on-demand self-test.
- **Are any DTCs present?**

YES : For DTCs B2044 or B2046, go to I6.

For DTCs B2045 or B2047, go to I8.

NO : Go to I5.

I5 CHECK CIRCUIT CCB08 (VT/WH) FOR AN OPEN (TO SJB)

- Disconnect: SJB C2280b

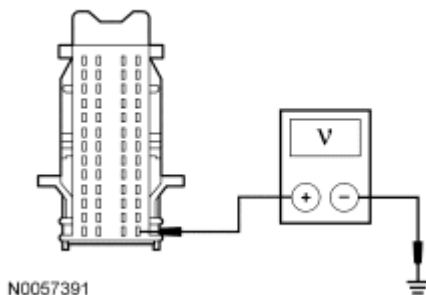


Fig. 51: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- While applying the brake pedal, measure the voltage between the SJB C2280b-40, circuit CCB08 (VT/WH), harness side and ground.

- **Is the voltage greater than 10 volts?**

YES : Go to I11.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

I6 CHECK THE BULB HOLDER

- Disconnect: LH Rear Lamp C412 (DTC B2044)
- Disconnect: RH Rear Lamp C415 (DTC B2046)
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB Self-Test
- Repeat the SJB on-demand self-test.
- **Is on-demand DTC B2044 or B2046 retrieved again?**

YES : Go to I7.

NO : INSTALL a new bulb holder. CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

I7 CHECK CIRCUIT CLS18 (GY/BN) OR CIRCUIT CLS19 (VT/OG) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280d

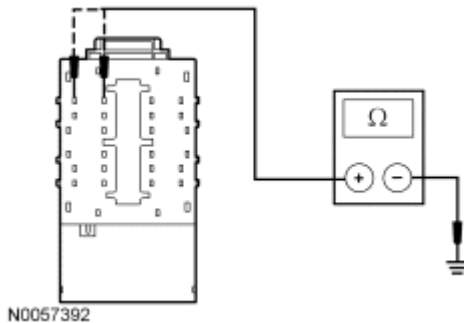


Fig. 52: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the SJB C2280d-25, circuit CLS18 (GY/BN), harness side and ground; or between the SJB C2280d-26, circuit CLS19 (VT/OG), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to I11.

NO : REPAIR the circuit in question. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

I8 CHECK CIRCUIT GD149 (BK/GY) (LH STOPLAMP) OR CIRCUIT GD151 (BK/GN) (RH STOPLAMP) FOR AN OPEN

- Disconnect: Inoperative Rear Lamp

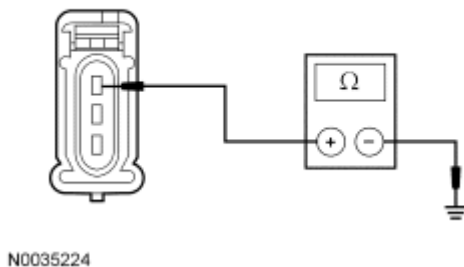


Fig. 53: Measuring Resistance
Courtesy of FORD MOTOR CO.

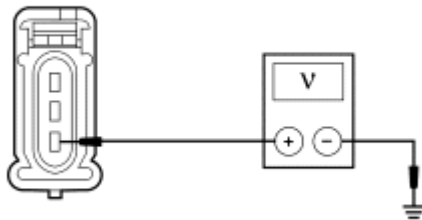
- Measure the resistance between the LH rear lamp C412-1, circuit GD149 (BK/GY), harness side and ground; or between the RH rear lamp C415-1, circuit GD151 (BK/GN), harness side and ground.

- Is the resistance less than 5 ohms?

YES : Go to I9.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

I9 CHECK FOR VOLTAGE TO THE BULB HOLDER



N0064675

Fig. 54: Measuring Voltage Between LH Rear Lamp C412-3 & C415-3, Circuit CLS18 (GY/BN) & CLS19 (VT/OG)

Courtesy of FORD MOTOR CO.

- While applying the brake pedal, measure the voltage between the LH rear lamp C412-3, circuit CLS18 (GY/BN), harness side and ground; or between the RH rear lamp C415-3, circuit CLS19 (VT/OG), harness side and ground.

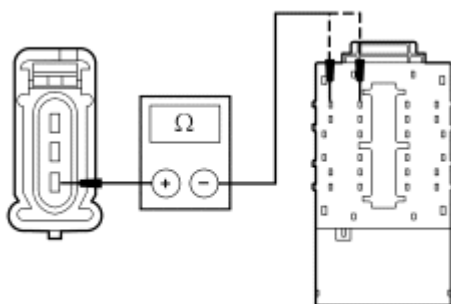
- Is the voltage greater than 10 volts?

YES : INSTALL a new bulb holder. CLEAR the DTCs. REPEAT the self-test.

NO : Go to I10.

I10 CHECK THE CIRCUIT CLS18 (GY/BN) OR CIRCUIT CLS19 (VT/OG) FOR AN OPEN

- Disconnect: SJB C2280d



N0062219

Fig. 55: Measuring Resistance Between LH Rear Lamp C412-3, Circuit CLS18 (GY/BN) & SJB C2280D-25

Courtesy of FORD MOTOR CO.

- Measure the resistance between the LH rear lamp C412-3, circuit CLS18 (GY/BN), harness side and the SJB C2280d-25, circuit CLS18 (GY/BN), harness side; or between the RH rear lamp C415-3, circuit CLS19 (VT/OG), harness side and the SJB C2280d-26, circuit CLS19 (VT/OG), harness

side.

- **Is the resistance less than 5 ohms?**

YES : Go to I11.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

I11 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test J: The Stoplamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

When the brake pedal is applied, the stoplamp switch routes voltage to the smart junction box (SJB), the high mounted stoplamp and the PCM through circuit CCB08 (VT/WH). When the SJB detects the brake pedal applied, the SJB provides voltage to the LH and RH rear stoplamps through circuits CLS18 (GY/BN) and CLS19 (VT/OG), respectively. If equipped with trailer tow, voltage is also supplied to the respective trailer tow stop/turn relay.

- DTC B1485 (Brake Pedal Input Short to Battery) - is an on-demand DTC that sets when the SJB detects a short to voltage from the stoplamp switch input circuit.
- DTC B2045 (Left Rear Stop Lamp Circuit Open) - is an on-demand DTC that sets when the SJB detects a short to voltage on the LH rear stop/turn lamp voltage supply circuit.
- DTC B2047 (Right Rear Stop Lamp Circuit Open) - is an on-demand DTC that sets when the SJB detects a short to voltage on the RH rear stop/turn lamp voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Stoplamp switch
- PCM
- SJB

PINPOINT TEST J: THE STOPLAMPS ARE ON CONTINUOUSLY**J1 DETERMINE IF ALL THE STOPLAMPS ARE ILLUMINATED**

- Key in OFF position.
- Observe the stoplamps with the brake pedal released.
- **Are all of the stoplamps illuminated?**

YES : Go to J3.

NO : Go to J2.

J2 CHECK THE STOPLAMPS VOLTAGE SUPPLY CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: SJB C2280d
- **Do the stoplamps continue to illuminate?**

YES : If the LH rear lamp continues to illuminate, REPAIR circuit CLS18 (GY/BN). CLEAR the DTCs. REPEAT the self-test.

If the RH rear lamp continues to illuminate, REPAIR circuit CLS19 (VT/OG). CLEAR the DTCs. REPEAT the self-test.

NO : Go to J7.

J3 CHECK THE STOPLAMP SWITCH

- Disconnect: Stoplamp Switch C2064
- **Do the stoplamps continue to illuminate?**

YES : Go to J4.

NO : INSTALL a new stoplamp switch. REFER to Stoplamp Switch. CLEAR the DTCs. REPEAT the self-test.

J4 CHECK THE PCM

- Disconnect: PCM C175b
- **Do the stoplamps continue to illuminate?**

YES : Go to J5.

NO : Go to J6.

J5 CHECK CIRCUIT CCB08 (VT/WH) FOR A SHORT TO VOLTAGE

- Disconnect: SJB C2280b
- **Does the high mounted stoplamp continue to illuminate?**

YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

NO : Go to J7.

J6 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
 - corrosion
 - damaged pins

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- pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new PCM. REFER to **ELECTRONIC ENGINE CONTROLS** article.
CLEAR the DTCs. REPEAT the self-test.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

J7 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

TURN SIGNAL AND HAZARD LAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
	Flex Probe Kit	105-R025C or equivalent

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Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

The SJB monitors the multifunction switch position by sending multiple voltage reference signals to the multifunction switch. When the multifunction switch is in the LH or RH TURN positions, or the hazard lamp switch is pressed, that input signal is routed to ground.

When the SJB receives a request for a turn signal or hazard lamps, the SJB supplies voltage to the appropriate turn lamps.

The timed on/off cycle is determined by the SJB and is set to flash approximately 80 times per minute if both the front and rear turn lamps operate correctly. If an individual turn signal lamp is inoperative, the SJB flashes the remaining turn lamp approximately 160 times per minute.

The timed on/off cycle for the hazard lamps is set to flash approximately 80 times per minute.

Field-Effect Transistor (FET) Protection

The SJB utilizes a FET protective circuit strategy for many of its outputs (for example, the headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault is detected. A continuous DTC is stored at that time of the fault. The circuit then resets after a customer demand of the function (switching the component on, battery saver being energized). When an excessive circuit load occurs several times, the module shuts down the output until a repair procedure is carried out. At the same time, the continuous DTC that was stored on the first failure, does not clear by a command to clear the continuous DTCs. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful on-demand self-test proves that the fault has been repaired. After the on-demand self-test has successfully completed (no on-demand DTCs present), the continuous DTC clears and the circuit function returns.

Each circuit has 3 predefined levels of short circuit tolerance established in the module based on each circuit's capability. When the first or second level is reached, the continuous DTC associated with the circuit sets along with DTC B106E. These DTCs may be cleared using the Clear DTC operation on the scan tool as long as the fault itself has been corrected. If any of the circuits are shorted past the third level, then DTCs B106F and B1342 set along with the associated continuous DTC. This DTC cannot be cleared and the module must be replaced.

The SJB FET protected output circuits for the turn lamp system are for the LH front turn lamp output and the RH front turn lamp output circuits. The rear turn lamps utilize the stoplamp circuitry and are also protected. For

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information regarding the rear stoplamps, refer to **Stoplamps**.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Multifunction switch • Hazard flasher lamp switch	• Bulb(s) • Wiring, terminals or connectors • Smart junction box (SJB)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.

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- If the network test passes, retrieve and record the continuous memory DTCs.
- 8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
- 9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
- 10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B2071	Hazard Switch Signal Short to Ground	Go to Pinpoint Test N .
B2281	Right Turn Switch Short to Ground	Go to Pinpoint Test L .
B2282	Left Turn Switch Short to Ground	Go to Pinpoint Test L .
B2A27	Right Front Turn Lamp Circuit Open	Go to Pinpoint Test M .
B2A28	Right Front Turn Lamp Circuit Short to Ground	Go to Pinpoint Test M .
B2A29	Left Front Turn Lamp Circuit Open	Go to Pinpoint Test M .
B2A2A	Left Front Turn Lamp Circuit Short to Ground	Go to Pinpoint Test M .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• No communication with the smart junction box (SJB)	• Fuse • Wiring, terminals or connectors • SJB	• REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
• The turn signal lamps are inoperative	• Wiring, terminals or connectors • Multifunction switch • SJB	• Go to Pinpoint Test K .
• The turn signal lamps are always on	• Wiring, terminals or connectors • Multifunction switch	• Go to Pinpoint Test L .

	• SJB	
• One turn signal/hazard lamp is inoperative/always on	• Wiring, terminals or connectors • Joint connector • Bulb holder (Edge) • Headlamp assembly (MKX) • SJB	• Go to Pinpoint Test M .
• The hazard lamps are inoperative/always on	• Wiring, terminals or connectors • Joint connector • Hazard flasher lamp switch • SJB	• Go to Pinpoint Test N .

Pinpoint Tests

Pinpoint Test K: The Turn Signal Lamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends reference signals to the multifunction switch through circuit CLS39 (VT/WH) (LH turn input) and circuit CLS41 (GY/YE) (RH turn input). When the multifunction switch is switched to the LH or RH TURN positions, the voltage signal is routed to ground. When the SJB detects the multifunction switch in the LH or RH TURN position, the SJB sends voltage to the appropriate turn lamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Multifunction switch
- SJB

PINPOINT TEST K: THE TURN SIGNAL LAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

K1 CHECK FOR CORRECT HIGH BEAM OPERATION

- Place the headlamp switch in the HEADLAMPS ON position.

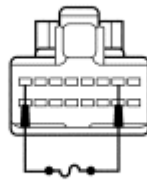
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- Place the multifunction switch in the HIGH BEAM position while observing the high beam headlamps.
- **Do the high beams operate correctly?**
YES : PLACE the headlamp switch in the OFF position. Go to K2.
NO : REFER to **Headlamps**.

K2 CHECK THE MULTIFUNCTION SWITCH

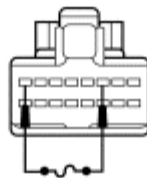
- Key in OFF position.
- Disconnect: Multifunction Switch C202



N0069012

Fig. 56: Connecting Fused (5A) Jumper Wire Between Multifunction Switch C202-2 & C202-8, CLS39 (VT/WH) & GD116 (BK/VT)
Courtesy of FORD MOTOR CO.

- For an inoperative LH turn signal, connect a fused (5A) jumper wire between the multifunction switch C202-2, circuit CLS39 (VT/WH), harness side and the multifunction switch C202-8, circuit GD116 (BK/VT), harness side.



N0069013

Fig. 57: Connecting Fused (5A) Jumper Wire Between Multifunction Switch C202-3, CLS41 (GY/YE) & GD116 (BK/VT)
Courtesy of FORD MOTOR CO.

- For an inoperative RH turn signal, connect a fused (5A) jumper wire between the multifunction switch C202-3, circuit CLS41 (GY/YE), harness side and the multifunction switch C202-3, circuit GD116 (BK/VT), harness side.
- Key in ON position.

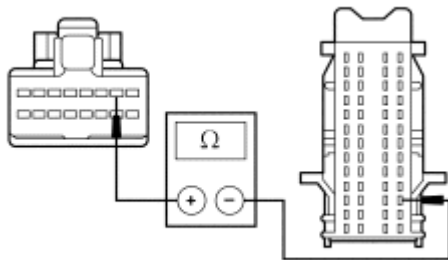
- **Do the turn signals in question flash on and off?**

YES : REMOVE the jumper wire. INSTALL a new multifunction switch. REFER to **STEERING COLUMN SWITCHES** article. TEST the system for normal operation.

NO : REMOVE the jumper wire. Go to K3.

K3 CHECK CIRCUIT CLS39 (VT/WH) (LH TURN SIGNAL) OR CLS41 (GY/YE) (RH TURN SIGNAL) FOR AN OPEN

- Disconnect: SJB C2280b

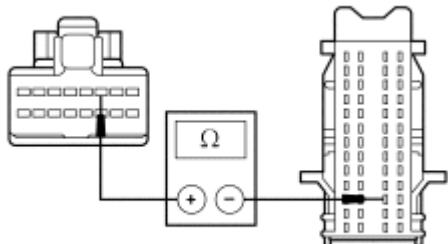


N0072831

Fig. 58: Measuring Resistance Between Multifunction Switch C202-2, Circuit CLS39 (VT/WH) & SJB C2280B-42

Courtesy of FORD MOTOR CO.

- For an inoperative LH turn signal, measure the resistance between the multifunction switch C202-2, circuit CLS39 (VT/WH), harness side and the SJB C2280b-42, circuit CLS39 (VT/WH), harness side.



N0072832

Fig. 59: Measuring Resistance Between Multifunction Switch C202-3, Circuit CLS41 (GY/YE) & SJB C2280B-29

Courtesy of FORD MOTOR CO.

- For an inoperative RH turn signal, measure the resistance between the multifunction switch C202-3, circuit CLS41 (GY/YE), harness side and the SJB C2280b-29, circuit CLS41 (GY/YE), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to K4.

NO : REPAIR the circuit in question. TEST the system for normal operation.

K4 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test L: The Turn Signal Lamps Are Always On

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends reference signals to the multifunction switch through circuits CLS39 (VT/WH) (LH turn input) and CLS41 (GY/YE) (RH turn input). When the multifunction switch is switched to the LH or RH TURN positions, the voltage signal is routed to ground. When the SJB detects the multifunction switch in the LH or RH TURN position, the SJB sends voltage to the appropriate turn lamps.

- DTC B2281 (Right Turn Switch Short to Ground) - is an on-demand DTC that sets when the SJB detects the RH turn signal switch input is short to ground.
- DTC B2282 (Left Turn Switch Short to Ground) - is an on-demand DTC that sets when the SJB detects the LH turn signal switch input is short to ground.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Multifunction switch
- SJB

PINPOINT TEST L: THE TURN SIGNAL LAMPS ARE ALWAYS ON

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

L1 CHECK THE MULTIFUNCTION SWITCH

- Key in OFF position.
- Disconnect: Multifunction Switch C202
- Key in ON position.
- **Do the turn lamps continue to flash on and off?**

YES : Go to L2.

NO : INSTALL a new multifunction switch. REFER to **STEERING COLUMN SWITCHES** article. CLEAR the DTCs. REPEAT the self-test.

L2 CHECK CIRCUIT CLS39 (VT/WH) (LH TURN SIGNAL) OR CIRCUIT CLS41 (GY/YE) (RH TURN SIGNAL) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280b

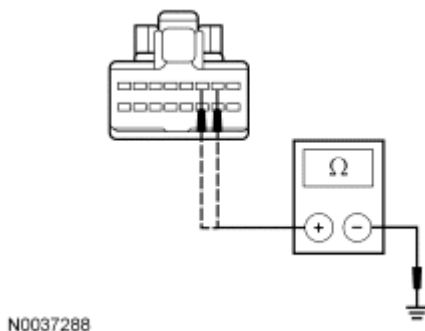


Fig. 60: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the multifunction switch C202-2 (LH turn lamps), circuit CLS39 (VT/WH), harness side and ground; or between the multifunction switch C202-3 (RH turn lamps), circuit CLS41 (GY/YE), harness side and ground.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to L3.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

L3 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

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YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DTCs. REPEAT the self-test.

Pinpoint Test M: One Turn Signal/Hazard Lamp Is Inoperative/Always On

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

When the smart junction box (SJB) detects the multifunction switch in the LH TURN position, the SJB provides voltage through circuit CLS21 (BU/GN) to the LH front turn lamp. When the SJB detects the multifunction switch in the RH TURN position, the SJB provides voltage through circuit CLS25 (YE/VT) to the RH front turn lamp. Circuits GD123 (BK/GY) and GD120 (BK/GN) provide ground for the LH and RH front turn lamps, respectively. The rear turn lamps are controlled utilizing the rear stoplamp circuitry.

DTC Description	Fault Trigger Conditions
B2A27 - Right Front Turn Lamp Circuit Open	An on-demand DTC that sets when the SJB detects an open or short to voltage on the RH front turn lamp voltage supply circuit.
B2A28 - Right Front Turn Lamp Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the RH front turn lamp voltage supply circuit.
B2A29 - Left Front Turn Lamp Circuit Open	An on-demand DTC that sets when the SJB detects an open or short to voltage on the LH front turn lamp voltage supply circuit.
B2A2A - Left Front Turn Lamp Circuit Short to Ground	A continuous and on-demand DTC that sets when the SJB detects a short to ground on the LH front turn lamp voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector
- Bulb holder (Edge)
- Headlamp assembly (MKX)
- SJB

PINPOINT TEST M: ONE TURN SIGNAL/HAZARD LAMP IS INOPERATIVE/ALWAYS ON

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

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NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

M1 CHECK THE STOPLAMPS

- Apply and release the brake pedal while observing the stoplamps.
- **Do the stoplamps operate correctly?**

YES : Go to M2.

NO : REFER to Stoplamps.

M2 DETERMINE IF A TURN LAMP IS ALWAYS ON

- Key in ON position.
- Monitor the turn lamps.
- **Is either front turn lamp always on?**

YES : Go to M3.

NO : Go to M4.

M3 CHECK CIRCUIT CLS21 (BU/GN) (LH TURN) OR CIRCUIT CLS25 (YE/VT) (RH TURN) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: SJB C2280f
- Key in ON position.
- **Does the front turn lamp in question continue to illuminate?**

YES : For the LH front turn lamp, REPAIR circuit CLS21 (BU/GN). CLEAR the DTCs. REPEAT the self-test.

For the RH front turn lamp, REPAIR circuit CLS25 (YE/VT). CLEAR the DTCs. REPEAT the self-test.

NO : Go to M11.

M4 USE THE RECORDED DTCs FROM THE SJB SELF-TEST

- Check the recorded results from the SJB on-demand self-test.
- **Was DTC B2A28 or B2A2A present?**

YES : Go to M5.

NO : Go to M7.

M5 CHECK THE BULB HOLDER FOR A SHORT TO GROUND

- Disconnect: Inoperative Front Turn Lamp
- Enter the following diagnostic mode on the diagnostic tool: SJB Self-Test
- Repeat the SJB on-demand self-test.
- **Is DTC B2A28 or B2A2A still present?**

YES : Go to M6.

NO : INSTALL a new bulb holder. TEST the system for normal operation.

M6 CHECK CIRCUIT CLS21 (BU/GN) (DTC B2A2A) OR CIRCUIT CLS25 (YE/VT) (DTC B2A28) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: SJB C2280f

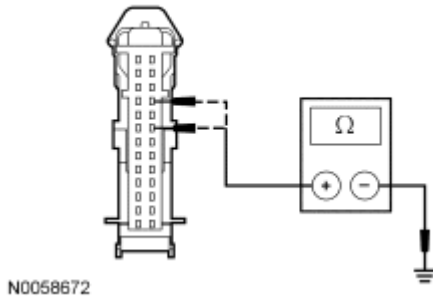


Fig. 61: Checking Circuits For Short To Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the SJB C2280f-21 (DTC B2A2A), circuit CLS21 (BU/GN), harness side and ground; or between the SJB C2280f-23 (DTC B2A28), circuit CLS25 (YE/VT), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to M11.

NO : REPAIR the circuit in question. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

M7 CHECK THE GROUND CIRCUIT FOR AN OPEN

NOTE: Make sure the headlamps operate correctly before continuing with diagnostics for the turn lamps.

- Key in OFF position.
- Disconnect: Inoperative Turn Lamp

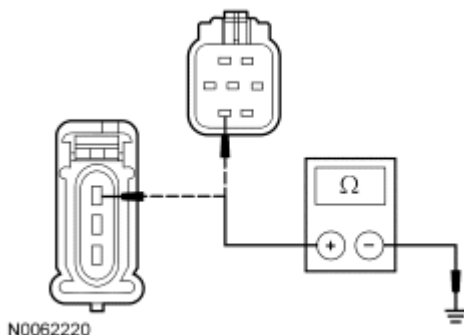


Fig. 62: Measuring Resistance Between Inoperative Turn Lamp, Harness Side & Ground

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Courtesy of FORD MOTOR CO.

- Measure the resistance between the inoperative turn lamp, harness side and ground as follows

Inoperative Turn Lamp	Connector-Pin	Circuit
Edge		
LH front	C1023-1	GD123 (BK/GY)
RH front	C1043-1	GD120 (BK/GN)
MKX		
LH front headlamp	C1021a-6	GD123 (BK/GY)
RH front headlamp	C1041a-6	GD123 (BK/GY)

- Is the resistance less than 5 ohms?

YES : Go to M9.

NO : For MKX, REPAIR the ground circuit in question for an open. CLEAR the DTCs. REPEAT the self-test.

For Edge LH front turn lamp, REPAIR circuit GD123 (BK/GY) for an open. CLEAR the DTCs. REPEAT the self-test.

For the Edge RH front turn lamp, go to M8.

M8 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN

- Disconnect: Joint Connector 7 C2200

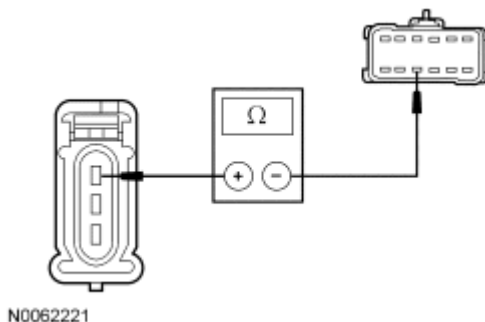


Fig. 63: Measuring Resistance Between RH Front Turn Lamp C1043-1, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-C
Courtesy of FORD MOTOR CO.

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- Measure the resistance between the RH front turn lamp C1043-1, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-C, circuit GD120 (BK/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. CLEAR the DTCs. REPEAT the self-test.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

M9 CHECK FOR VOLTAGE TO THE TURN LAMP

- Connect: SJB C2280e
- Turn the hazard lamps on.

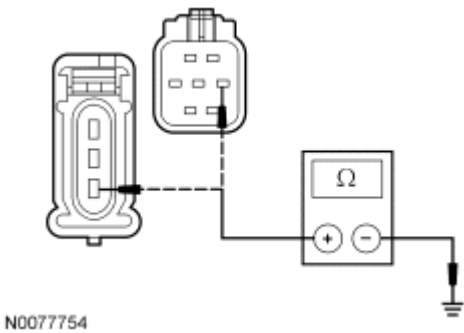


Fig. 64: Measuring Resistance Between Inoperative Turn Lamp, Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between the inoperative turn lamp, harness side and ground as follows:

Inoperative Turn Lamp Connector-Pin	Circuit
Edge	
LH front C1023-3	CLS21 (BU/GN)
RH front C1043-3	CLS25 (YE/VT)
MKX	
LH front headlamp C1021a-5	CLS21 (BU/GN)
RH front headlamp C1041a-5	CLS25 (YE/VT)

- **Does the voltage alternate from 0 volts to greater than 10 volts?**
YES : For Edge, INSTALL a new bulb holder. CLEAR the DTCs. REPEAT the self-test.

For MKX, REPAIR or INSTALL a new headlamp. REFER to **Headlamp Assembly**. CLEAR the DTCs. REPEAT the self-test.

NO : Go to M10.

M10 CHECK CIRCUIT CLS21 (BU/GN) (LH TURN) OR CIRCUIT CLS25 (YE/VT) (RH TURN) FOR AN OPEN

- Disconnect: SJB C2280f

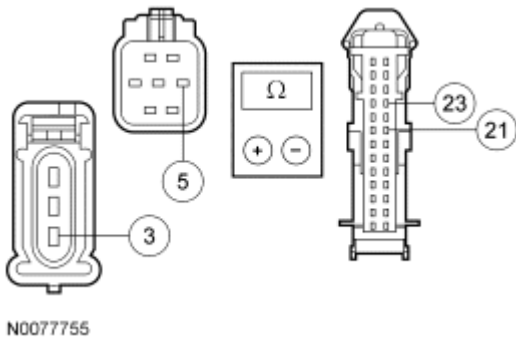


Fig. 65: Measuring Resistance Between Inoperative Turn Lamp, Harness Side & SJB
Courtesy of FORD MOTOR CO.

- Measure the resistance between the inoperative turn lamp, harness side and the SJB, harness side as follows:

Inoperative Turn Lamp Connector-Pin	SJB Connector-Pin	Circuit
Edge		
LH front C1023-3	C2280f-21	CLS21 (BU/GN)
RH front C1043-3	C2280f-23	CLS25 (YE/VT)
MKX		
LH front headlamp C1021a-5	C2280f-21	CLS21 (BU/GN)
RH front headlamp C1041a-5	C2280f-23	CLS25 (YE/VT)

- **Is the resistance less than 5 ohms?**

YES : Go to M11.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test

M11 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test N: The Hazard Lamps Are Inoperative/Always On

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a voltage reference signal to the hazard lamp switch through circuit CLS32 (BN/YE). The hazard flasher lamp switch is a momentary contact switch. When the hazard flasher lamp switch is engaged, the voltage signal is routed to ground through circuit GD116 (BK/VT). When the SJB detects a request for the hazard lamps, the SJB provides on/off voltage to all the turn lamps.

- DTC B2071 (Hazard Switch Signal Short to Ground) - is an on-demand DTC that sets when the SJB detects a short to ground from the hazard flasher lamp switch input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector
- Hazard flasher lamp switch
- SJB

PINPOINT TEST N: THE HAZARD LAMPS ARE INOPERATIVE/ALWAYS ON

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when

checking ground circuits for continuity to ground.

N1 CHECK THE RECORDED DTCs FROM THE SJB SELF-TEST

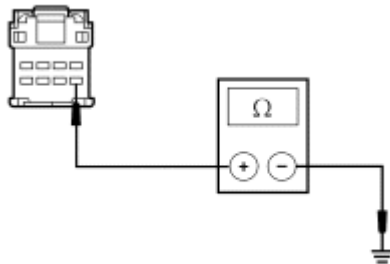
- Check the recorded DTCs from the SJB self-test.
- **Is DTC B2071 present?**
YES : Go to N2.
NO : Go to N4.

N2 CHECK THE HAZARD FLASHER LAMP SWITCH

- Disconnect: Hazard Flasher Lamp Switch C2355
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB Self-Test
- Repeat the SJB self-test.
- **Is DTC B2071 still present?**
YES : Go to N3.
NO : INSTALL a new hazard flasher lamp switch. REFER to Hazard Flasher Lamp Switch.
 CLEAR the DTCs. REPEAT the self-test.

N3 CHECK CIRCUIT CLS32 (BN/YE) FOR A SHORT TO GROUND

- Disconnect: SJB C2280b



N0062225

Fig. 66: Measuring Resistance Between Hazard Flasher Lamp Switch C2355-5, Circuit CLS32 (BN/YE) & Ground

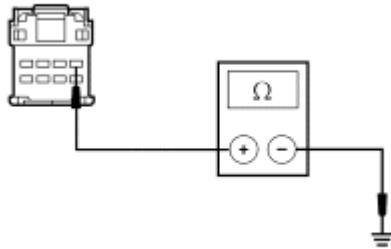
Courtesy of FORD MOTOR CO.

- Measure the resistance between the hazard flasher lamp switch C2355-5, circuit CLS32 (BN/YE), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to N8.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

N4 CHECK CIRCUIT GD116 (BK/VT) FOR CONTINUITY TO GROUND

NOTE: Make sure the headlamps operate correctly before continuing with diagnostics for the hazard lamps.

- Key in OFF position.
- Disconnect: Hazard Flasher Lamp Switch C2355



N0062223

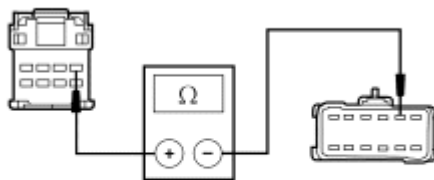
Fig. 67: Measuring Resistance Between Hazard Flasher Lamp Switch C2355-1, Circuit GD116 (BK/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the hazard flasher lamp switch C2355-1, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to N6.
NO : Go to N5.

N5 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN

NOTE: Make sure the headlamps operate correctly before continuing with diagnostics for the hazard lamps.

- Key in OFF position.
- Disconnect: Joint Connector 6 C2365



N0062224

Fig. 68: Measuring Resistance Between Hazard Flasher Lamp Switch C2355-1, GD116 (BK/VT) & Joint Connector 6 C2365-H
Courtesy of FORD MOTOR CO.

- Measure the resistance between the hazard flasher lamp switch C2355-1, circuit GD116 (BK/VT),

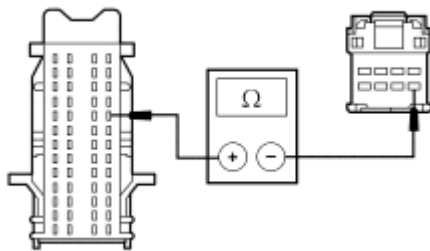
harness side and the joint connector 6 C2365-H, circuit GD116 (BK/VT), harness side.

- **Is the resistance less than 5 ohms?**

YES : INSTALL a new joint connector. CLEAR the DTCs. REPEAT the self-test.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

N6 CHECK CIRCUIT CLS32 (BN/YE) FOR AN OPEN



N0062226

Fig. 69: Measuring Resistance Between SJB C2280B-48, Circuit CLS32 (BN/YE) & Hazard Flasher Lamp Switch C2355-5

Courtesy of FORD MOTOR CO.

- Measure the resistance between the SJB C2280b-48, circuit CLS32 (BN/YE), harness side and the hazard flasher lamp switch C2355-5, circuit CLS32 (BN/YE), harness side.

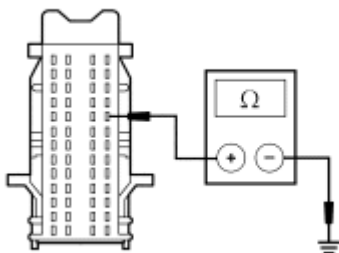
- **Is the resistance less than 5 ohms?**

YES : Go to N7.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

N7 CHECK THE HAZARD FLASHER LAMP SWITCH

- Connect: Hazard Flasher Lamp Switch C2355



N0058676

Fig. 70: Checking Hazard Flasher Lamp Switch

Courtesy of FORD MOTOR CO.

- While pressing and releasing the hazard flasher lamp switch, measure the resistance between the SJB C2280b-48, circuit CLS32 (BN/YE), harness side and ground.

- **Is the resistance less than 5 ohms with the switch pressed and greater than 10,000 ohms with the switch released?**

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YES : Go to N8.

NO : INSTALL a new hazard flasher lamp switch. REFER to **Hazard Flasher Lamp Switch**. CLEAR the DTCs. REPEAT the self-test.

N8 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

PARKING, REAR AND LICENSE PLATE LAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

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The SJB monitors the headlamp switch position by sending voltage reference signals on multiple circuits to the headlamp switch. There is one circuit for each headlamp switch position. At any given time, one of the signal circuits is routed to ground.

If the SJB does not detect any of the inputs to the headlamp switch are active (routed to ground) for 5 seconds, the SJB turns on the exterior lights and keeps them on for 10 minutes after the ignition switch is turned off (or 10 minutes from the time the SJB does not detect any headlamp switch input if the ignition switch was already off). Additionally, if the SJB detects multiple circuits short to ground, the SJB implements a planned strategy depending on the multiple inputs received.

If either of these situations occur, the SJB should **NOT** be ruled immediately as being at fault. This is normal behavior of the SJB design as it has detected a fault with the inputs from the headlamp switch.

When the SJB receives an input from the headlamp switch indicating a request for the parking lamps, the SJB energizes the parking lamp relay (internal to the SJB), which supplies voltage to the exterior lamps.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Headlamp switch	• Smart junction box (SJB) fuse 22 (15A) • Bulb(s) • Bulb holder(s) • Wiring, terminals or connectors • SJB

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

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5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B1578	Lamp Park Input Circuit Short to Ground	Go to Pinpoint Test P .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• No communication with the smart junction box (SJB)	• Fuse • Wiring, terminals or connectors • SJB	• REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
• One or more parking, rear	• Fuse • Wiring, terminals or connectors • Bulb Holder	

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or license plate lamps is inoperative	• Liftgate lamp • Liftgate release handle • Headlamp assembly • SJB	• Go to Pinpoint Test O .
• The parking, rear or license plate lamps are on continuously	• Wiring, terminals or connectors • Headlamp switch • SJB	• Go to Pinpoint Test P .

Pinpoint Tests

Pinpoint Test O: One Or More Parking, Rear or License Plate Lamps Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Parking, Rear and License Lamps for schematic and connector information.

Normal Operation

When the headlamp switch is placed in the PARKING LAMPS ON position, the smart junction box (SJB) provides a ground for the parking lamp relay coil (integrated into the SJB). When the parking lamp relay is energized, voltage is routed through circuit CLS30 (VT/WH) to the rear and front parking lamps. Ground is provided to the rear parking, liftgate and license plate lamps through circuit GD149 (BK/GY) or circuit GD151 (BK/GN). Ground is provided to the front parking and side marker lamps through circuit GD120 (BK/GN) or circuit GD123 (BK/GY).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Bulb Holder
- Liftgate lamp
- Liftgate release handle
- Headlamp assembly
- SJB

PINPOINT TEST O: ONE OR MORE PARKING, REAR OR LICENSE PLATE LAMPS IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

O1 DETERMINE IF ALL THE PARKING LAMPS ARE INOPERATIVE

- Key in OFF position.
- Place the headlamp switch to the PARKING LAMPS ON position.
- **Are all the parking lamps inoperative?**

YES : PLACE the headlamp switch in the OFF position. VERIFY the SJB fuse 22 (15A) is OK. If OK, go to O6.

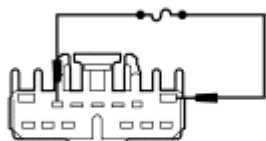
NO : If all the front parking lamps are inoperative, go to O2.

If all the rear parking and license plate lamps are inoperative, go to O3.

For all others, go to O4.

O2 CHECK CIRCUIT CLS30 (VT/WH) FOR AN OPEN (FRONT LAMPS)

- Disconnect: SJB C2280e



N0058677

Fig. 71: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (5A) jumper wire between the SJB C2280e-1, circuit CBB53 (GY/BN), harness side and the SJB C2280e-6, circuit CLS30 (VT/WH), harness side.
- Key in ON position.
- **Do the front parking lamps illuminate?**
YES : REMOVE the jumper wire. Go to O6.
NO : REMOVE the jumper wire. REPAIR the circuit. TEST the system for normal operation.

O3 CHECK CIRCUIT CLS30 (VT/WH) FOR AN OPEN (REAR LAMPS)

- Disconnect: SJB C2280d

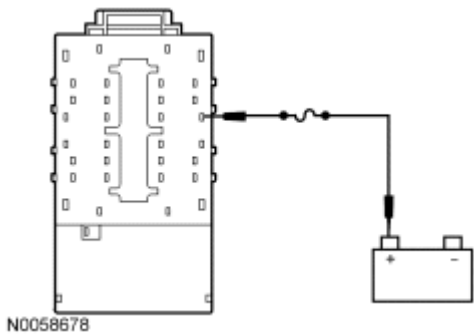


Fig. 72: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (5A) jumper wire between the positive battery terminal and the SJB C2280d-20, circuit CLS30 (VT/WH), harness side.
- **Do the rear parking lamps illuminate?**
YES : REMOVE the jumper wire. Go to O6.
NO : REMOVE the jumper wire. REPAIR the circuit. TEST the system for normal operation.

O4 CHECK THE PARKING LAMP GROUND CIRCUIT FOR AN OPEN

NOTE: If a front parking lamp is inoperative, make sure the front turn signal operates correctly before continuing diagnostics for the front parking lamp.

- Disconnect: Inoperative Lamp
- Measure the resistance between the inoperative lamp, harness side and ground as follows:

EDGE

Inoperative Lamp	Connector-Pin	Circuit
LH front	C1023-1	GD123 (BK/GY)
RH front	C1043-1	GD120 (BK/GN)
LH rear	C412-1	GD149 (BK/GY)
RH rear	C415-1	GD151 (BK/GN)
LH rear side marker	C4136-1	GD149 (BK/GY)
RH rear side marker	C4138-1	GD151 (BK/GN)
Liftgate release		GD149

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handle (license plate lamps)	C4353-2	(BK/GY)
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MKX

Inoperative Lamp	Connector-Pin	Circuit
LH front (headlamp)	C1021a-6	GD123 (BK/GY)
RH front (headlamp)	C1041a-6	GD123 (BK/GY)
LH front side marker	C151-2	GD120 (BK/GN)
RH front side marker	C161-2	GD123 (BK/GY)
LH rear	C412-1	GD149 (BK/GY)
RH rear	C415-1	GD151 (BK/GN)
Liftgate (rear applique)	C4352-2	GD149 (BK/GY)
LH rear side marker	C4355-2	GD149 (BK/GY)
RH rear side marker	C4356-2	GD149 (BK/GY)
LH license plate	C452-2	GD149 (BK/GY)
RH license plate	C462-2	GD149 (BK/GY)

- **Is the resistance less than 5 ohms?**

YES : Go to O5.

NO : REPAIR the ground circuit in question. TEST the system for normal operation.

O5 CHECK CIRCUIT CLS30 (VT/WH) FOR AN OPEN

- Place the headlamp switch in the PARKING LAMPS ON position.
- Measure the voltage between the inoperative lamp, harness side and ground as follows:

EDGE

Inoperative Lamp	Connector-Pin	Circuit
LH front	C1023-2	CLS30 (VT/WH)

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RH front	C1043-2	CLS30 (VT/WH)
LH rear	C412-2	CLS30 (VT/WH)
RH rear	C415-2	CLS30 (VT/WH)
LH rear side marker	C4136-2	CLS30 (VT/WH)
RH rear side marker	C4138-2	CLS30 (VT/WH)
Liftgate release handle (license plate lamps)	C4353-1	CLS30 (VT/WH)

MKX

Inoperative Lamp	Connector- Pin	Circuit
LH front (headlamp)	C1021a-7	CLS30 (VT/WH)
RH front (headlamp)	C1041a-7	CLS30 (VT/WH)
LH front side marker	C151-1	CLS30 (VT/WH)
RH front side marker	C161-1	CLS30 (VT/WH)
LH rear	C412-2	CLS30 (VT/WH)
RH rear	C415-2	CLS30 (VT/WH)
Liftgate (rear applique)	C4352-1	CLS30 (VT/WH)
LH rear side marker	C4355-1	CLS30 (VT/WH)
RH rear side marker	C4356-1	CLS30 (VT/WH)
LH license plate	C452-1	CLS30 (VT/WH)
RH license plate	C462-1	CLS30 (VT/WH)

- Is the voltage greater than 10 volts?

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YES : REPAIR or INSTALL a new bulb holder or component in question. REFER to Headlamp Assembly, Rear Lamp Assembly - Liftgate, or HANDLES, LOCKS, LATCHES AND ENTRY SYSTEMS article for the liftgate release handle. TEST the system for normal operation.

NO : REPAIR the circuit. TEST the system for normal operation.

O6 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to MULTIFUNCTION ELECTRONIC MODULES article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test P: The Parking, Rear Or License Plate Lamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Parking, Rear and License Lamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) monitors the parking lamp request input from the headlamp switch through circuit CLS34 (GY). The SJB provides a ground for the parking lamp relay coil (integrated into the SJB) when a request for the parking lamps is received. When the parking lamp relay is energized, voltage is routed through circuit CLS30 (VT/WH) to the rear and front parking lamps.

- DTC B1578 (Lamp Park Input Circuit Short to Ground) - is an on-demand DTC that sets when the SJB detects a short to ground in the parking lamps on input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp switch
- SJB

PINPOINT TEST P: THE PARKING, REAR OR LICENSE PLATE LAMPS ARE ON CONTINUOUSLY

P1 USE THE RECORDED DTCs FROM THE SJB SELF-TEST

- Check the recorded results from the SJB self-test.

• **Was DTC B1578 recorded?**

YES : Go to P2.

NO : Go to P4.

P2 CHECK THE HEADLAMP SWITCH

- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Carry out the headlamp switch component test. Refer to appropriate COMPONENT TESTING article.

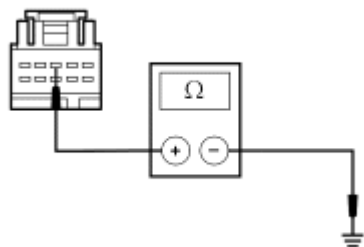
• **Is the headlamp switch OK?**

YES : Go to P3.

NO : INSTALL a new headlamp switch. REFER to **Headlamp Switch**. CLEAR the DTCs. REPEAT the self-test.

P3 CHECK CIRCUIT CLS34 (GY) FOR A SHORT TO GROUND

- Disconnect: SJB C2280b



N0056262

Fig. 73: Checking Circuit For A Short To Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-3, circuit CLS34 (GY), harness side and ground.

• **Is the resistance greater than 10,000 ohms?**

YES : Go to P6.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

P4 CHECK CIRCUIT CLS30 (VT/WH) FOR A SHORT TO VOLTAGE (TO REAR LAMPS)

- Key in OFF position.
- Disconnect: SJB C2280d
- Key in ON position.
- **Do the rear parking lamps continue to illuminate?**

YES : REPAIR the circuit. TEST the system for normal operation.

NO : Go to P5.

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P5 CHECK CIRCUIT CLS30 (VT/WH) FOR A SHORT TO VOLTAGE (TO FRONT LAMPS)

- Key in OFF position.
- Disconnect: SJB C2280e
- Key in ON position.
- **Do the front parking lamps continue to illuminate?**
YES : REPAIR the circuit. TEST the system for normal operation.
NO : Go to P6.

P6 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**
YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

FOG LAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

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Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

The SJB monitors the headlamp switch position by sending a voltage reference signal to the headlamp switch. When the fog lamp switch (part of the headlamp switch) is engaged, the voltage reference signal is routed to ground.

The fog lamps can be turned on when the following conditions are met:

- The ignition switch is in the RUN or START position.
- The low beam headlamps or the parking lamps are on.
- The high beams are off.

When the SJB receives an input from the headlamp switch indicating a request for the fog lamps, the SJB provides voltage to the fog lamp relay. When the fog lamp relay is energized, voltage is routed to the fog lamps.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

VISUAL INSPECTION CHART

Electrical
• Smart junction box (SJB) fuse 21 (15A) • Wiring, terminals or connectors • Bulb(s) • Headlamp switch • SJB

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is in the LOW BEAM position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power

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and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

SMART JUNCTION BOX (SJB) DTC CHART

DTC	Description	Action
B2254	Front Fog Lamp Switch Failure	Go to Pinpoint Test S .
All other DTCs	-	REFER to <u>MULTIFUNCTION ELECTRONIC MODULES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• No communication with the smart junction box (SJB)	• Fuse • Wiring, terminals or connectors • SJB	• REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
• Both fog lamps are inoperative	• Fuse • Wiring, terminals or connectors • Headlamp switch	• Go to Pinpoint Test Q .

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	• SJB	
• An individual fog lamp is inoperative • The fog lamps are on continuously	• Wiring, terminals or connectors • Joint connector • Wiring, terminals or connectors • Headlamp switch • SJB	NOTE: Make sure the fog lamp bulb is OK before carrying out diagnostics. • Go to Pinpoint Test R . • Go to Pinpoint Test S .
• The fog lamps on indicator is inoperative	• Wiring, terminals or connectors • Headlamp switch • SJB	• Go to Pinpoint Test T .

Pinpoint Tests

Pinpoint Test Q: Both Fog Lamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fog Lamps for schematic and connector information.

Normal Operation

The smart junction box (SJB) sends a voltage reference signal to the headlamp switch through circuit CLF21 (GY/VT) to monitor the fog lamp request. When the fog lamp switch is engaged, the voltage signal is routed to ground.

When the SJB detects the fog lamps on request, the SJB provides ground to the fog lamp relay coil (integrated into the SJB). When the fog lamp relay is energized, voltage is routed through circuit CLF12 (BN/YE) to the fog lamps.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Headlamp switch
- SJB

PINPOINT TEST Q: BOTH FOG LAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

Q1 CHECK THE SJB FOG LAMP SWITCH PID

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB Data Logger
- Monitor the SJB fog lamp PID (FOG SW) while engaging the fog lamp switch.
- **Does the PID indicate the fog lamp switch is on?**

YES : Go to Q4.

NO : Go to Q2.

Q2 CHECK THE HEADLAMP SWITCH

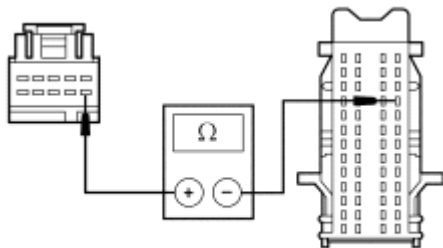
- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Carry out the headlamp switch component test. Refer to appropriate COMPONENT TESTING article.
- **Is the headlamp switch OK?**

YES : Go to Q3.

NO : INSTALL a new headlamp switch. REFER to **Headlamp Switch**. TEST the system for normal operation.

Q3 CHECK CIRCUIT CLF21 (GY/VT) FOR AN OPEN

- Disconnect: SJB C2280b



N0062227

Fig. 74: Measuring Resistance Between Headlamp Switch C205-6, Circuit CLF21 (GY/VT) & SJB C2280B-49

Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-6, circuit CLF21 (GY/VT), harness side and the SJB C2280b-49, circuit CLF21 (GY/VT), harness side.

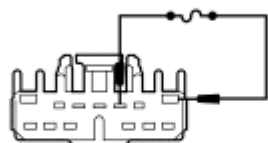
- **Is the resistance less than 5 ohms?**

YES : Go to Q5.

NO : REPAIR the circuit. TEST the system for normal operation.

Q4 CHECK CIRCUIT CLF12 (BN/YE) FOR AN OPEN

- Key in OFF position.
- Disconnect: SJB C2280e



N0058681

Fig. 75: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (15A) jumper wire between the SJB C2280e-1, circuit CBB53 (GY/BN), harness side and the SJB C2280e-3, circuit CLF12 (BN/YE), harness side.
- Key in ON position.
- **Do the fog lamps illuminate?**
YES : REMOVE the jumper wire. VERIFY the SJB fuse 21 (15A) is OK. If OK, go to Q5.
NO : REPAIR the circuit. TEST the system for normal operation.

Q5 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**
YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test R: An Individual Fog Lamp Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fog Lamps for schematic and connector information.

Normal Operation

When the fog lamp relay is energized, voltage is routed through circuit CLF12 (BN/YE) to the fog lamps. Ground is provided on circuits GD120 (BK/GN) and GD123 (BK/GY) to the LH and RH fog lamps, respectively.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Joint connector

PINPOINT TEST R: AN INDIVIDUAL FOG LAMP IS INOPERATIVE

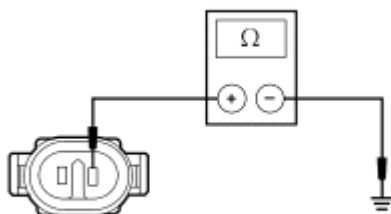
CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

NOTE: Verify that the headlamps operate correctly before proceeding with this pinpoint test.

R1 CHECK THE GROUND CIRCUIT FOR CONTINUITY

- Key in OFF position.
- Disconnect: Inoperative Fog Lamp



N0062299

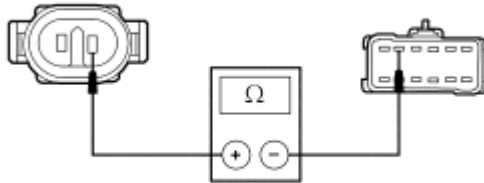
Fig. 76: Measuring Resistance Between LH Fog Lamp C152-B & C162-B, Circuit GD120 (BK/GN) & GD123 (BK/GY)
Courtesy of FORD MOTOR CO.

- Measure the resistance between the LH fog lamp C152-B, circuit GD120 (BK/GN), harness side and ground; or between the RH fog lamp C162-B, circuit GD123 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : REPAIR circuit CLF12 (BN/YE) for an open. TEST the system for normal operation.
NO : For the RH fog lamp, REPAIR circuit GD123 (BK/GY) for an open. TEST the system for normal operation.

For the LH fog lamp, go to R2.

R2 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN

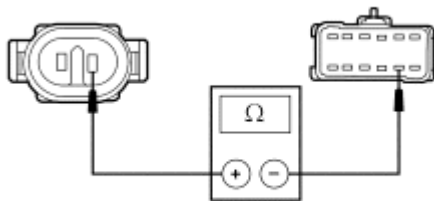
- Disconnect: Joint Connector 7 C2200



N0062228

Fig. 77: Measuring Resistance Between LH Fog Lamp C152-B, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-K
Courtesy of FORD MOTOR CO.

- For Edge, measure the resistance between the LH fog lamp C152-B, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-K, circuit GD120 (BK/GN), harness side and ground.



N0062229

Fig. 78: Measuring Resistance Between LH Fog Lamp C152-B, Circuit GD120 (BK/GN) & Joint Connector 7 C2200-E
Courtesy of FORD MOTOR CO.

- For MKX, measure the resistance between the LH fog lamp C152-B, circuit GD120 (BK/GN), harness side and the joint connector 7 C2200-E, circuit GD120 (BK/GN), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test S: The Fog Lamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fog Lamps for schematic and connector information.

Normal Operation

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The smart junction box (SJB) sends a voltage reference signal to the headlamp switch through circuit CLF21 (GY/VT) to monitor the fog lamp request. When the fog lamp switch is engaged, the voltage signal is routed to ground.

When the SJB detects the fog lamps on request, the SJB provides ground to the fog lamp relay coil (integrated into the SJB). When the fog lamp relay is energized, voltage is routed through circuit CLF12 (BN/YE) to the fog lamps, and on a separate output, circuit CLF12 (BN/YE) to the headlamp switch for the fog lamps on indicator.

- DTC B2254 (Front Fog Lamp Switch Failure) - is an on-demand DTC that sets when the SJB detects a short to ground on the fog lamp on input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp switch
- SJB

PINPOINT TEST S: THE FOG LAMPS ARE ON CONTINUOUSLY

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

S1 CHECK FOR RECORDED SJB DTCs

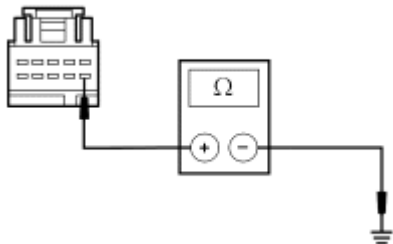
- Check the recorded results from the SJB self-test.
- **Is DTC B2254 present?**
YES : Go to S2.
NO : Go to S4.

S2 CHECK THE HEADLAMP SWITCH

- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Carry out the headlamp switch component test. Refer to appropriate COMPONENT TESTING article
- **Is the headlamp switch OK?**
YES : Go to S3.
NO : INSTALL a new headlamp switch. REFER to **Headlamp Switch**. CLEAR the DTCs. REPEAT the self-test.

S3 CHECK CIRCUIT CLF21 (GY/VT) FOR A SHORT TO GROUND

- Disconnect: SJB C2280b



N0038825

Fig. 79: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-6, circuit CLF21 (GY/VT), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to S6.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

S4 CHECK CIRCUIT CLF12 (BN/YE) FOR A SHORT TO VOLTAGE (TO HEADLAMP SWITCH)

- Key in OFF position.
- Disconnect: SJB C2280a
- Key in ON position.
- **Do the fog lamps continue to illuminate?**
YES : Go to S5.
NO : REPAIR the circuit. TEST the system for normal operation.

S5 CHECK CIRCUIT CLF12 (BN/YE) FOR A SHORT TO VOLTAGE (TO THE FOG LAMPS)

- Key in OFF position.
- Disconnect: SJB C2280e
- Key in ON position.
- **Do the fog lamps continue to illuminate?**
YES : REPAIR the circuit. TEST the system for normal operation.
NO : Go to S6.

S6 CHECK FOR CORRECT SJB OPERATION

- Key in OFF position.
- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.

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- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DTCs. REPEAT the self-test.

Pinpoint Test T: The Fog Lamps On Indicator Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fog Lamps for schematic and connector information.

Normal Operation

When the smart junction box (SJB) detects the fog lamps on request, the SJB provides ground to the fog lamp relay coil (integrated into the SJB). When the fog lamp relay is energized, voltage is routed through circuit CLF12 (BN/YE) to the fog lamps, and on a separate output, circuit CLF12 (BN/YE) to the headlamp switch for the fog lamps on indicator.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp switch
- SJB

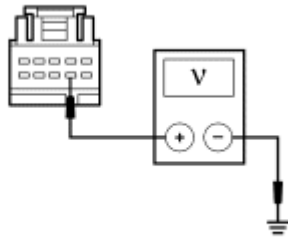
PINPOINT TEST T: THE FOG LAMPS ON INDICATOR IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

T1 CHECK THE HEADLAMP SWITCH

NOTE: Make sure the headlamp switch is in the OFF position.

- Key in OFF position.
- Disconnect: Headlamp Switch C205
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: SJB DataLogger
- Select the SJB fog lamps (FOG_LMP) active command. Command the fog lamps on.



N0062231

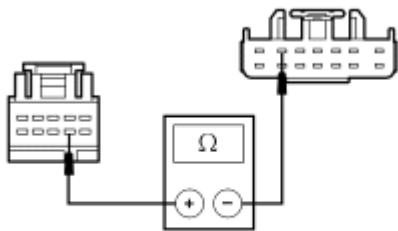
Fig. 80: Measuring Voltage Between Headlamp Switch C205-7, Circuit CLF12 (BN/YE) & Ground
Courtesy of FORD MOTOR CO.

NOTE: Make sure the fog lamps are illuminated when taking a reading.

- Measure the voltage between the headlamp switch C205-7, circuit CLF12 (BN/YE), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : INSTALL a new headlamp switch. REFER to Headlamp Switch. TEST the system for normal operation.
NO : Go to T2.

T2 CHECK CIRCUIT CLF12 (BN/YE) FOR AN OPEN

- Key in OFF position.
- Disconnect: SJB C2280a



N0062232

Fig. 81: Measuring Resistance Between Headlamp Switch C205-7, Circuit CLF12 (BN/YE) & SJB C2280A-6
Courtesy of FORD MOTOR CO.

- Measure the resistance between the headlamp switch C205-7, circuit CLF12 (BN/YE), harness side and the SJB C2280a-6, circuit CLF12 (BN/YE), harness side.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to T3.
NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

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T3 CHECK FOR CORRECT SJB OPERATION

- Disconnect all the SJB connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DTCs. REPEAT the self-test.

REVERSING LAMPS

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

The transaxle is hardwired to the PCM which monitors the transmission gear status. When the PCM detects the transaxle is in REVERSE (R), the PCM provides ground for the reversing lamps relay coil. When the reversing lamps relay is energized, voltage is routed to the reversing lamps.

Inspection and Verification

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1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

VISUAL INSPECTION CHART

Electrical
• Battery junction box (BJB) fuse 44 (10A) • Bulb(s) • Bulb holder(s) • Wiring, terminals or connectors • PCM

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: **Make sure to use the latest scan tool software release.**

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: **The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the PCM.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

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PCM DTC CHART

DTC	Description	Action
P1910	Reverse Lamp Control Circuit/Open	If the reversing lamps are inoperative, go to Pinpoint Test U . If the reversing lamps are always on, go to Pinpoint Test W .
All other DTCs	-	REFER to MULTIFUNCTION ELECTRONIC MODULES article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
The reversing lamps are inoperative	- Fuse - Wiring, terminals or connectors - Reversing lamps relay - Battery junction box (BJB) - PCM	Go to Pinpoint Test U .
The individual reversing lamp is inoperative	- Wiring, terminals or connectors - Bulb holder - Rear lamps assembly	Go to Pinpoint Test V .
The reversing lamps are on continuously	- Wiring, terminals or connectors - Reversing lamps relay - Auto-dimming interior mirror - PCM	Go to Pinpoint Test W .

Pinpoint Tests

Pinpoint Test U: The Reversing Lamps Are Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fuse and Relay Information for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Reversing Lamps for schematic and connector information.

Normal Operation

The PCM monitors the transaxle gear status. When the PCM detects the transaxle in REVERSE (R), the PCM

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provides ground for the reversing lamps relay coil through circuit CLS28 (BU/WH). Voltage is supplied from the battery junction box (BJB) to the coil and the switched side of the reversing lamps relay. When the reversing lamps relay is energized, voltage is routed to the reversing lamps through circuit CLS10 (GN/BN).

- DTC P1910 (Reverse Lamp Control Circuit/Open) - sets when the PCM detects no voltage on the reversing lamp relay coil ground control circuit (if the coil voltage supply circuit or the coil ground control circuit is open).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Reversing lamps relay
- BJB
- PCM

PINPOINT TEST U: THE REVERSING LAMPS ARE INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

U1 CHECK FOR PCM TRANSMISSION RANGE (TR) SENSOR DTCs FROM THE SELF-TEST

- Review the DTCs from the PCM KOEO self-test.
- **Are any TR sensor DTCs present?**
YES : REFER to **AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50** article.
NO : Go to U2.

U2 CHECK THE REVERSING LAMPS RELAY

- Key in OFF position.
- Disconnect: Reversing Lamps Relay
- Key in ON position.
- Substitute a known good relay and recheck the reversing lamps operation.
- **Do the reversing lamps operate correctly?**
YES : REMOVE the known good relay. INSTALL a new reversing lamps relay. TEST the system for normal operation.
NO : REMOVE the known good relay. Go to U3.

U3 CHECK THE BJB

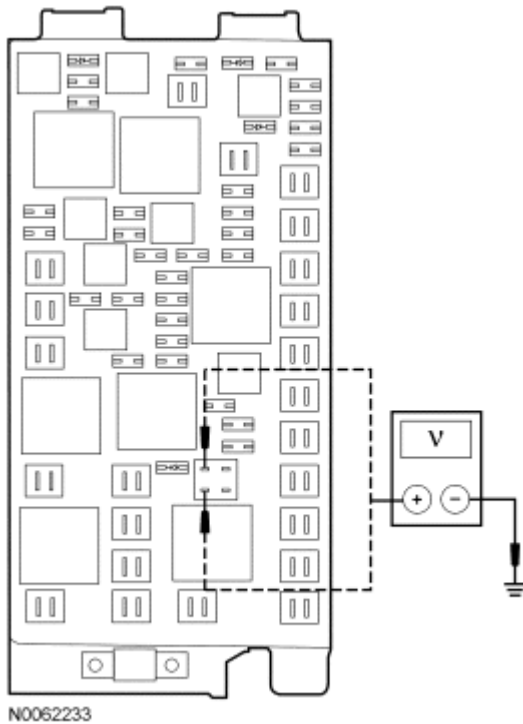


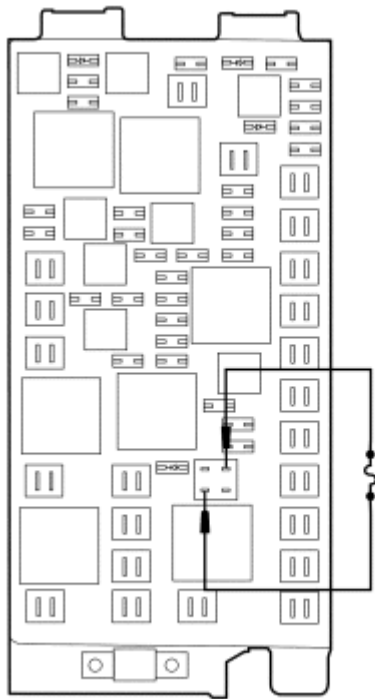
Fig. 82: Measuring Voltage Between Reversing Lamps Relay Pin 5, BJB Face Side & Reversing Lamps Relay Pin 1
Courtesy of FORD MOTOR CO.

- Measure the voltage between the reversing lamps relay pin 5, BJB face side and ground; and between the reversing lamps relay pin 1, BJB face side and ground.
- **Are the voltages greater than 10 volts?**

YES : Go to U4.

NO : INSTALL a new BJB. TEST the system for normal operation.

U4 CHECK CIRCUIT CLS10 (GN/BN) FOR AN OPEN



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Fig. 83: Connecting Voltage Between Reversing Lamps Relay Pin 5, BJB Face Side & Reversing Lamps Relay Pin 3

Courtesy of FORD MOTOR CO.

- Connect a fused (5A) jumper wire between the reversing lamps relay pin 5, BJB face side and the reversing lamps relay pin 3, circuit CLS10 (GN/BN), BJB face side.

- **Do the reversing lamps illuminate?**

YES : REMOVE the jumper wire. Go to U5.

NO : VERIFY the BJB fuse 44 (10A) is OK. If OK, REPAIR the circuit. TEST the system for normal operation.

U5 CHECK CIRCUIT CLS28 (BU/WH) FOR A SHORT TO VOLTAGE

- Disconnect: PCM C175t
- Key in ON position.

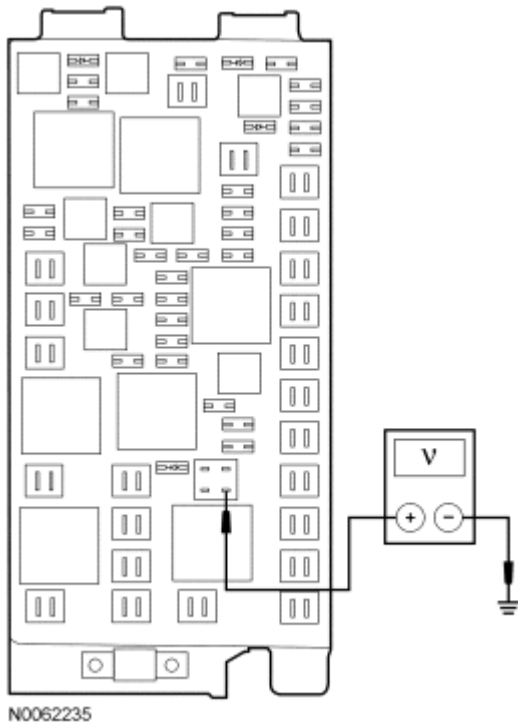


Fig. 84: Measuring Voltage Between Reversing Lamps Relay Pin 2, Circuit CLS28 (BU/WH), BJB Face Side & Ground
 Courtesy of FORD MOTOR CO.

- Measure the voltage between the reversing lamps relay pin 2, circuit CLS28 (BU/WH), BJB face side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
NO : Go to U6.

U6 CHECK CIRCUITS CLS28 (BU/WH) FOR AN OPEN

- Key in OFF position.
- Connect: Reversing Lamps Relay

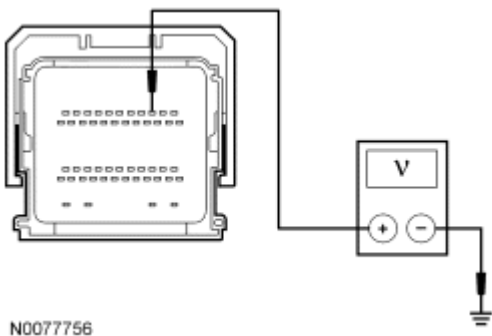


Fig. 85: Measuring Voltage Between PCM C175T-3, Circuit CLS28 (BU/WH), Harness Side & Ground

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Courtesy of FORD MOTOR CO.

- Measure the voltage between the PCM C175t-3, circuit CLS28 (BU/WH), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to U7.

NO : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

U7 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new PCM. REFER to **ELECTRONIC ENGINE CONTROLS** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DTCs. REPEAT the self-test.

Pinpoint Test V: The Individual Reversing Lamp Is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Reversing Lamps for schematic and connector information.

When the reversing lamps relay is energized, voltage is routed to the reversing lamps through circuit CLS10 (GN/BN). The LH and RH reversing lamps are provided ground through circuits GD149 (BK/GY) and GD151 (BK/GN), respectively.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Bulb holder
- Rear lamp assembly

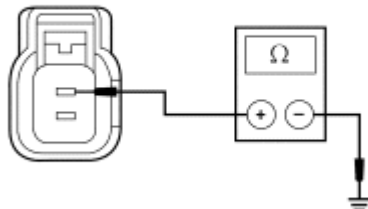
PINPOINT TEST V: THE INDIVIDUAL REVERSING LAMP IS INOPERATIVE

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

V1 CHECK CIRCUIT GD149 (BK/GY) OR CIRCUIT GD151 (BK/GN) FOR AN OPEN

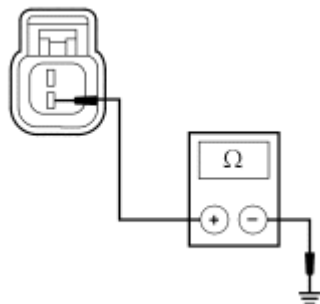
- Key in OFF position.
- Disconnect: Inoperative Reversing Lamp



N0062237

Fig. 86: Measuring Resistance Between LH Reversing Lamp C451-1 & C461-1, Circuit GD149 (BK/GY) & GD151 (BK/GN)
Courtesy of FORD MOTOR CO.

- For Edge, measure the resistance between the LH reversing lamp C451-1, circuit GD149 (BK/GY), harness side and ground, or between the RH reversing lamp C461-1, circuit GD151 (BK/GN), harness side and ground.



N0062238

Fig. 87: Measuring Resistance Between LH Reversing Lamp C487-2 & C488-2, Circuit GD149 (BK/GY) & GD151 (BK/GN)
Courtesy of FORD MOTOR CO.

- For MKX, measure the resistance between the LH reversing lamp C487-2, circuit GD149 (BK/GY), harness side and ground, or between the RH reversing lamp C488-2, circuit GD151 (BK/GN), harness side and ground.

- **Is the resistance less than 5 ohms?**

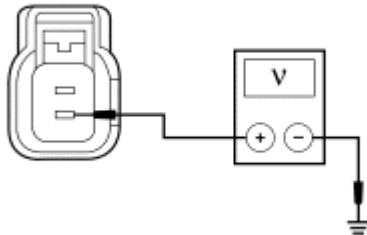
YES : Go to V2.

NO : REPAIR the ground circuit in question. TEST the system for normal operation.

V2 CHECK CIRCUIT CLS10 (GN/BN) FOR AN OPEN

- Apply the parking brake.
- Select REVERSE.

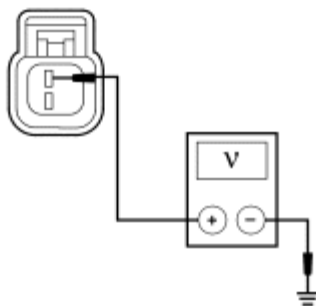
- Key in ON position.



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Fig. 88: Measuring Voltage Between LH Reversing Lamp C451-2, Circuit CLS10 (GN/BN) & RH Reversing Lamp C461-2
Courtesy of FORD MOTOR CO.

- For Edge, measure the voltage between the LH reversing lamp C451-2, circuit CLS10 (GN/BN), harness side and ground, or between the RH reversing lamp C461-2, circuit CLS10 (GN/BN), harness side and ground.



N0062239

Fig. 89: Measuring Voltage Between LH Reversing Lamp C487-1, Circuit CLS10 (GN/BN) & RH Reversing Lamp C488-1
Courtesy of FORD MOTOR CO.

- For MKX, measure the voltage between the LH reversing lamp C487-1, circuit CLS10 (GN/BN), harness side and ground, or between the RH reversing lamp C488-1, circuit CLS10 (GN/BN), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : INSTALL a new bulb holder or rear lamp assembly. REFER to **Rear Lamp Assembly**.
 TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test W: The Reversing Lamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Reversing Lamps for schematic and connector information.

Normal Operation

The PCM monitors the transaxle gear status. When the PCM detects the transaxle in REVERSE (R), the PCM provides ground for the reversing lamps relay coil through circuit CLS28 (BU/WH). Voltage is supplied from the battery junction box (BJB) to the coil and the switched side of the reversing lamps relay. When the reversing lamps relay is energized, voltage is routed to the reversing lamps through circuit CLS10 (GN/BN).

- DTC P1910 (Reverse Lamp Control Circuit/Open) - sets when the PCM detects no voltage on the reversing lamp relay coil ground control circuit (if the coil ground control circuit is short to ground).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Reversing lamps relay
- Auto-dimming interior mirror
- PCM

PINPOINT TEST W: THE REVERSING LAMPS ARE ON CONTINUOUSLY

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

W1 CHECK FOR PCM TRANSMISSION RANGE (TR) SENSOR DTCs FROM THE SELF-TEST

- Review the DTCs from the PCM KOEO self-test.
- **Are any TR sensor DTCs present?**
YES : REFER to **AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50** article.
NO : Go to W2.

W2 CHECK THE REVERSING LAMPS RELAY OUTPUT

- Key in OFF position.
- Disconnect: Reversing Lamps Relay
- Key in ON position.
- **Do the reversing lamps continue to illuminate?**
YES : Go to W3.
NO : Go to W4.

W3 CHECK CIRCUIT CLS10 (GN/BN) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Auto-Dimming Interior Mirror C911 or C9012
- Key in ON position.
- **Do the reversing lamps continue to illuminate?**
YES : INSTALL a new auto-dimming interior mirror. REFER to **REAR VIEW MIRRORS**

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article. TEST the system for normal operation.

NO : REPAIR the circuit. TEST the system for normal operation.

W4 CHECK THE REVERSING LAMPS RELAY

- Key in OFF position.
- Substitute a known good relay.
- Key in ON position.
- **Do the reversing lamps continue to illuminate?**

YES : REMOVE the known good relay. Go to W5.

NO : REMOVE the known good relay. INSTALL a new reversing lamps relay. TEST the system for normal operation.

W5 CHECK CIRCUIT CLS28 (BU/WH) FOR A SHORT TO GROUND

- Key in OFF position.
- Connect: Reversing Lamps Relay
- Disconnect: PCM C175t
- Key in ON position.
- **Do the reversing lamps continue to illuminate?**

YES : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

NO : Go to W6.

W6 CHECK FOR CORRECT PCM OPERATION

- Disconnect the PCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new PCM. REFER to **ELECTRONIC ENGINE CONTROLS** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DTCs. REPEAT the self-test.

TRAILER LAMPS

Special Tools

Illustration	Tool Name	Tool Number
	73III Automotive Meter	105-R0057 or equivalent

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ST1137-A



ST2574-A

Flex Probe Kit

105-R025C or equivalent

Principles of Operation

The trailer lamps are supplied power by the trailer tow park lamp relay, the LH trailer tow stop/turn relay and the RH trailer tow stop/turn relay. They are energized to correspond with the exterior lighting functions of the vehicle. A common ground is provided to the trailer tow connector for the trailer lamps.

Inspection and Verification

1. Verify the customer concern.
2. Verify the exterior lighting system of the vehicle is operating correctly. If not, refer to the appropriate pinpoint test in this service information.
3. Visually inspect for obvious signs of electrical damage.

VISUAL INSPECTION CHART

Electrical
• Battery junction box (BJB) fuse(s): ○ 9 (20A) (trailer tow parking lamps) ○ 24 (15A) (trailer tow LH stop/turn relay) ○ 46 (15A) (trailer tow RH stop/turn relay) • Wiring, terminals or connectors • Trailer tow stop/turn relays • Trailer tow parking lamp relay • Trailer

4. If the concern is not visually evident, verify the symptom. Go to **Symptom Chart**.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
	• Wiring, terminals or	

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• The trailer lamps are inoperative - all	- connectors • Trailer	• Go to Pinpoint Test X .
• The individual trailer lamp is inoperative - trailer stop/turn lamp	• Fuse • Wiring, terminals or connectors • Joint connector • Trailer tow stop/turn relay • Battery junction box (BJB) • Trailer	• Go to Pinpoint Test Y .
• The trailer lamps are inoperative - trailer parking lamps	• Fuse • Wiring, terminals or connectors • Joint connector • Trailer tow parking lamp relay • BJB • Trailer	• Go to Pinpoint Test Z .
• The trailer lamps are on continuously	• Wiring, terminals or connectors • Trailer tow relay • Trailer	• Go to Pinpoint Test AA .

Pinpoint Tests

Pinpoint Test X: The Trailer Lamps Are Inoperative - All

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The trailer tow connector receives ground through circuit RAT08 (WH)/GD149 (BK/GY) for all of the trailer lamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Trailer

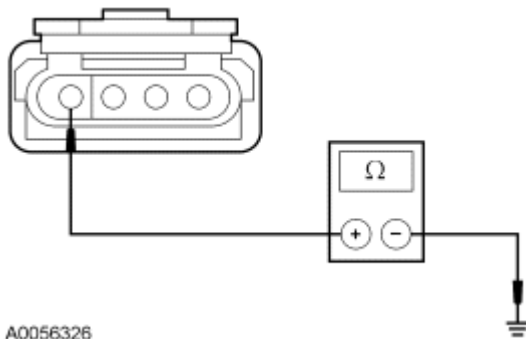
PINPOINT TEST X: THE TRAILER LAMPS ARE INOPERATIVE - ALL

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

X1 CHECK CIRCUIT RAT08 (WH)/GD149 (BK/GY) FOR AN OPEN

- Disconnect: Trailer Tow C439



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Fig. 90: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

- Measure the resistance between the trailer tow C439-4, circuit RAT08 (WH), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : The vehicle is operating correctly. SEND the trailer to an authorized camper/trailer repair facility.
NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test Y: The Individual Trailer Lamp Is Inoperative - Trailer Stop/Turn Lamp

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fuse and Relay Information for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The battery junction box (BJB) fuse 24 (15A) supplies voltage to the LH trailer tow stop/turn relay and BJB fuse 46 (15A) supplies voltage to the RH trailer tow stop/turn relay. The trailer tow stop/turn relay coils are grounded through circuit GD120 (BK/GN).

The trailer tow LH stop/turn relay coil receives voltage from circuit CLS18 (GY/BN) when the vehicle LH rear stoplamp is illuminated. When the trailer tow LH stop/turn relay is energized, voltage is routed to the trailer tow connector through circuit CAT06 (YE).

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The trailer tow RH stop/turn relay coil receives voltage through circuit CLS19 (VT/OG) when the vehicle RH rear stoplamp is illuminated. When the trailer tow RH stop/turn relay is energized, voltage is routed to the trailer tow connector through circuit CAT09 (GN).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Joint connector
- Trailer tow stop/turn relay
- BJB
- Trailer

PINPOINT TEST Y: THE INDIVIDUAL TRAILER LAMP IS INOPERATIVE - TRAILER STOP/TURN LAMP

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

Y1 CHECK CIRCUIT CAT06 (YE) OR CIRCUIT CAT09 (GN) FOR VOLTAGE

- Key in OFF position.

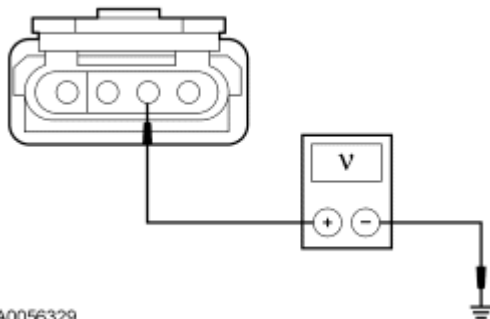
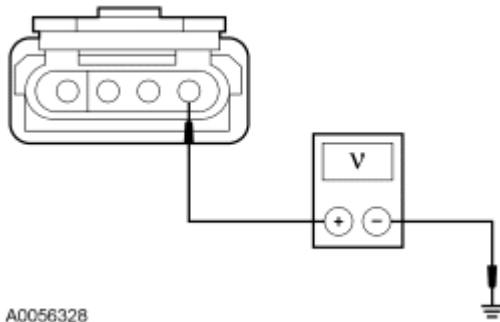


Fig. 91: Measuring Voltage Between Trailer Tow C439-2, Circuit CAT06 (YE), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- For the LH trailer turn/stoplamp, while applying the brake pedal, measure the voltage between the trailer tow C439-2, circuit CAT06 (YE), harness side and ground.



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Fig. 92: Measuring Voltage Between Trailer Tow C439-1, Circuit CAT09 (GN), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- For the RH trailer turn/stoplamp, while applying the brake pedal, measure the voltage between the trailer tow C439-1, circuit CAT09 (GN), harness side and ground.

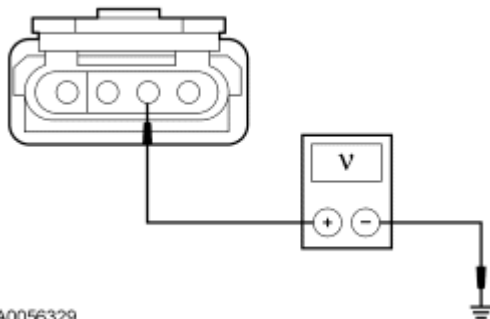
- **Is the voltage greater than 10 volts?**

YES : The vehicle is operating correctly. SEND the trailer to an authorized camper/trailer repair facility.

NO : Go to Y2.

Y2 CHECK THE TRAILER TOW STOP/TURN RELAY

- Disconnect: Suspect Trailer Tow Stop/Turn Relay
- Substitute a known good relay.



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Fig. 93: Measuring Voltage Between Trailer Tow C439-2, Circuit CAT06 (YE), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- For the LH trailer turn/stoplamp, while applying the brake pedal, measure the voltage between the trailer tow C439-2, circuit CAT06 (YE), harness side and ground.

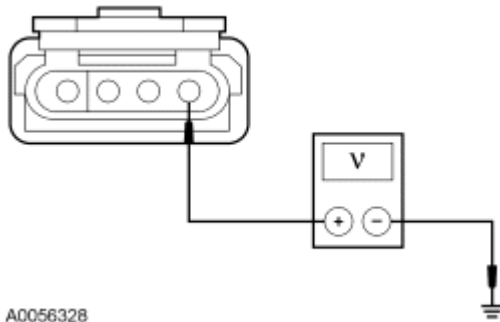


Fig. 94: Measuring Voltage Between Trailer Tow C439-1, Circuit CAT09 (GN), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- For the RH trailer turn/stoplamp, while applying the brake pedal, measure the voltage between the trailer tow C439-1, circuit CAT09 (GN), harness side and ground.

- **Is the voltage greater than 10 volts?**

YES : REMOVE the known good relay. INSTALL a new trailer tow stop/turn relay. TEST the system for normal operation.

NO : REMOVE the known good relay. Go to Y3.

Y3 CHECK CIRCUIT CLS18 (GY/BN) OR CIRCUIT CLS19 (VT/OG) FOR AN OPEN

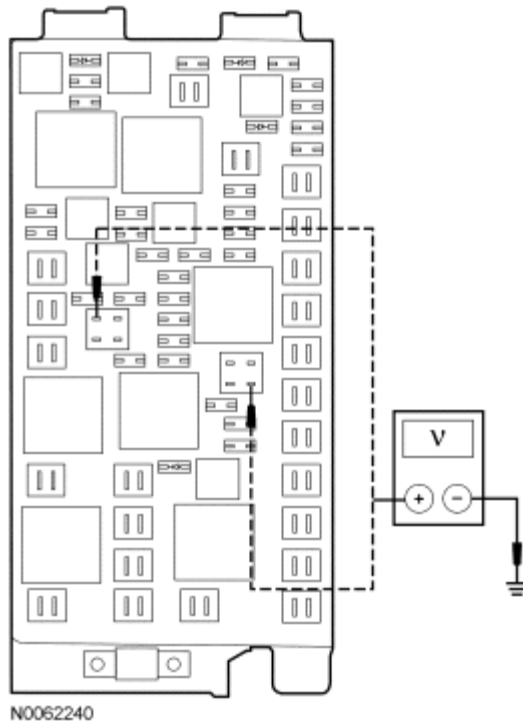


Fig. 95: Measuring Voltage Between Trailer Tow LH Stop/Turn Relay Pin 1 & Pin 2, CLS18 (GY/BN) & CLS19 (VT/OG)

Courtesy of FORD MOTOR CO.

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- While applying the brake pedal, measure the voltage between the trailer tow LH stop/turn relay pin 1, circuit CLS18 (GY/BN), BJB face side and ground; or between the trailer tow RH stop/turn relay pin 2, circuit CLS19 (VT/OG), BJB face side and ground.

- **Is the voltage greater than 10 volts?**

YES : Go to Y4.

NO : REPAIR the circuit in question. TEST the system for normal operation.

Y4 CHECK THE BJB

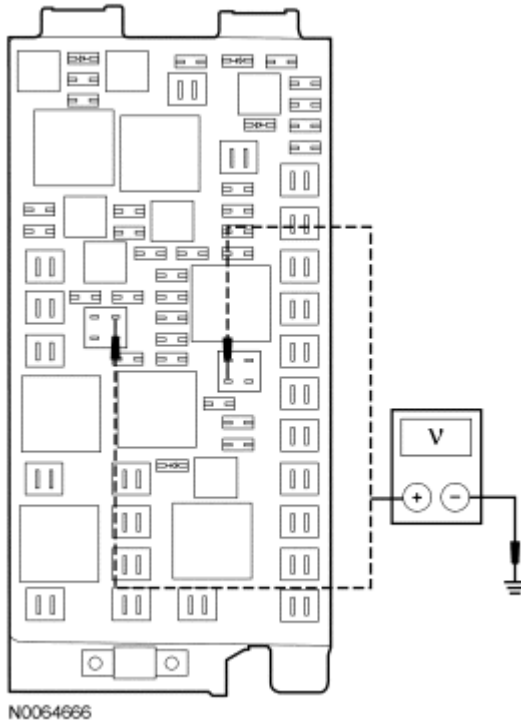


Fig. 96: Measuring Voltage Between Trailer Tow LH Stop/Turn Relay Pin 3, BJB Face Side & RH Stop/Turn Relay Pin 5,
Courtesy of FORD MOTOR CO.

- Measure the voltage between the trailer tow LH stop/turn relay pin 3, BJB face side and ground; or between the trailer tow RH stop/turn relay pin 5, BJB face side and ground.

- **Is the voltage greater than 10 volts?**

YES : Go to Y5.

NO : VERIFY the BJB fuse 24 (15A) (LH stop/turn relay) or fuse 46 (15A) (RH stop/turn relay) is OK. If OK, INSTALL a new BJB. TEST the system for normal operation.

Y5 CHECK CIRCUIT GD120 (BK/GN) FOR CONTINUITY TO GROUND

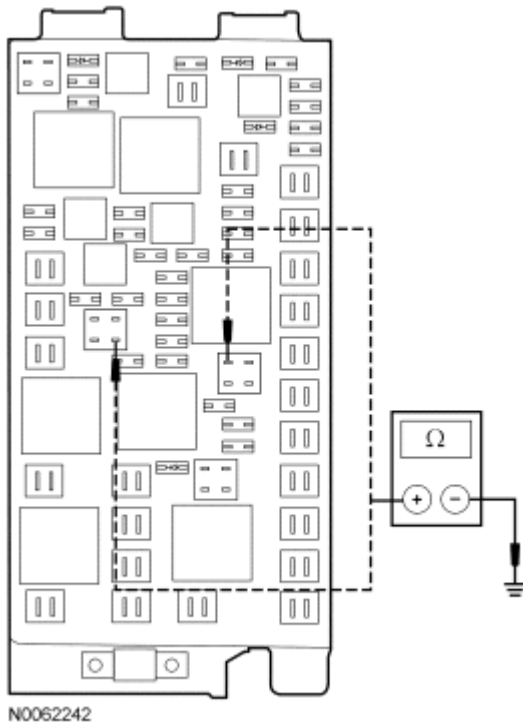


Fig. 97: Measuring Resistance Between Trailer Tow LH Turn Relay Pin 2, Circuit GD120 (BK/GN) & RH Turn Relay Pin 1
Courtesy of FORD MOTOR CO.

- Measure the resistance between the trailer tow LH turn relay pin 2, circuit GD120 (BK/GN), BJB face side and ground; or between the trailer tow RH turn relay pin 1, circuit GD120 (BK/GN), BJB face side and ground.

- **Is the resistance less than 5 ohms?**

YES : REPAIR circuit CAT06 (YE) (LH trailer stop/turn) or circuit CAT09 (GN) (RH trailer stop/turn) for an open. TEST the system for normal operation.

NO : Go to Y6.

Y6 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN

- Disconnect: Joint Connector 7 C2200

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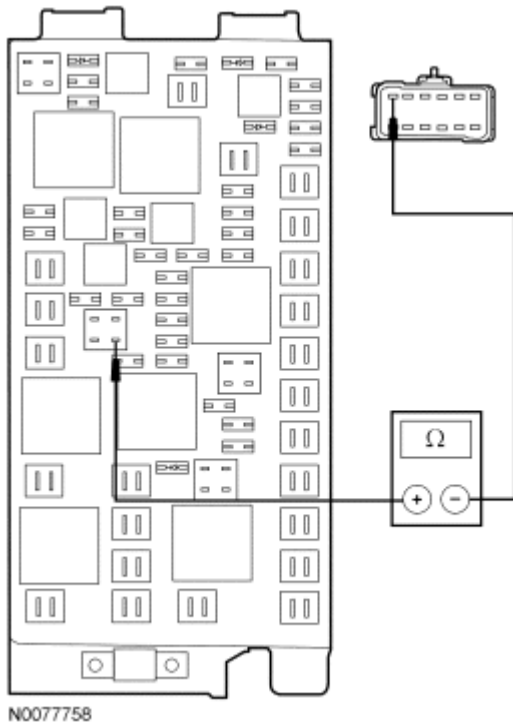


Fig. 98: Measuring Resistance Between Trailer Tow LH Stop/Turn Relay Pin 2, GD120 (BK/GN) & Joint Connector 7 C2200-M
Courtesy of FORD MOTOR CO.

- For the LH trailer turn, measure the resistance between the trailer tow LH stop/turn relay pin 2, circuit GD120 (BK/GN), BJB face side and the joint connector 7 C2200-M, circuit GD120 (BK/GN), harness side.

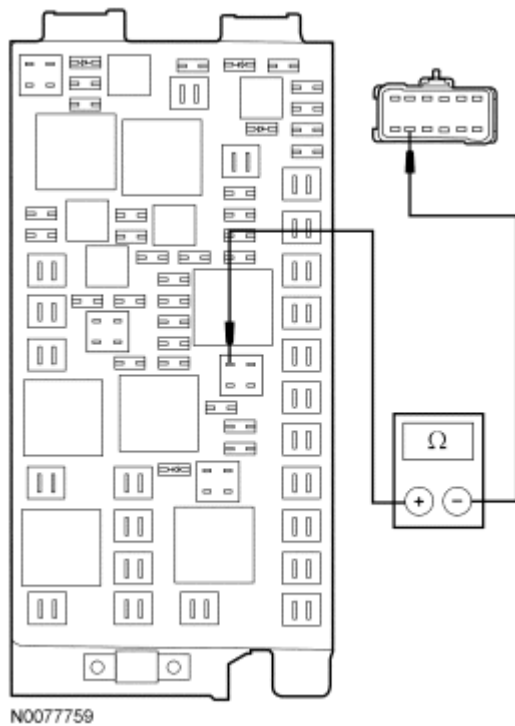


Fig. 99: Measuring Resistance Between Trailer Tow RH Stop/Turn Relay Pin 1, GD120 (BK/GN) & Joint Connector 7 C2200-B
 Courtesy of FORD MOTOR CO.

- For the RH trailer turn, measure the resistance between the trailer tow RH stop/turn relay pin 1, circuit GD120 (BK/GN), BJB face side and the joint connector 7 C2200-B, circuit GD120 (BK/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test Z: The Trailer Lamps Are Inoperative - Trailer Parking Lamps

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Fuse and Relay Information for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The trailer tow parking lamp relay coil receives voltage through circuit CLS30 (VT/WH) when the vehicle parking lamps are illuminated. The trailer tow parking lamp relay coil is grounded through circuit GD120 (BK/GN). The battery junction box (BJB) fuse 9 (20A) supplies voltage to the trailer tow parking lamp relay switch side. When the trailer tow parking lamp relay is energized, the voltage is routed to the trailer tow

connector through circuit CAT11 (BN).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Joint connector
- Trailer tow parking lamp relay
- BJB
- Trailer

PINPOINT TEST Z: THE TRAILER LAMPS ARE INOPERATIVE - TRAILER PARKING LAMPS

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: The battery ground cable must be disconnected for accurate test results when checking ground circuits for continuity to ground.

Z1 CHECK CIRCUIT CAT11 (BN) FOR VOLTAGE

- Key in OFF position.
- Place the headlamp switch in the PARKING LAMPS ON position.

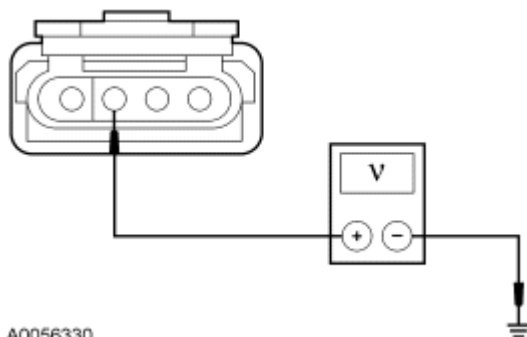


Fig. 100: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the trailer tow C439-3, circuit CAT11 (BN), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : The vehicle is operating correctly. SEND the trailer to an authorized camper/trailer repair facility.

NO : Go to Z2.

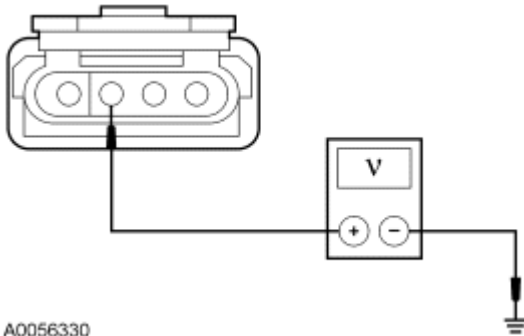
Z2 CHECK THE TRAILER TOW PARKING LAMP RELAY

- Place the headlamp switch in the OFF position.

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- Disconnect: Trailer Tow Parking Lamp Relay
- Substitute a known good relay.
- Place the headlamp switch in the PARKING LAMPS ON position.



A0056330

Fig. 101: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the trailer tow C439-3, circuit CAT11 (BN), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : REMOVE the known good relay. Go to Z3.
NO : REMOVE the known good relay. INSTALL a new trailer tow parking lamp relay. TEST the system for normal operation.

Z3 CHECK CIRCUIT CLS30 (VT/WH) FOR AN OPEN

NOTE: Make sure the parking lamps are still on.

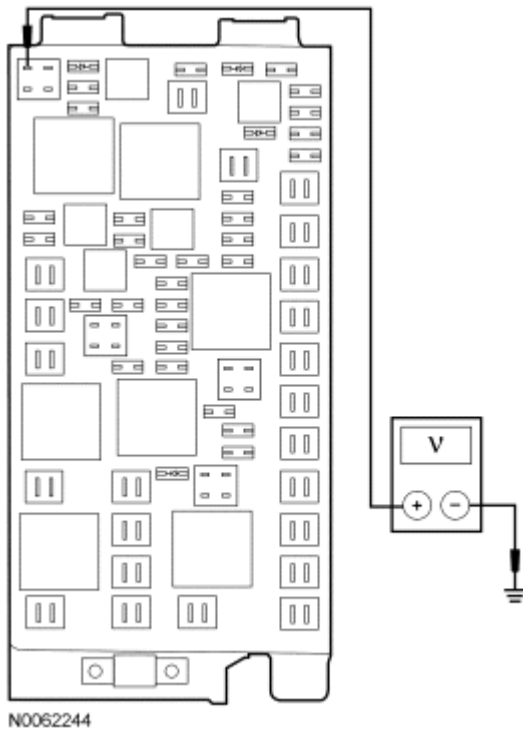


Fig. 102: Measuring Voltage Between Trailer Tow Parking Lamp Relay Pin 1, Circuit CLS30 (VT/WH) & Ground

Courtesy of FORD MOTOR CO.

- Measure the voltage between the trailer tow parking lamp relay pin 1, circuit CLS30 (VT/WH), BJB face side and ground.
- **Is the voltage greater than 10 volts?**
YES : Go to Z4.
NO : REPAIR the circuit. TEST the system for normal operation.

Z4 CHECK THE BJB

- Place the headlamp switch in the OFF position.

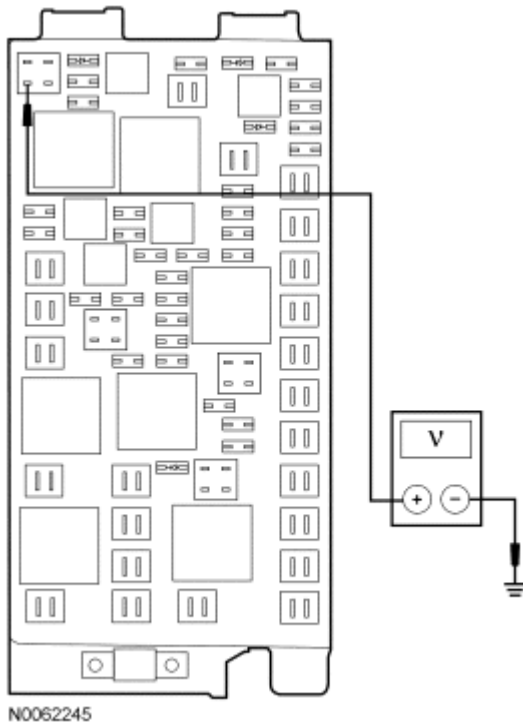


Fig. 103: Measuring Voltage Between Trailer Tow Parking Lamp Relay Pin 5, BJB Face Side & Ground

Courtesy of FORD MOTOR CO.

- Measure the voltage between the trailer tow parking lamp relay pin 5, BJB face side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to Z5.

NO : VERIFY the BJB fuse 9 (20A) is OK. If OK, INSTALL a new BJB. TEST the system for normal operation.

Z5 CHECK CIRCUIT GD120 (BK/GN) FOR CONTINUITY TO GROUND

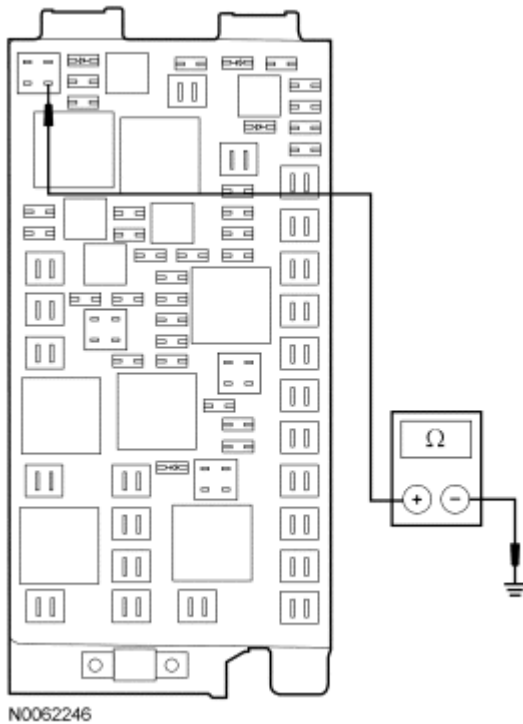


Fig. 104: Measuring Resistance Between Trailer Tow Parking Lamp Relay Pin 2, Circuit GD120 (BK/GN) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the trailer tow parking lamp relay pin 2, circuit GD120 (BK/GN), BJB face side and ground.
- **Is the resistance less than 5 ohms?**
YES : REPAIR circuit CAT11 (BN) for an open. TEST the system for normal operation.
NO : Go to Z6.

Z6 CHECK CIRCUIT GD120 (BK/GN) FOR AN OPEN

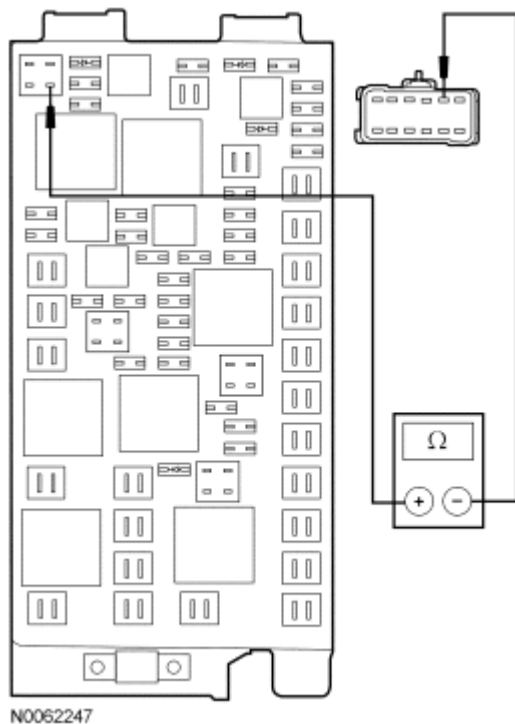


Fig. 105: Measuring Resistance Between Trailer Tow Parking Lamp Relay Pin 2, GD120 (BK/GN) & Joint Connector 7 C2200-H
 Courtesy of FORD MOTOR CO.

- Measure the resistance between the trailer tow parking lamp relay pin 2, circuit GD120 (BK/GN), BJB face side and the joint connector 7 C2200-H, circuit GD120 (BK/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new joint connector. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test AA: The Trailer Lamps Are On Continuously

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Trailer/Camper Adapter for schematic and connector information.

Normal Operation - Trailer Tow Stop/Turn Lamp

The trailer tow LH stop/turn relay coil receives voltage through circuit CLS18 (GY/BN) when the vehicle LH rear stoplamp is illuminated. When the trailer tow LH stop/turn relay is energized, voltage is routed to the trailer tow connector through circuit CAT06 (YE).

The trailer tow RH stop/turn relay coil receives voltage through circuit CLS19 (VT/OG) when the vehicle RH rear stoplamp is illuminated. When the trailer tow RH stop/turn relay is energized, voltage is routed to the trailer tow connector through circuit CAT09 (GN).

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Trailer Tow Parking

The trailer tow parking lamp relay coil receives voltage through circuit CLS30 (VT/WH) when the vehicle parking lamps are illuminated. When the trailer tow parking lamp relay is energized, voltage is routed to the trailer tow connector through circuit CAT11 (BN).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Trailer tow relay
- Trailer

PINPOINT TEST AA: THE TRAILER LAMPS ARE ON CONTINUOUSLY

CAUTION: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AA1 CHECK FOR VOLTAGE AT THE TRAILER TOW CONNECTOR

- Key in ON position.
- Measure the voltage the between trailer tow connector, harness side and ground as follows:

Connector-Pin	Circuit
C439-2	CAT06 (YE)
C439-1	CAT09 (GN)
C439-3	CAT11 (BN)

- **Is any voltage present?**

YES : Go to AA2.

NO : The vehicle is operating correctly. SEND the trailer to an authorized camper/trailer repair facility.

AA2 CHECK THE SUSPECT RELAY

- Key in OFF position.
- Disconnect: Suspect Trailer Tow Relay
- Measure the voltage the between trailer tow connector, harness side and ground as follows:

Connector-Pin	Circuit
C439-2	CAT06

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	(YE)
C439-1	CAT09 (GN)
C439-3	CAT11 (BN)

- **Is any voltage present?**

YES : REPAIR circuit CAT06 (YE) (LH turn), circuit CAT09 (GN) (RH turn), or circuit CAT11 (BN) (parking lamps) for a short to voltage. TEST the system for normal operation.

NO : INSTALL a new trailer tow relay in question. TEST the system for normal operation.

ADAPTIVE HEADLIGHTING

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

The headlamp control module (HCM) receives information from the communication network such as vehicle speed, steering wheel angle and headlamps on status. Based on the information received, the HCM determines if the headlamps need to be angled left or right as well as the degree of angle.

The HCM does not operate the adaptive headlamps until the vehicle speed is above 5 km/h (3 mph).

When the ignition is placed in the RUN/START position, the HCM runs an initialization check to determine the

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straight-ahead position of the headlamps. The lamps track left to right, then back to center to signify that the system is operating correctly.

If the HCM does not receive the required information over the communication network or determines that there is a fault with the circuitry to the sensors or control motors, it defaults the headlamps to the straight-ahead position and sets DTC(s).

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Headlamp assembly	• Battery junction box (BJB) fuse 51 (5A) • Wiring, terminals or connectors • Headlamp control module (HCM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure to use the latest scan tool software release.

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

5. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.

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7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the HCM.

9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.

10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

DTC Charts

HEADLAMP CONTROL MODULE (HCM) DTC CHART

DTC	Description	Action
B1342	ECU Is Faulted	INSTALL a new HCM. REFER to <u>Headlamp Control Module (HCM)</u> . TEST the system for normal operation.
B2609	ECU Output to Left Headlamp Stepper Motor Circuit Failure	Go to Pinpoint Test AF .
B2612	ECU Output to Right Headlamp Stepper Motor Circuit Failure	Go to Pinpoint Test AF .
B100B	Left Headlamp Swivel Control Failure	Go to Pinpoint Test AF .
B100C	Right Headlamp Swivel Control Failure	Go to Pinpoint Test AF .
C1278	Steering Wheel Angle 1 and 2 Signal Faulted	Go to Pinpoint Test AF .
U0073	Control Module Communication Bus A Off	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0100	Lost Communication With ECM/PCM	Go to Pinpoint Test AB .
U0126	Lost Communication With Steering Angle Sensor Module	Go to Pinpoint Test AC .
U0155	Lost Communication With Instrument Panel Cluster (IC) Control Module	Go to Pinpoint Test AD .
U2050	No Application Present	CARRY OUT the programmable module installation (PMI). REFER to <u>MODULE CONFIGURATION</u> article. TEST the system for normal operation.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action

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• No communication with the adaptive front lighting module	• Fuse • Wiring, terminals or connectors • Headlamp control module (HCM)	• REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
• The adaptive front lighting is inoperative/does not operate correctly	• Wiring, terminals or connectors • Headlamp assembly • HCM	• Go to Pinpoint Test AF .

Pinpoint Tests

Pinpoint Test AB: DTC U0100 - Lost Communication With ECM/PCM

Normal Operation

The headlamp control module (HCM) receives vehicle speed information from the PCM. If the HCM does not receive communication from the PCM, DTC U0100 is set.

This pinpoint test is intended to diagnose the following:

- HCM
- PCM

PINPOINT TEST AB: DTC U0100 - LOST COMMUNICATION WITH ECM/PCM

AB1 REPEAT THE ON-DEMAND SELF-TEST

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: HCM Self-Test
- Clear the DTCs and repeat the HCM on-demand self-test.
- **Is DTC U0100 retrieved again?**

YES : Go to AB2.

NO : The vehicle is operating correctly at this time. The HCM may have temporarily lost communication with the PCM.

AB2 CHECK OTHER MODULES FOR DTC U0100

- Enter the following diagnostic mode on the diagnostic tool: Instrument Cluster (IC) Self-Test
- Carry out the on-demand self-test for other modules on the high speed controller area network (HS-CAN). The IC, ABS module and restraints control module (RCM), for example, communicate on the HS-CAN.
- **Is DTC U0100 retrieved from any other module?**

YES : Go to AB3.

NO : Go to AB4.

AB3 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new PCM. REFER to **ELECTRONIC ENGINE CONTROLS** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

AB4 CHECK FOR CORRECT HCM OPERATION

- Disconnect the HCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AC: DTC U0126 - Lost Communication With Steering Wheel Angle Sensor Module**Normal Operation**

The headlamp control module (HCM) receives steering wheel angle information from the steering angle sensor module (SASM). If the HCM does not receive communication from the SASM, DTC U0126 is set.

This pinpoint test is intended to diagnose the following:

- HCM
- SASM

PINPOINT TEST AC: DTC U0126 - LOST COMMUNICATION WITH STEERING WHEEL ANGLE

SENSOR MODULE**AC1 REPEAT THE ON-DEMAND SELF-TEST**

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: HCM Self-Test
- Clear the DTCs and repeat the HCM on-demand self-test.
- **Is DTC U0126 retrieved again?**

YES : Go to AC2.

NO : The vehicle is operating correctly at this time. The HCM may have temporarily lost communication with the SASM.

AC2 CHECK THE ABS MODULE FOR DTC U0126

- Enter the following diagnostic mode on the diagnostic tool: ABS Module Self-Test
- Carry out the on-demand self-test for the ABS module.
- **Is DTC U0126 retrieved?**

YES : REFER to **MODULE COMMUNICATIONS NETWORK** article.

NO : Go to AC3.

AC3 CHECK FOR CORRECT HCM OPERATION

- Disconnect the HCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AD: DTC U0155 - Lost Communication With Instrument Panel Cluster (IC) Control Module**Normal Operation**

The smart junction box (SJB) communicates the headlamps on status to the instrument cluster (IC) over the medium speed controller area network (MS-CAN). The IC acts as a gateway module and relays the information through the high speed controller area network (HS-CAN). The headlamp control module (HCM) receives headlamps on status information from the IC gateway. If the HCM does not receive communication from the IC, DTC U0155 is set.

This pinpoint test is intended to diagnose the following:

- HCM
- IC

PINPOINT TEST AD: DTC U0155 - LOST COMMUNICATION WITH INSTRUMENT PANEL CLUSTER (IC) CONTROL MODULE

AD1 REPEAT THE ON-DEMAND SELF-TEST

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: HCM Self-Test
- Clear the DTCs and repeat the HCM on-demand self-test.
- **Is DTC U0155 retrieved again?**
YES : Go to AD2.
NO : The vehicle is operating correctly at this time. The HCM may have temporarily lost communication with the PCM.

AD2 CHECK OTHER MODULES FOR DTC U0155

- Enter the following diagnostic mode on the diagnostic tool: ABS Module Self-Test
- Carry out the on-demand self-test for other modules on the HS-CAN. The PCM, ABS module and restraints control module (RCM), for example, communicate on the HS-CAN.
- **Is DTC U0155 retrieved from any other module?**
YES : REFER to **MODULE COMMUNICATIONS NETWORK** article.
NO : Go to AD3.

AD3 CHECK FOR CORRECT HCM OPERATION

- Disconnect the HCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**
YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AE: DTC C1278 - Steering Wheel Angle 1 and 2 Signal Faulted

Normal Operation

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The headlamp control module (HCM) receives steering wheel angle information from the steering angle sensor module (SASM). If the HCM receives faulty steering wheel angle data from the SASM, DTC C1278 is set.

This pinpoint test is intended to diagnose the following:

- HCM
- SASM

PINPOINT TEST AE: DTC C1278 - STEERING WHEEL ANGLE 1 AND 2 SIGNAL FAULTED

AE1 REPEAT THE ON-DEMAND SELF-TEST

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: HCM Self-Test
- Clear the DTCs and repeat the HCM on-demand self-test.
- **Is DTC C1278 retrieved again?**

YES : Go to AE2.

NO : The vehicle is operating correctly at this time. The HCM may have temporarily received incorrect data from the steering wheel angle sensor module.

AE2 CHECK THE ABS MODULE FOR DTC C1278

- Carry out the on-demand self-test for the ABS module.
- **Is DTC C1278 retrieved?**

YES : Go to AE3.

NO : Go to AE4.

AE3 CHECK FOR CORRECT SASM OPERATION

- Disconnect the SASM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the SASM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SASM. REFER to Steering Wheel Rotation Sensor in **VEHICLE DYNAMIC SYSTEMS** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

AE4 CHECK FOR CORRECT HCM OPERATION

- Disconnect the HCM connector.
- Check for:
 - corrosion

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- damaged pins
- pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AF: The Adaptive Front Lighting Is Inoperative/Does Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Headlamps/Autolamps for schematic and connector information.

Normal Operation

The headlamp control module (HCM) receives information from the high speed controller area network (HS-CAN), such as steering wheel angle, vehicle speed, and headlamps on status. Based on the information received, the HCM determines what angle (left to right of straight ahead) the headlamps should be.

The HCM does not operate the adaptive headlamps until the headlamps are on and the vehicle speed is above 5 km/h (3 mph). Depending on vehicle speed and steering wheel angle, the HCM determines and positions the headlamps at an angle to improve visibility while the vehicle is in a turn. At low and high vehicle speeds (8 km/h [5 mph] and 97 km/h [60 mph] for example), the angle of the headlamps is changed minimally. At vehicle speeds of 32-40 km/h (20-25 mph), the headlamp angle change is greater.

The HCM monitors the position of the headlamps and controls the motors, which changes the angle of the headlamps.

DTC Description	Fault Trigger Conditions
B2609 - ECU Output to Left Headlamp Stepper Motor Circuit Failure	A continuous and on-demand DTC that sets when the HCM detects a failure from the LH headlamp stepper motor or circuitry.
B2612 - ECU Output to Right Headlamp Stepper Motor Circuit Failure	A continuous and on-demand DTC that sets when the HCM detects a failure from the RH headlamp stepper motor or circuitry.
B100B - Left Headlamp Swivel Control Failure	A continuous and on-demand DTC that sets when the HCM detects a failure from the LH headlamp position sensor or circuitry.
B100C - Right Headlamp Swivel Control Failure	A continuous and on-demand DTC that sets when the HCM detects a failure from the RH headlamp position sensor or circuitry.

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This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp assembly
- HCM

PINPOINT TEST AF: THE ADAPTIVE FRONT LIGHTING IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

AF1 CHECK THE RECORDED DTCs FROM THE HCM SELF-TEST

- Check the recorded results from the HCM self-test.
- **Are any DTCs recorded?**

YES : For DTC B2612 or B100C, go to AF3.

For DTC B2609 or B100B, go to AF11.

For all other DTCs, REFER to DTC Charts in this service information.

NO : Go to AF2.

AF2 ACTIVE COMMAND THE HEADLAMP ANGLE

- Key in ON position.
- Turn the headlamps on.
- Select the HCM headlamp angle control (BEAM_CTRL_LFT, BEAM_CTRL_RGT) active commands. Command the headlamps to the full left and full right positions while observing the headlamp beam patterns on a wall.
- **Do the headlamps move left and right?**

YES : Go to AF10.

NO : Go to AF11.

AF3 CHECK THE CIRCUITRY BETWEEN THE LH AND RH HEADLAMPS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: LH Headlamp C1021b and RH Headlamp C1041b
- Key in ON position.
- Measure the voltage between the LH headlamp, harness side and ground as follows:

Connector-Pin	Circuit
C1021b-1	CLF46 (YE/GY)

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C1021b-2	CLF45 (VT/GY)
C1021b-4	LLF43 (GN/VT)
C1021b-5	LLF44 (BU/GY)
C1021b-7	CLF48 (GN/BN)
C1021b-8	CLF47 (BU/GN)
C1021b-10	RLF45 (GY/VT)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

NO : Go to AF4.

AF4 CHECK THE CIRCUITRY BETWEEN THE LH AND RH HEADLAMPS FOR A SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the LH headlamp, harness side and ground as follows:

Connector-Pin	Circuit
C1021b-1	CLF46 (YE/GY)
C1021b-2	CLF45 (VT/GY)
C1021b-4	LLF43 (GN/VT)
C1021b-5	LLF44 (BU/GY)
C1021b-7	CLF48 (GN/BN)
C1021b-8	CLF47 (BU/GN)
C1021b-10	RLF45 (GY/VT)

- **Are the resistances greater than 10,000 ohms?**

YES : Go to AF5.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

AF5 CHECK THE CIRCUITRY BETWEEN THE LH AND RH HEADLAMPS FOR AN OPEN

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- Measure the resistance between the LH headlamp, harness side and RH headlamp, harness side:

LH Headlamp Connector-Pin	RH Headlamp Connector-Pin	Circuit
C1021b-1	C1041b-3	CLF46 (YE/GY)
C1021b-2	C1041b-2	CLF45 (VT/GY)
C1021b-4	C1041b-4	LLF43 (GN/VT)
C1021b-5	C1041b-5	LLF44 (BU/GY)
C1021b-7	C1041b-9	CLF48 (GN/BN)
C1021b-8	C1041b-8	CLF47 (BU/GN)
C1021b-10	C1041b-10	RLF45 (GY/VT)

- Are the resistances less than 5 ohms?

YES : Go to AF6.

NO : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

AF6 CHECK FOR SHORTED TOGETHER CIRCUITS BETWEEN THE LH AND RH HEADLAMPS

- Measure the resistance between each circuit on the LH headlamp, harness side as follows:

Connector-Pin Circuit	Connector-Pin Circuit
C1021b-1 CLF46 (YE/GY)	C1021b-2 CLF45 (VT/GY) C1021b-4 LLF43 (GN/VT) C1021b-5 LLF44 (BU/GY) C1021b-7 CLF48 (GN/BN) C1021b-8 CLF47

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	(BU/GN) C1021b-10 RLF45 (GY/VT)
C1021b-2 CLF45 (VT/GY)	C1021b-4 LLF43 (GN/VT) C1021b-5 LLF44 (BU/GY) C1021b-7 CLF48 (GN/BN) C1021b-8 CLF47 (BU/GN) C1021b-10 RLF45 (GY/VT)
C1021b-4 LLF43 (GN/VT)	C1021b-5 LLF44 (BU/GY) C1021b-7 CLF48 (GN/BN) C1021b-8 CLF47 (BU/GN) C1021b-10 RLF45 (GY/VT)
C1021b-5 LLF44 (BU/GY)	C1021b-7 CLF48 (GN/BN) C1021b-8 CLF47 (BU/GN) C1021b-10 RLF45 (GY/VT)
C1021b-7 CLF48 (GN/BN)	C1021b-8 CLF47 (BU/GN) C1021b-10 RLF45 (GY/VT)
C1021b-8	C1021b-10

CLF47 (BU/GN)	RLF45 (GY/VT)
------------------	------------------

- Are the resistances greater than 10,000 ohms between each circuit?

YES : Go to AF7.

NO : REPAIR the circuits in question. CLEAR the DTCs. REPEAT the self-test.

AF7 CHECK THE RH STEPPER MOTOR CIRCUITS FOR AN OPEN

- Remove the LH headlamp. Refer to Headlamp Assembly.
- Disconnect: HCM

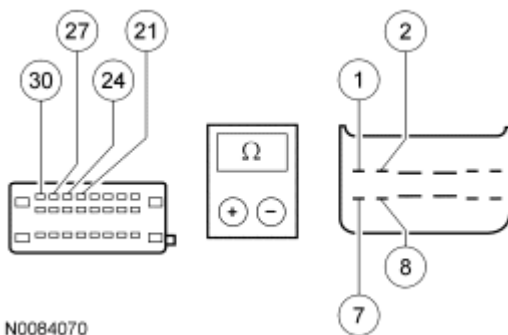


Fig. 106: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the HCM, harness side and the LH headlamp, component side as follows:

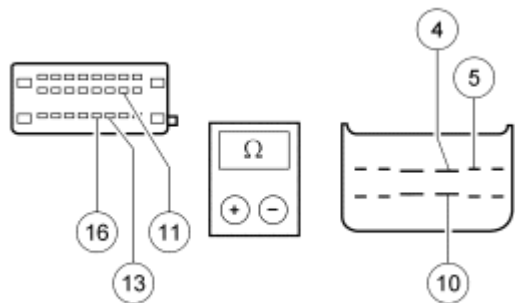
HCM Connector	LH Headlamp Connector	Circuit
Pin 21	Pin 1	CLF46
Pin 30	Pin 2	CLF45
Pin 24	Pin 7	CLF48
Pin 27	Pin 8	CLF47

- Are the resistances less than 5 ohms?

YES : Go to AF8.

NO : REPAIR or INSTALL a new LH headlamp. REFER to Headlamp Assembly. CLEAR the DTCs. REPEAT the self-test.

AF8 CHECK THE RH SENSOR CIRCUITS FOR AN OPEN



N0084071

Fig. 107: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the HCM, harness side and the LH headlamp, component side as follows:

HCM Connector	LH Headlamp Connector	Circuit
Pin 13	Pin 4	LLF43
Pin 11	Pin 5	LLF44
Pin 16	Pin 10	RLF45

- **Are the resistances less than 5 ohms?**

YES : Go to AF9.

NO : REPAIR or INSTALL a new LH headlamp. REFER to **Headlamp Assembly**. CLEAR the DTCs. REPEAT the self-test.

AF9 CHECK FOR CORRECT HCM OPERATION (RH HEADLAMP DTCs)

- Disconnect the HCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.

If the concern remains, REINSTALL the original HCM. INSTALL a new RH headlamp. REFER to **Headlamp Assembly**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose

or corroded connector.

AF10 CHECK THE STEERING WHEEL ROTATION SENSOR FOR CORRECT INSTALLATION

NOTE: If the PID indicates that the steering wheel rotation sensor is out of synchronization by 360 degrees, this is an indication that the steering column shaft was disconnected and the steering wheel was rotated one full turn prior to reconnecting it. The clockspring may be damaged and should be checked for correct installation.

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Steering Angle Sensor Module (SASM) DataLogger
- Visually inspect the front wheels and make sure they are aligned to the straight-ahead position.

NOTE: The PID could read +360 degrees or -360 degrees. The PID should indicate 0 when the wheels are in the straight-ahead position.

- Monitor the SASM steering wheel angle (SWA_SEN) PID.
- **Does the PID indicate the steering wheel at approximately 0 degrees?**

YES : Go to AF11.

NO : REMOVE and correctly INSTALL the steering wheel rotation sensor. REFER to **VEHICLE DYNAMIC SYSTEMS** article. MAKE SURE the clockspring is also correctly indexed. TEST the system for normal operation.

AF11 CHECK FOR CORRECT HCM OPERATION (LH HEADLAMP DTCs)

- Disconnect the HCM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HCM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new HCM. REFER to **Headlamp Control Module (HCM)**. TEST the system for normal operation.

If the concern remains, INSTALL a new LH headlamp. REFER to **Headlamp Assembly**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

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DAYTIME RUNNING LAMPS (DRL)

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

NOTE: When installing a new SJB, it must be configured for this feature, if originally equipped. Refer to MODULE CONFIGURATION article.

The SJB controls the daytime running lamps (DRL) by using various inputs to determine whether or not the DRL should be illuminated. These inputs include:

- Ignition switch position
- Headlamp switch position
- Headlamps on or off
- Transaxle gear selection

When the DRL is active, the SJB provides a pulse-width modulated (PWM) voltage to the low beam headlamps. This illuminates the headlamps at a reduced intensity.

The SJB can be configured to turn the DRL on for this vehicle. Refer to MODULE CONFIGURATION article for additional information on this programmable parameter.

Inspection and Verification

1. Verify the customer concern.
2. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure the headlamp switch is in the OFF position.

NOTE: Make sure the multifunction switch is not in the HIGH BEAM position.

NOTE: Make sure to use the latest scan tool software release.

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3. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.

4. If the scan tool does not communicate with the VCM:
 - Check the VCM connection to the vehicle.
 - Check the scan tool connection to the VCM.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
5. If the scan tool does not communicate with the vehicle:
 - Verify the ignition key is in the ON position.
 - Verify the scan tool operation with a known good vehicle.
 - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
6. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record the continuous memory DTCs.
7. Check the smart junction box (SJB) and make sure the daytime running lamps (DRL) configuration is enabled.
8. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• The daytime running lamps (DRL) are inoperative	• Exterior lighting system input/output • Smart junction box (SJB) configuration • Ignition switch input • Transaxle gear input • SJB	• Go to Pinpoint Test AG .

Pinpoint Tests

Pinpoint Test AG: The Daytime Running Lamps (DRL) Are Inoperative

Normal Operation

When the ignition switch is in the RUN position, the headlamps are not on (from the autolamp system or

manual control from the headlamp switch), the transaxle is not in PARK (P), the smart junction switch (SJB) provides a pulse-width modulated (PWM) voltage to the low beam headlamps. This illuminates the headlamps at a reduced intensity.

This pinpoint test is intended to diagnose the following:

- Exterior lighting system input/output
- SJB configuration
- Ignition switch input
- Transaxle gear input
- SJB

PINPOINT TEST AG: THE DAYTIME RUNNING LAMPS (DRL) ARE INOPERATIVE

AG1 VERIFY THE DRL OPERATION

- Place the gear selector lever in any gear other than PARK (P).
- **Are the DRL illuminated?**

YES : The system is operating correctly. INFORM the customer of the conditions required for the DRL to operate correctly.

NO : Go to AG2.

AG2 CHECK THE OPERATION OF THE HEADLAMPS

- Place the headlamp switch in the HEADLAMPS ON position and observe the headlamps.
- **Do the headlamps operate correctly?**

YES : Go to AG3.

NO : REFER to **Headlamps**.

AG3 CHECK THE IGNITION SWITCH INPUT

- Place the headlamp switch in the OFF position.
- Enter the following diagnostic mode on the diagnostic tool: SJB DataLogger
- Monitor the SJB ignition switch PID (IGN SW) while cycling the ignition switch through all positions.
- **Does the PID agree with the ignition switch positions?**

YES : Go to AG4.

NO : REFER to **STEERING COLUMN SWITCHES** article to diagnose the input from the ignition switch.

AG4 CHECK THE TRANSAXLE GEAR INPUT

- Enter the following diagnostic mode on the diagnostic tool: PCM DataLogger
- Monitor the PCM transmission gear status PID (TR) while placing the gear selector lever through all positions.
- **Does the PID agree with the gear selection?**

YES : Go to AG5.

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NO : REFER to **AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50** article to diagnose the transmission range input.

AG5 CHECK SJB FOR DRL CONFIGURATION

- Enter the following diagnostic mode on the diagnostic tool: SJB Module Programming
- Check the SJB configuration and make sure the DRL feature is enabled.
- **Is the SJB DRL configuration enabled?**

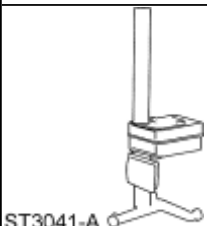
YES : INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

NO : ENABLE the DRL configuration. TEST the system for normal operation.

GENERAL PROCEDURES

HEADLAMP ADJUSTMENT

Special Tools

Illustration	Tool Name	Tool Number
 ST3041-A	Vision 100 Headlamp Aimer	196-00005 or equivalent

Headlamp Aiming

1. The headlamp aiming procedure depends on what type of beam pattern the headlamp is equipped with. Vehicles may come equipped with VOL or VOR headlamps. To identify the headlamp beam pattern, look on the headlamp lens. Molded in small letters on the headlamp lens is one of the following:
 - VOL and SAE
 - VOR and SAE
2. Once the headlamp beam pattern is identified, aim the headlamps using one of the following methods as applicable.
 - Photometric aimers can aim VOL- and VOR- headlamps. This is the preferred method of headlamp aiming.
 - Visual or screen method aiming can be used to aim VOL- and VOR- headlamps.
 - Mechanical aimers cannot be used with VOR- or VOL-type headlamps. Aerodynamic lamps that can be aimed mechanically have 3 nibs molded into the lens of the lamp.

Photometric Aiming

1. For the photometric aiming procedure, refer to the appropriate photometric headlamp aimer instruction manual.

Screen Method Aiming**All headlamp types**

- NOTE:** Horizontal aim is not necessary for VOL or VOR headlamps.
- NOTE:** Consult your state vehicle inspection manual for recommended tolerance ranges for visual aiming.
- NOTE:** The sight shield may need to be positioned or removed for access to the adjusters.

1. Before starting headlamp adjustment:

- check the tire inflation.
- check that no other load is in the vehicle other than a half tank of fuel.
- check that the headlamps are clean.
- check for correct headlamp operation.
- check that the vehicle is on level ground.
- if the vehicle is equipped with air suspension, make sure that the switch is on.

NOTE: The vertical wall or screen must be a minimum of 2.4-m (8-ft) wide.

2. Park the vehicle on a level surface approximately 7.6-m (25-ft) from the vertical wall or screen directly in front of it.

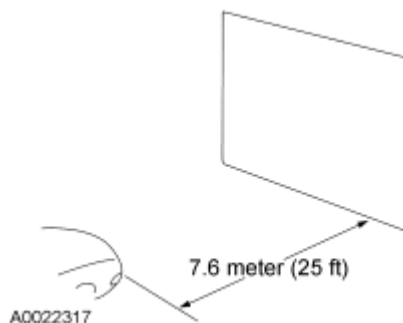


Fig. 108: Identifying Distance Between Vehicle And Vertical Wall Or Screen
Courtesy of FORD MOTOR CO.

NOTE: The center of the lamp is marked either on the lens (circle, crosshair or other mark) or on the bulb shield internal to the lamp (crosshair or other mark).

3. Mark a horizontal reference line on the vertical wall or screen.

1. Measure the center of the headlamp height to ground and record the measurement.

2. Make a 2.4-meter (8-ft) horizontal mark (using masking tape) on the vertical wall or screen at the same distance from the ground as previously recorded.

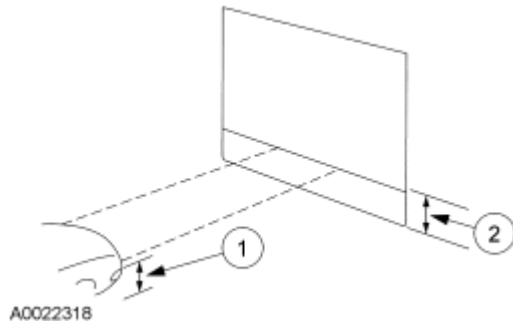


Fig. 109: Aiming Headlights
Courtesy of FORD MOTOR CO.

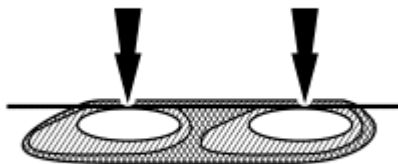
NOTE: This procedure should be done in a dark environment to effectively see the headlamp beam pattern.

4. Turn on the low beam headlamps to illuminate the wall or screen and open the hood.
5. On the wall or screen, locate the high intensity area of the beam pattern.

VOR-type headlamps

NOTE: The appearance of the VOR beam pattern may vary between vehicles.

6. Identify at the top edge of this high intensity area a distinct horizontal cutoff in the beam pattern. If the top edge of this cutoff is not even with the horizontal reference line, the headlamp beam needs to be adjusted using the headlamp adjusting screw.



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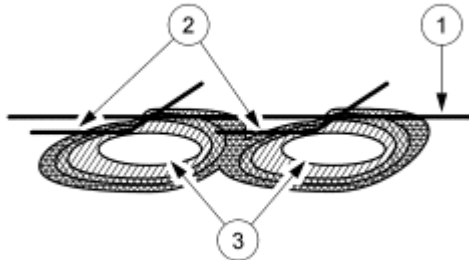
Fig. 110: Identifying High Intensity Area (VOR Type Headlamps)
Courtesy of FORD MOTOR CO.

VOL-type headlamps

7. For VOL-type headlamps, there is a distinct cutoff in the left portion of the beam pattern. The edge of this

cutoff should be positioned 50.2 mm (2 in) below the horizontal reference line. Adjust the headlamp as necessary using the headlamp adjusting screw.

1. Horizontal reference line.
2. Top edge of the beam pattern.
3. High intensity zone.



A0042249

Fig. 111: Identifying VOL-Type Headlamps Beam Pattern
Courtesy of FORD MOTOR CO.

FRONT FOG LAMP ADJUSTMENT

NOTE: Horizontal aim is not required for this vehicle and is not adjustable. Consult your state vehicle inspection center for recommended tolerance ranges for visual aiming.

1. Before starting the fog lamp assembly adjustment:
 - check the tire inflation.
 - make sure there are no other loads in the vehicle other than a half tank of fuel.
 - make sure the vehicle is on level ground.
 - make sure the fog lamps and headlamps are clean.
 - make sure the headlamps are operating and are correctly aimed.

NOTE: The vertical wall screen must be a minimum of 2.4-m (8-ft) wide.

2. Park the vehicle on a level surface approximately 7.6 m (25 ft) from the vertical wall or screen directly in front of it.

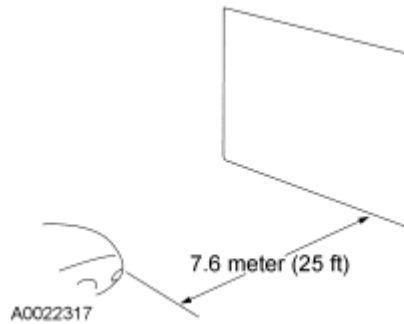


Fig. 112: Identifying Distance Between Vehicle And Vertical Wall Or Screen
Courtesy of FORD MOTOR CO.

- The correct visual aim for the fog lamps is with the top edge of the high-intensity zone 10 cm (4 in) below the horizontal center of the fog lamps.

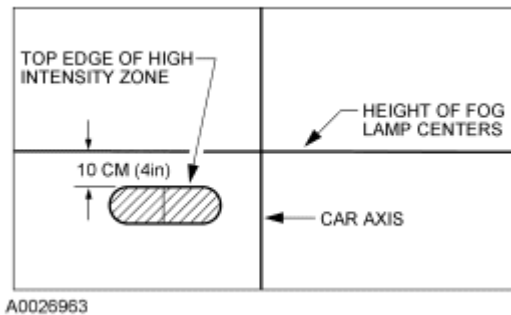


Fig. 113: Identifying Correct Visual Aim For Fog Lamps
Courtesy of FORD MOTOR CO.

AUTOLAMPS TIME DELAY ADJUSTMENT

NOTE: The time delay can be programmed through the instrument cluster message center using the message center buttons (if equipped) or using the following steps.

NOTE: Steps 2 through 5 must be carried out within a 10-second period.

- Start with the ignition switch off and the headlamp switch in the AUTOLAMPS ON position.
- Place the headlamp switch in the OFF position.
- Place the ignition switch in the RUN position.
- Place the ignition switch in the OFF position.
- Place the headlamp switch in the AUTOLAMPS ON position. The exterior lamps turn on at this point.
- Wait the desired amount of time and place the headlamp switch in the OFF position (maximum of 3 minutes). The exterior lamps turn off and the autolamp time delay is now set.

REMOVAL AND INSTALLATION

HEADLAMP ASSEMBLY

NOTE: Edge shown, MKX similar.



N0062279

Fig. 114: Identifying Headlamp Assembly
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W711676	Headlamp assembly bolts (4 required)
2	13005/13006	Headlamp assembly (LH/RH)
3	-	Headlamp electrical connectors (part of 14290)

REMOVAL AND INSTALLATION

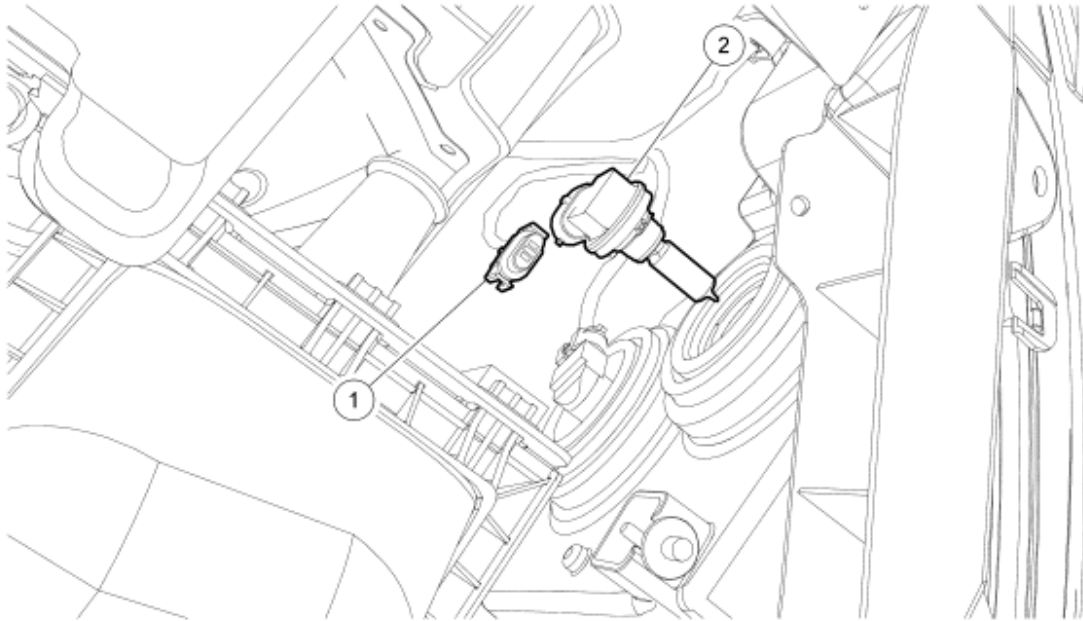
NOTE: Make sure the headlamp switch and the ignition switch are in the OFF position.

1. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
2. Remove the 4 bolts and the headlamp assembly.
 - Disconnect the electrical connectors.
3. To install, reverse the removal procedure.

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HEADLAMP BULB



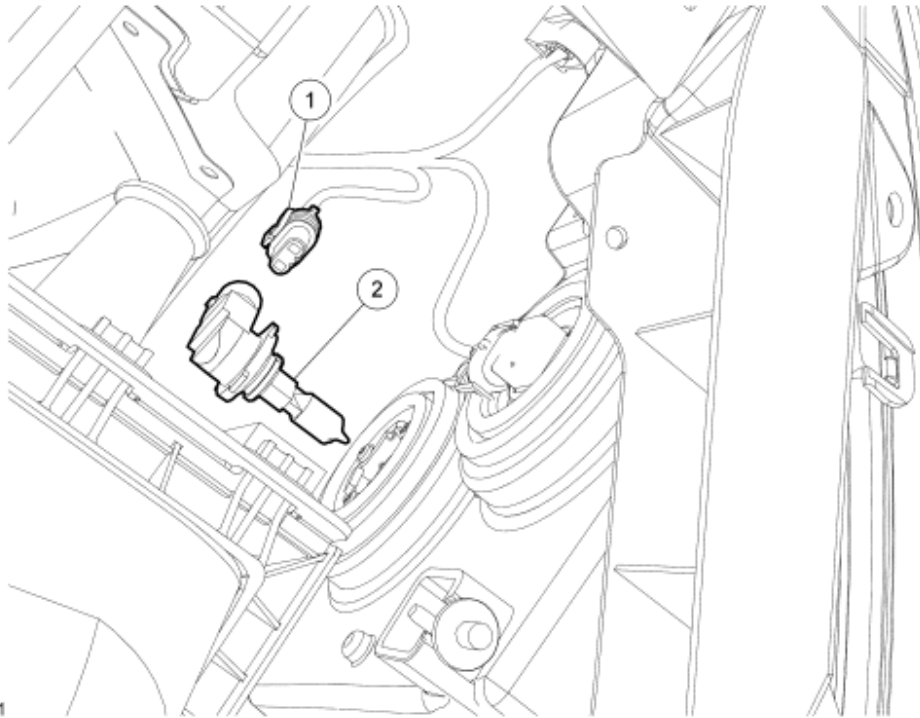
N0062280

Fig. 115: Exploded View Of Headlamp Bulb Edge Low Beam
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp bulb electrical connector (part of 14290)
2	13N021	Headlamp bulb

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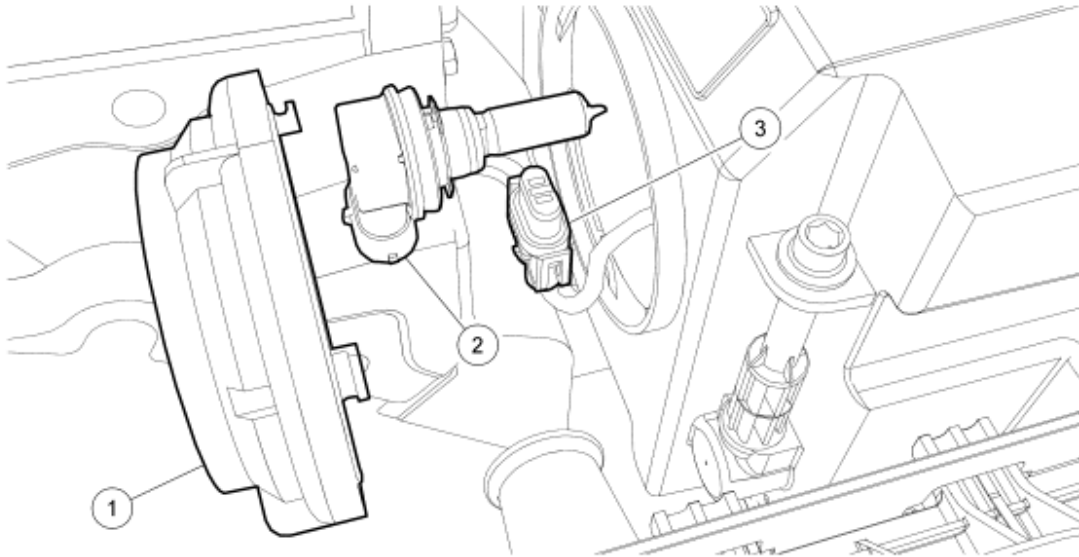
N0062281

Fig. 116: Identifying Edge High Beam
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp bulb electrical connector (part of 14290)
2	13N021	Headlamp bulb

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N0062282

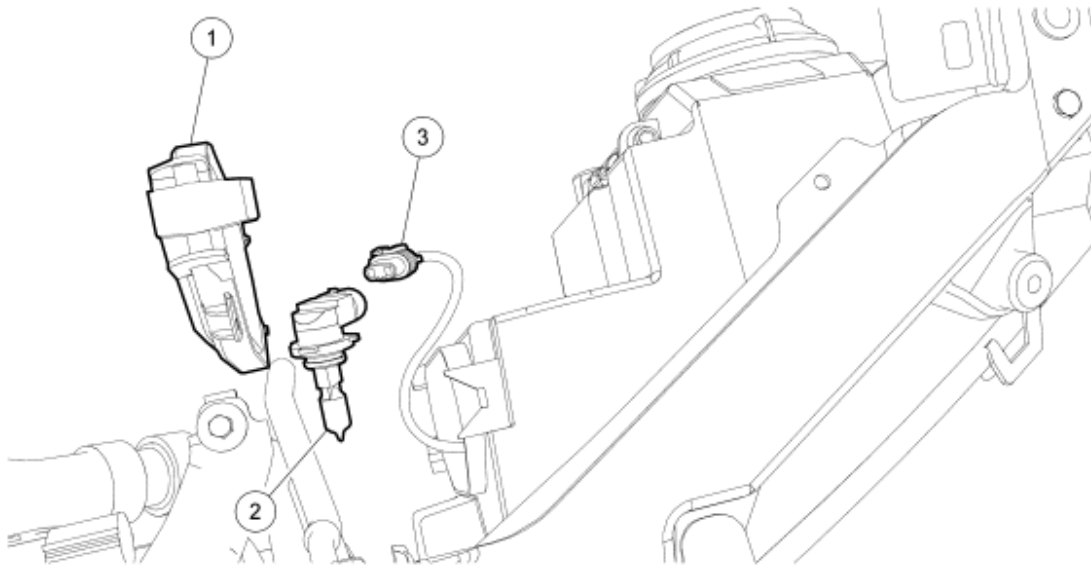
Fig. 117: Identifying MKX Low Beam
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp bulb access cover (part of 13005/13006)
2	13N021	Headlamp bulb
3	-	Headlamp bulb electrical connector (part of 13005/13006)

NOTE: The air cleaner assembly is shown removed for clarity.

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N0082042

Fig. 118: Identifying MKX High Beam
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp bulb access cover (part of 13005/13006)
2	13N021	Headlamp bulb
3	-	Headlamp bulb electrical connector (part of 13005/13006)

REMOVAL AND INSTALLATION

WARNING: The bulb contains gas under pressure. The bulb may shatter if the glass envelope is scratched or if the bulb is dropped. Handle the bulb only by its base. Avoid touching the glass envelope. Failure to follow these instructions may result in personal injury.

NOTE: The headlamp bulb should not be removed from the headlamp until just before a new bulb is installed. Removing the bulb for an extended period of time may affect headlamp bulb performance. Contaminants may enter the headlamp where they can settle on the lens and reflector. Never turn on the headlamps with the bulb removed from the headlamp.

NOTE: Make sure the headlamp switch and the ignition switch are in the OFF position.

MKX

1. Remove the headlamp bulb access cover.

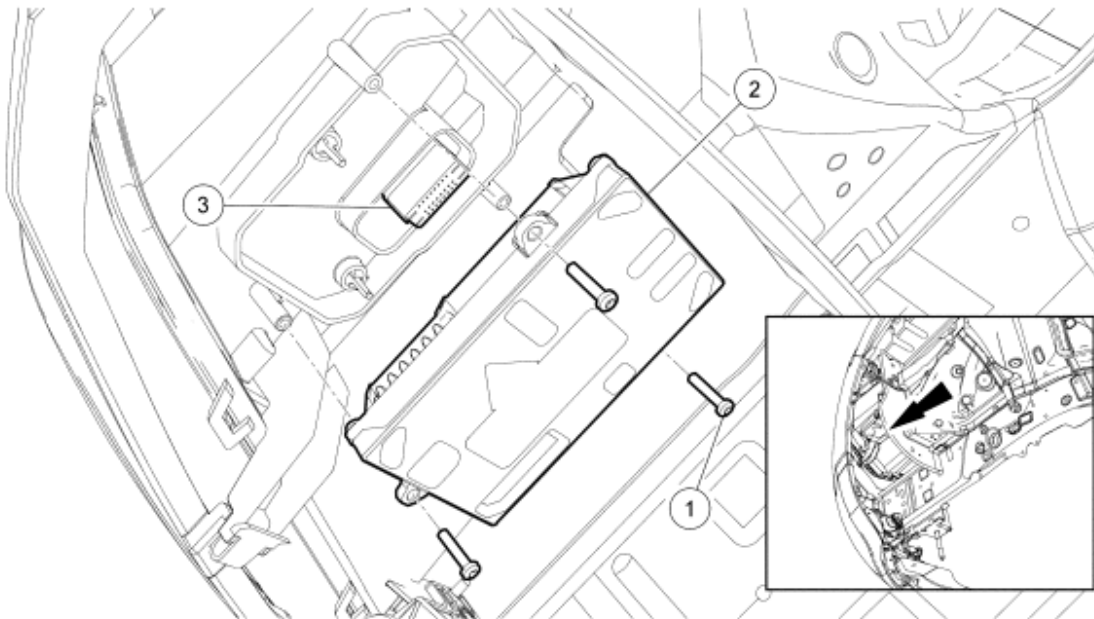
All vehicles

NOTE: To remove, rotate the headlamp bulb approximately 1/8 turn counterclockwise.

2. Remove the headlamp bulb.
 - Disconnect the electrical connector.
3. To install, reverse the removal procedure.

HEADLAMP CONTROL MODULE (HCM)

NOTE: The headlamp control module (HCM) can be accessed from under the vehicle.



N0062284

Fig. 119: Exploded View Of Headlamp Control Module (HCM)

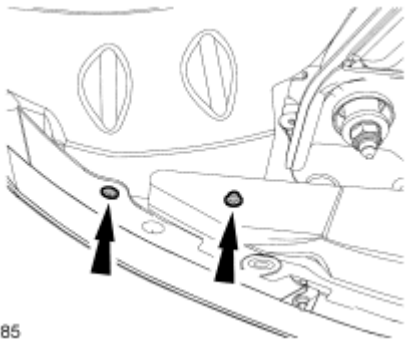
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp control module (HCM) screws (3 required) (part of 13005)
2	13C148	HCM

3	-	HCM electrical connector (part of 13005)
---	---	--

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the 2 screws and position the fender splash shield aside.

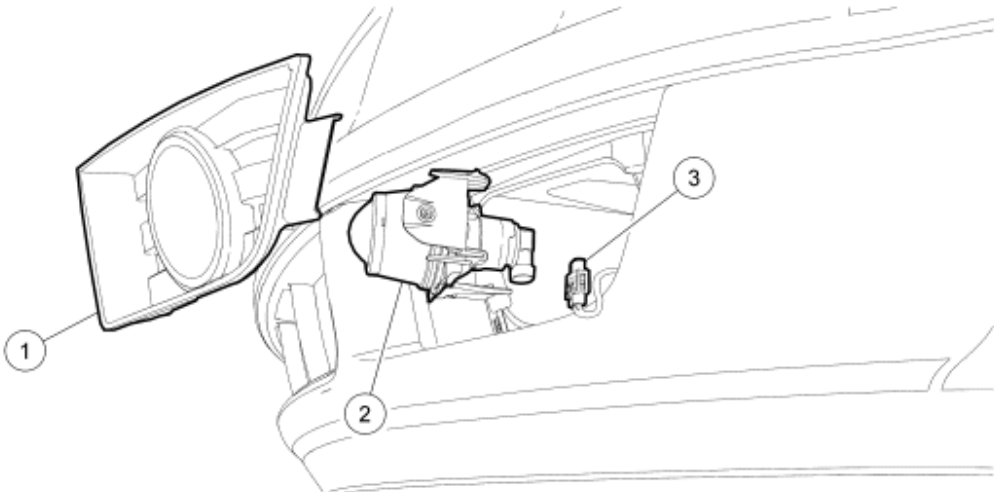


N0062285

Fig. 120: Locating Screws On Fender Splash Shield
 Courtesy of FORD MOTOR CO.

3. Remove the 3 screws and the HCM.
 - Disconnect the electrical connector.
4. To install, reverse the removal procedure.

FOG LAMP



N0062286

Fig. 121: Exploded View Of Fog Lamp
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description

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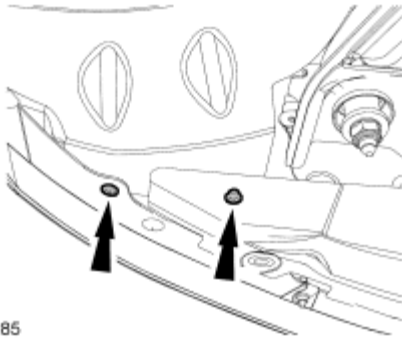
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1	17E811	Fog lamp bezel
2	15K200	Fog lamp
3	-	Fog lamp electrical connector (part of 14290)

REMOVAL AND INSTALLATION

NOTE: Make sure the headlamp switch and the ignition switch are in the OFF position.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the 2 screws and position the fender splash shield aside.



N0062285

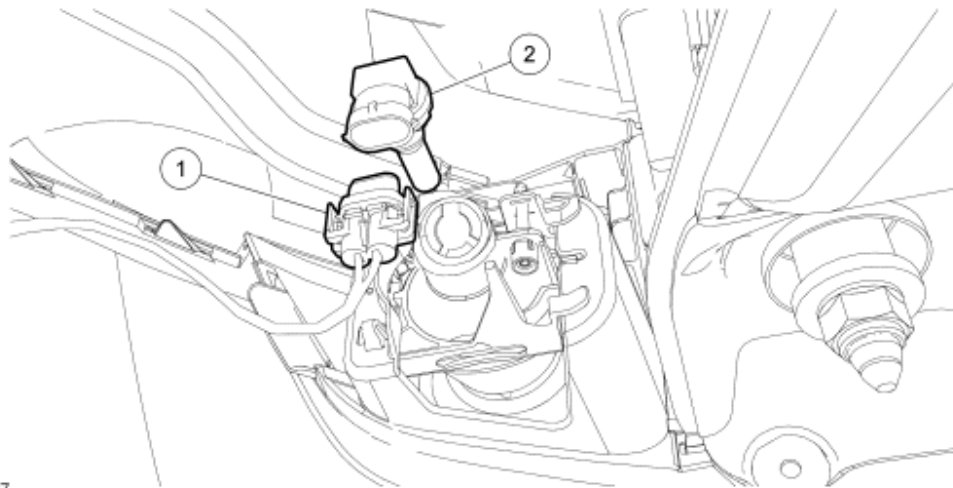
Fig. 122: Locating Screws On Fender Splash Shield
Courtesy of FORD MOTOR CO.

3. Release the tabs and remove the fog lamp bezel.
4. Release the locking tabs and remove the fog lamp.
 - Disconnect the electrical connector.
5. To install, reverse the removal procedure.

FOG LAMP BULB

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N0062287

Fig. 123: Exploded View Of Fog Lamp Bulb
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Fog lamp electrical connector (part of 14290)
2	13N021	Fog lamp bulb

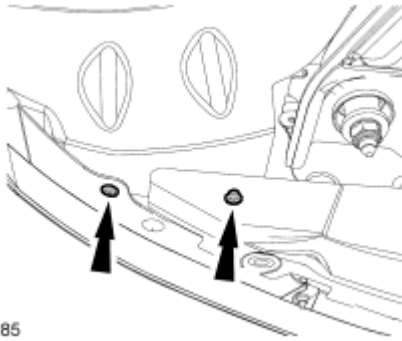
REMOVAL AND INSTALLATION

WARNING: The bulb contains gas under pressure. The bulb may shatter if the glass envelope is scratched or if the bulb is dropped. Handle the bulb only by its base. Avoid touching the glass envelope. Failure to follow these instructions may result in personal injury.

NOTE: The fog lamp bulb should not be removed from the fog lamp until just before a new bulb is installed. Removing the bulb for an extended period of time may affect fog lamp bulb performance. Contaminants may enter the fog lamp where they can settle on the lens and reflector. Never turn on the fog lamps with the bulb removed.

NOTE: Make sure the headlamp switch and the ignition switch are in the OFF position.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the 2 screws and position the fender splash shield aside.

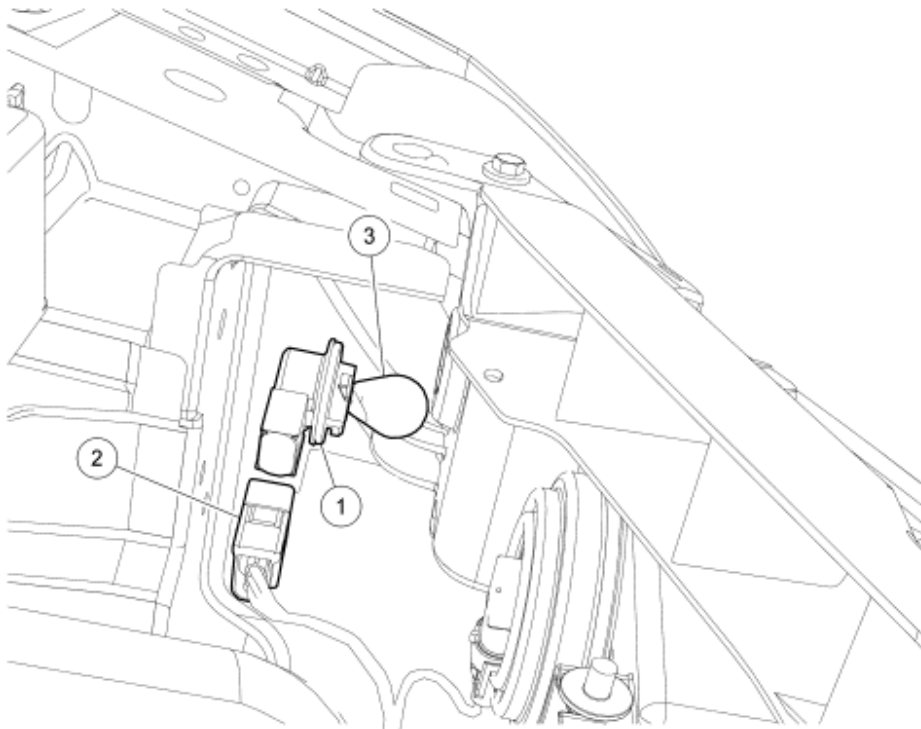


N0062285

Fig. 124: Locating Screws On Fender Splash Shield
Courtesy of FORD MOTOR CO.

3. Disconnect the electrical connector and remove the fog lamp bulb.
4. To install, reverse the removal procedure.

PARKING LAMP BULB - FRONT



N0062288

Fig. 125: Exploded View Of Front Parking Lamp Bulb - Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13411	Front parking lamp bulb holder
2	-	Front parking lamp electrical connector (part of 14290)

3	13466	Front parking lamp bulb
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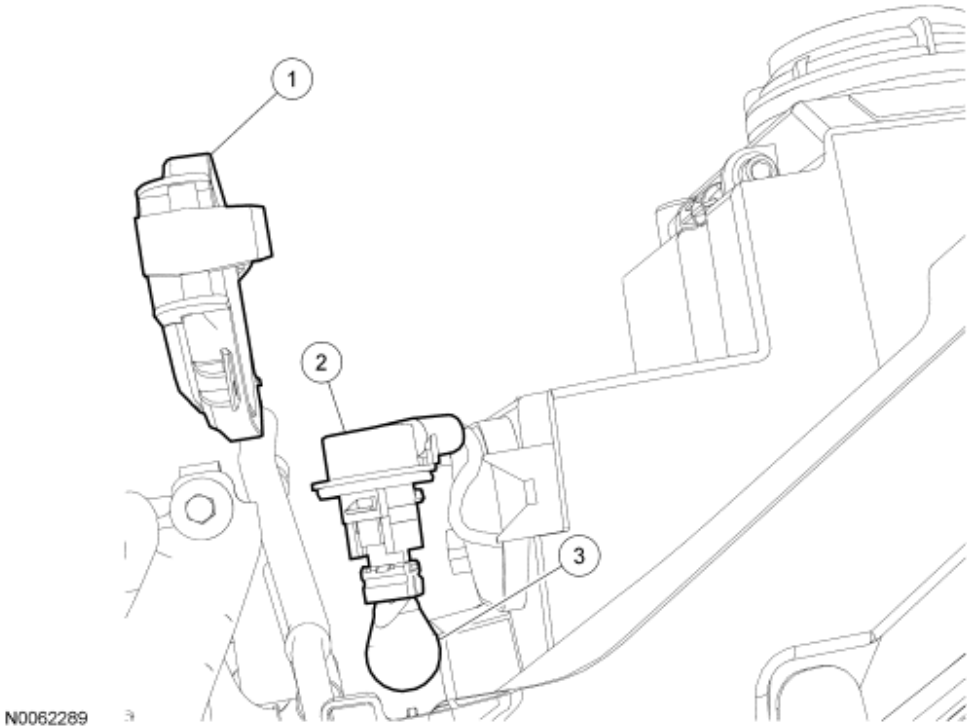


Fig. 126: Exploded View Of Front Parking Lamp Bulb - MKX
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Headlamp bulb access cover (part of 13005/13006)
2	-	Front parking lamp electrical connector (part of 13005/13006)
3	13466	Front parking lamp bulb

REMOVAL AND INSTALLATION

MKX

1. Remove the headlamp bulb access cover.

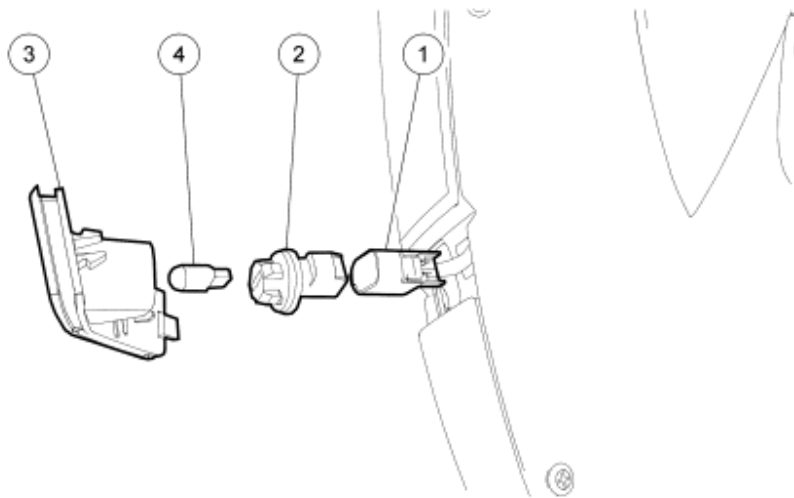
All vehicles

2. Remove the front parking lamp holder from the headlamp assembly.
 - Remove the front parking lamp bulb.
3. To install, reverse the removal procedure.

SIDE MARKER LAMP - FRONT

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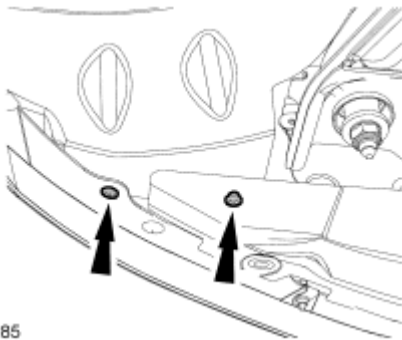
N0062290

Fig. 127: Exploded View Of Front Side Marker Lamp
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Side marker lamp electrical connector (part of 14290)
2	-	Side marker bulb holder
3	13200/13201	Side marker lamp (LH/RH)
4	-	Side marker bulb

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the 2 screws and position the fender splash shield aside.



N0062285

Fig. 128: Locating Screws On Fender Splash Shield
Courtesy of FORD MOTOR CO.

3. Release the tabs and remove the side marker lamp.
 - Disconnect the electrical connector.

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4. To install, reverse the removal procedure.

LIGHT SENSOR

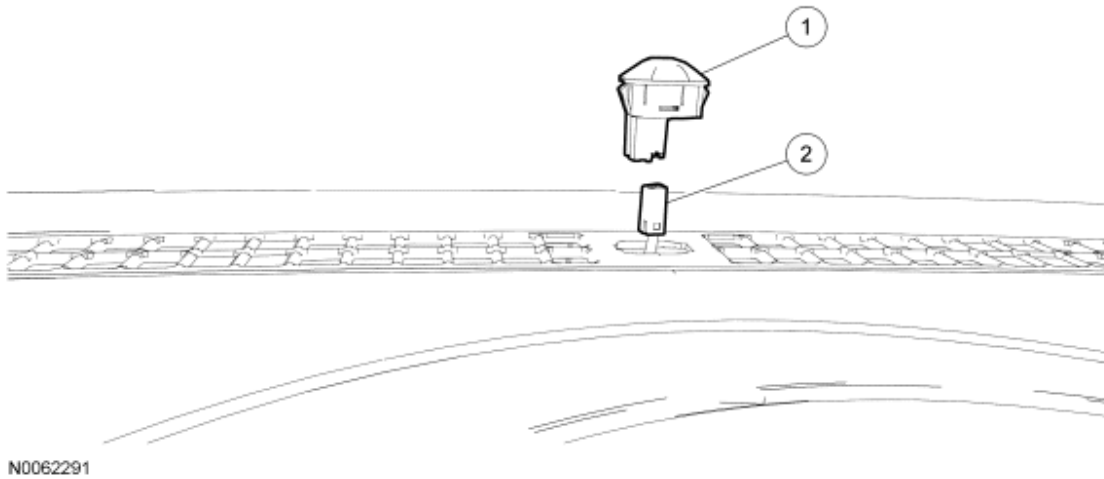


Fig. 129: Exploded View Of Light Sensor
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13A018	Light sensor
2	-	Light sensor electrical connector (part of 14401)

REMOVAL AND INSTALLATION

1. Using a suitable tool, remove the light sensor.
 - Disconnect the electrical connector.
2. To install, reverse the removal procedure.

STOPLAMP SWITCH

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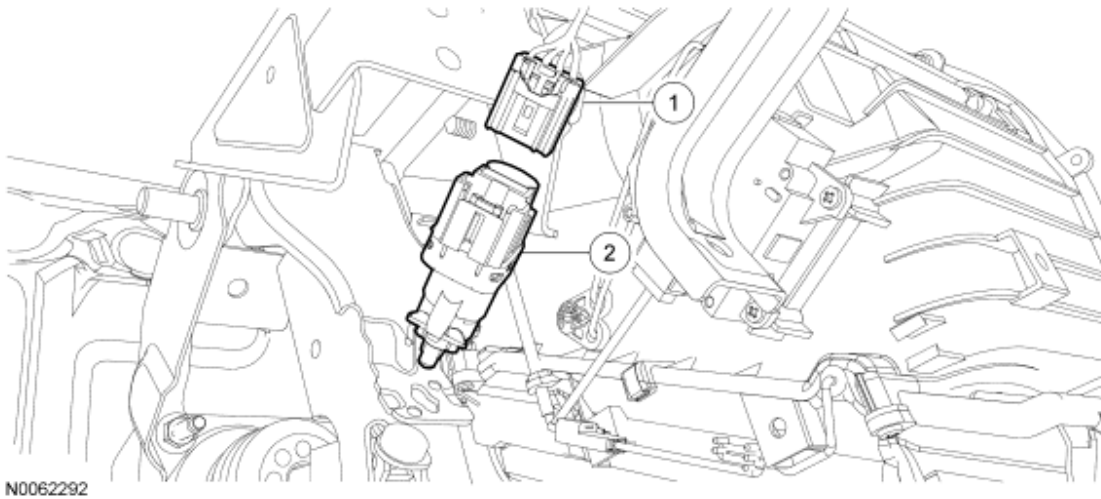


Fig. 130: Exploded View Of Stoplamp Switch
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Stoplamp switch electrical connector (part of 14290)
2	9G854	Stoplamp switch

REMOVAL

1. Disconnect the stoplamp switch electrical connector.

NOTE: The stoplamp switch will be difficult to remove if the switch plunger is not depressed. A slight rearward force should be adequate to depress the plunger.

2. With a light force (approximately 1-4 N [1-3 lb]), pull rearward on the brake pedal, rotate the stoplamp switch 45 degrees clockwise and remove the stoplamp switch.

INSTALLATION

CAUTION: It is very important to not press, pull or touch the brake pedal assembly during the installation of the stoplamp switch. Failure to follow these instructions may damage the stoplamp switch.

1. Without movement to the brake pedal assembly, install the stoplamp switch and rotate 45 degrees counterclockwise.
2. Connect the stoplamp switch electrical connector.

HEADLAMP SWITCH

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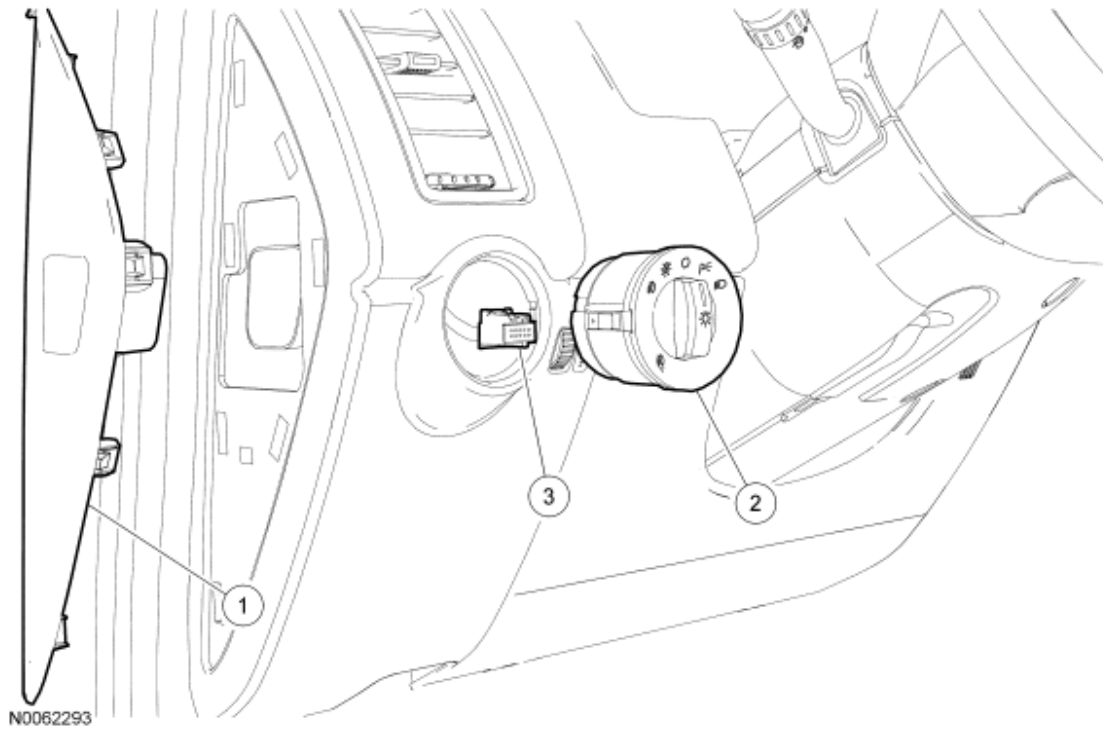


Fig. 131: Exploded View Of Headlamp Switch
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	7804481	Instrument panel side finish panel
2	11654	Headlamp switch
3	-	Headlamp switch electrical connector (part of 14401)

REMOVAL AND INSTALLATION

1. Remove the instrument panel side finish panel.

NOTE: The headlamp switch is removed by pushing from behind.

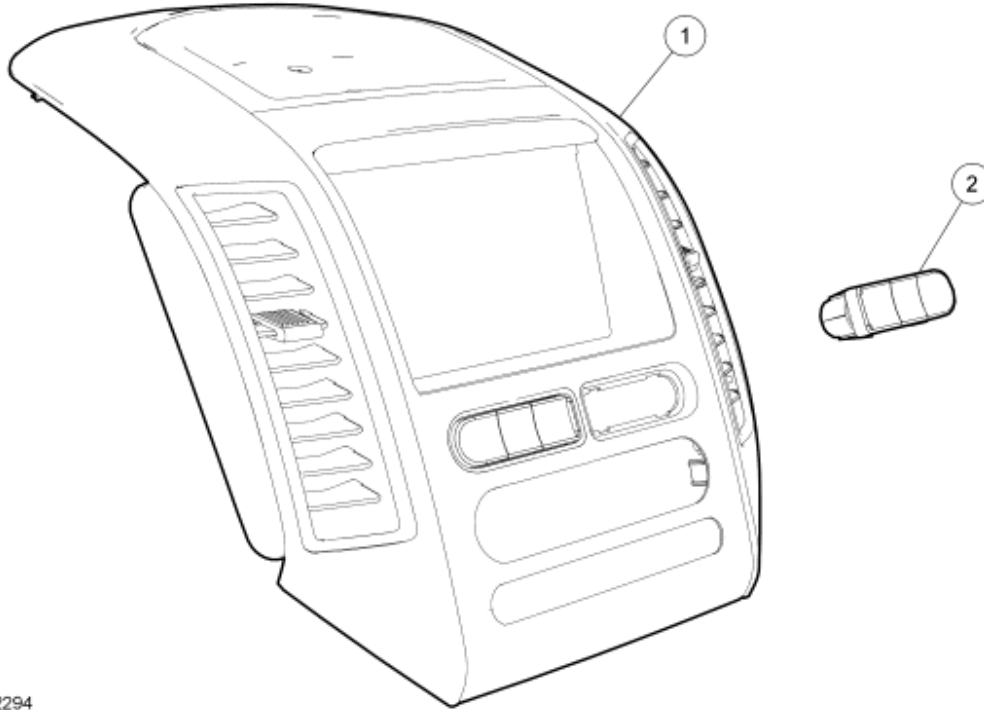
2. Release the tabs and remove the headlamp switch.
 - Disconnect the electrical connector.
3. To install, reverse the removal procedure.

HAZARD FLASHER LAMP SWITCH

NOTE: Edge shown, MKX similar.

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N0062294

Fig. 132: Exploded View Of Hazard Flasher Lamp Switch
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	78044D70	Instrument panel center finish panel
2	13D733	Hazard flasher lamp switch

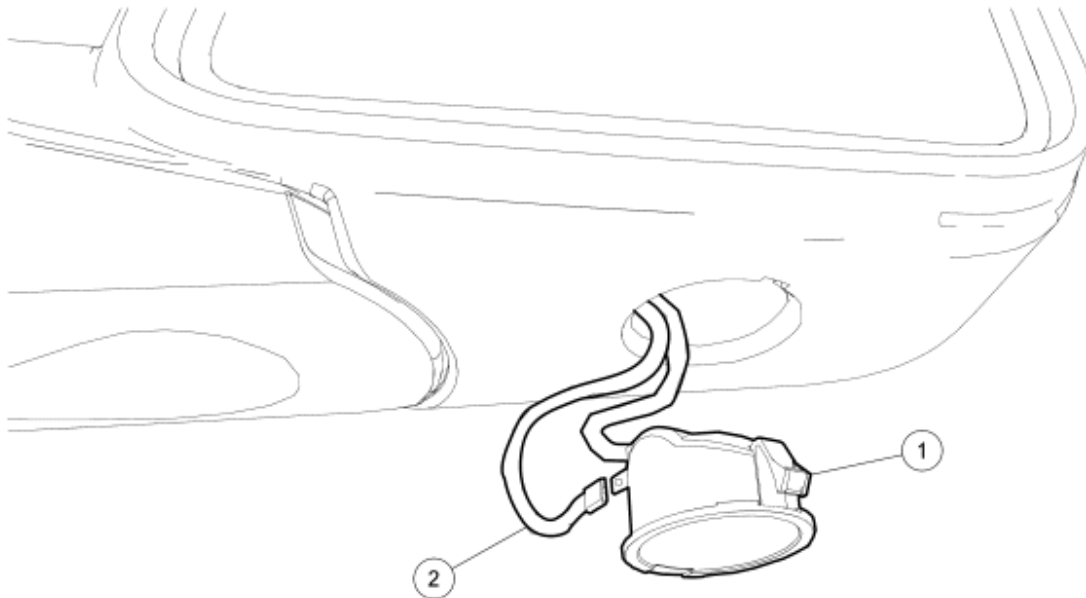
REMOVAL AND INSTALLATION

1. Remove the instrument panel center finish panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
2. Release the tabs and remove the hazard flasher lamp switch.
3. To install, reverse the removal procedure.

PUDDLE LAMP

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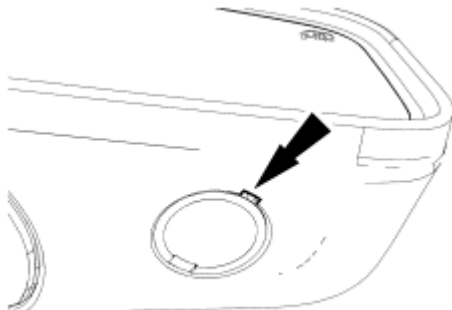
N0062295

Fig. 133: Exploded View Of Puddle Lamp
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13B375	Puddle lamp
2	-	Puddle lamp electrical connector (part of 17683)

REMOVAL AND INSTALLATION

1. Release the tab to remove the puddle lamp from the mirror housing.
 - Disconnect the electrical connector.



N0062296

Fig. 134: Locating Puddle Lamp
Courtesy of FORD MOTOR CO.

NOTE: An audible click can be heard when the puddle lamp is fully seated.

2. To install, reverse the removal procedure.

REAR LAMP ASSEMBLY

NOTE: Edge shown, MKX similar.

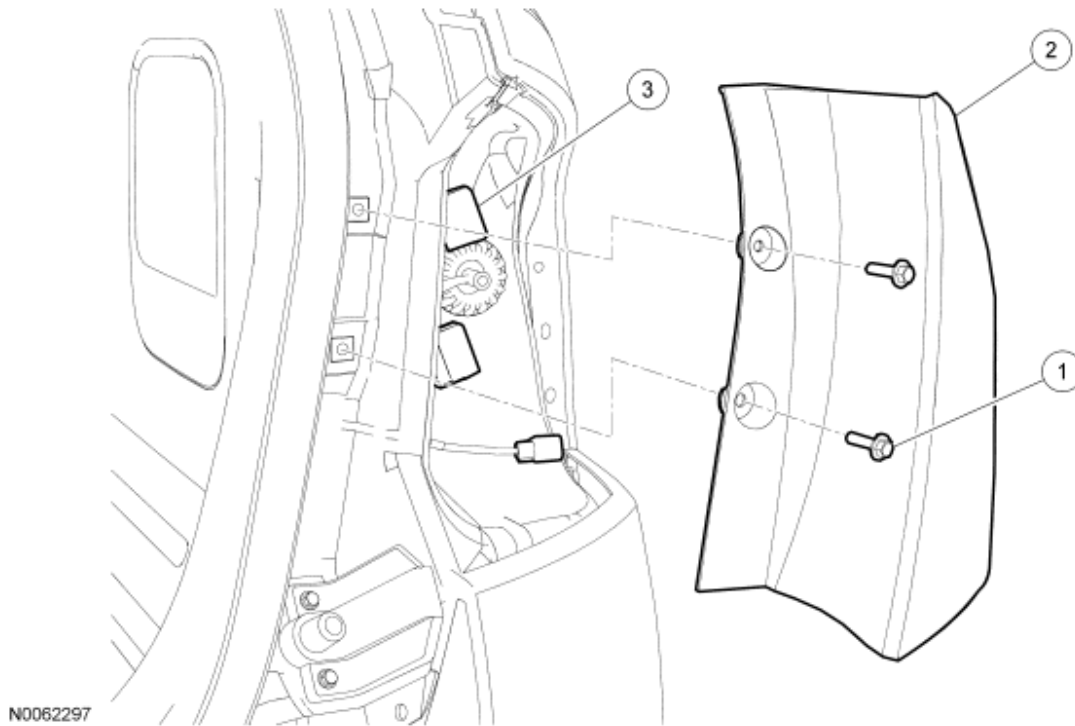


Fig. 135: Exploded View Of Rear Lamp Assembly
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W712801	Rear lamp assembly screws (2 required)
2	13A505/13A504	Rear lamp assembly (LH/RH)
3	-	Rear lamp assembly electrical connectors (part of 14A005)

REMOVAL AND INSTALLATION

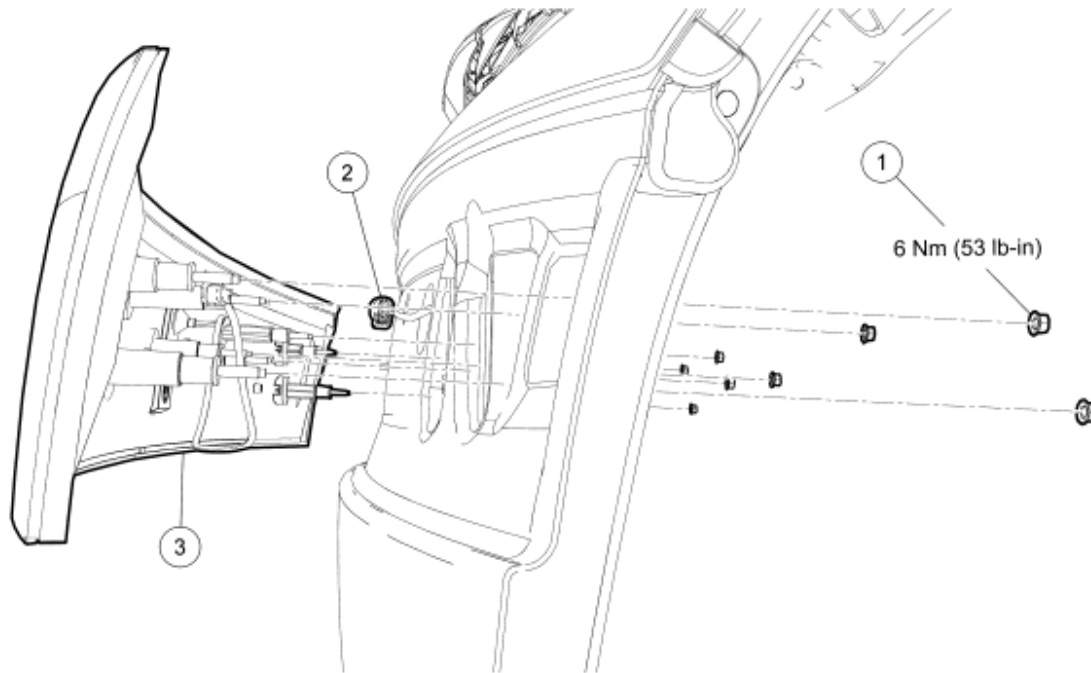
NOTE: The rear lamp assembly is held in place by snap-type retainers. The lamp should be pulled straight and rearward to remove.

1. Remove the 2 screws and the rear lamp assembly.
 - Disconnect the electrical connectors.

2. To install, reverse the removal procedure.

- An audible click should be heard when the rear lamp assembly is fully seated to the quarter panel.

REAR LAMP ASSEMBLY - LIFTGATE



N0082043

Fig. 136: Identifying Rear Lamp Assembly With Torque Specifications
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W704141	Liftgate lamp nuts (8 required)
2	-	Liftgate lamp electrical connector (part of 14A583)
3	13A565	Liftgate lamp

REMOVAL AND INSTALLATION

1. Remove the luggage trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
2. Disconnect the liftgate lamp electrical connector.
3. Remove the 8 nuts and the liftgate lamp.
 - To install, tighten to 6 Nm (53 lb-in).
4. To install, reverse the removal procedure.

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SPECIFICATIONS

TORQUE SPECIFICATIONS

Description	Nm	lb-in
Roof rack bolts	6	53
Spoiler nuts	6	53

REMOVAL AND INSTALLATION

RADIATOR GRILLE

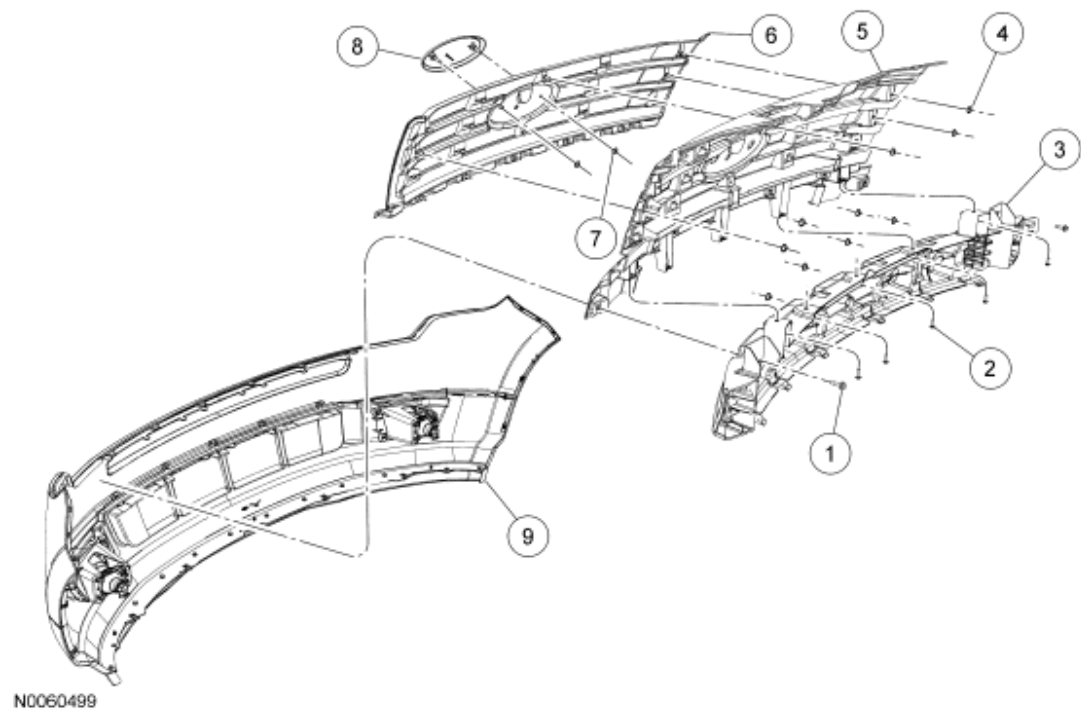


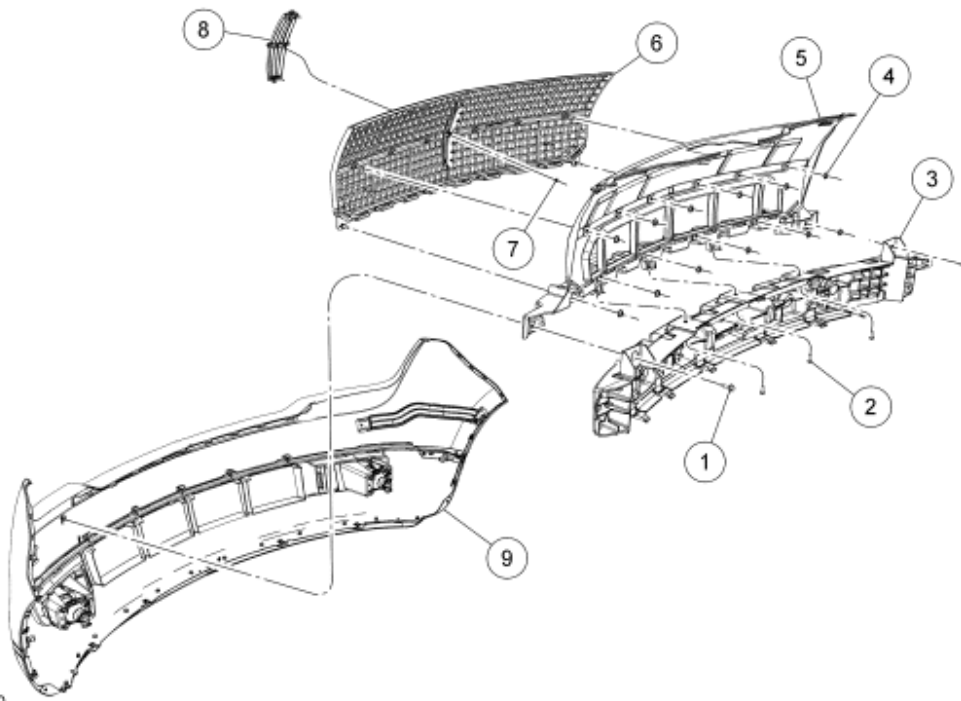
Fig. 1: Exploded View Of Edge Radiator Grille
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W506413-S	Isolator-to-bumper cover bolt (2 required)
2	W505003-S	Isolator-to-radiator grille reinforcement screw (5 required)
3	17E898	Isolator

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4	N800488-S	Radiator grille reinforcement-to-radiator grille nut (11 required)
5	8A164	Radiator grille reinforcement
6	16A912	Radiator grille
7	N800488-S	Emblem retainer
8	8B262	Emblem
9	17D957	Front bumper cover



N0060500

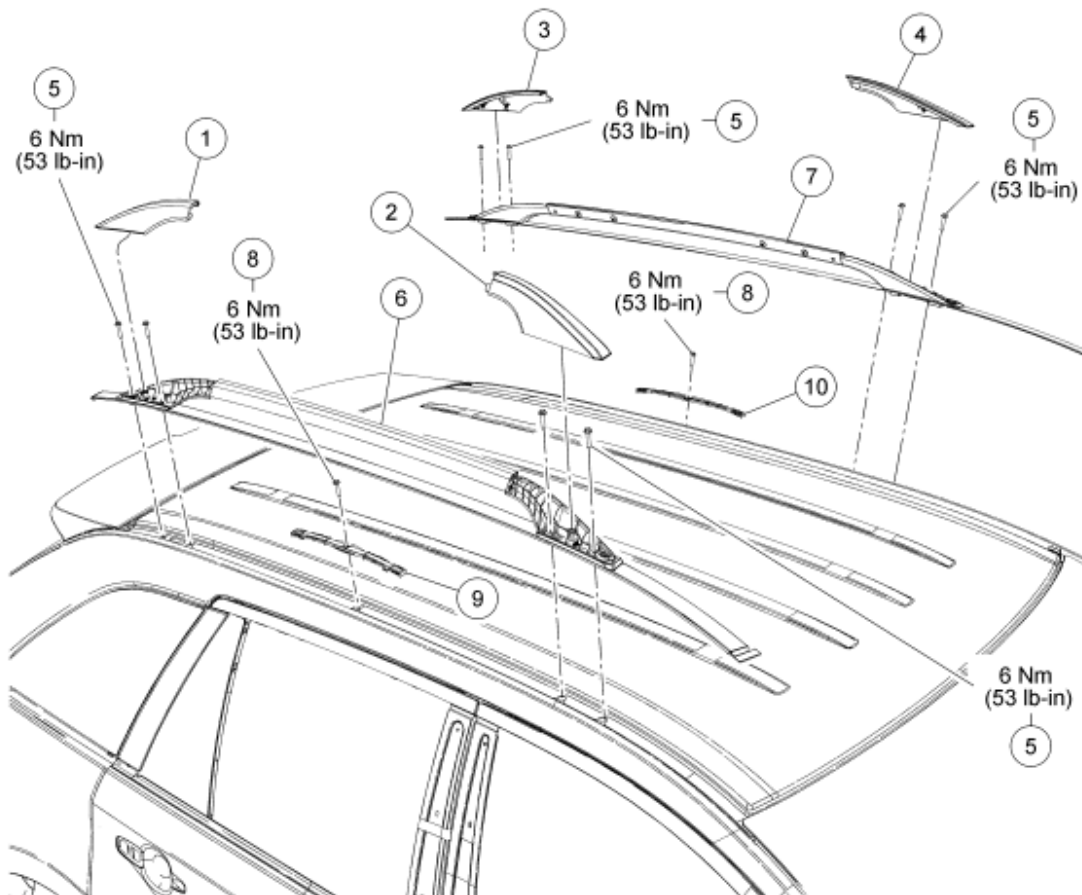
Fig. 2: Exploded View Of MKX Radiator Grille
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W506413-S	Isolator-to-bumper cover bolt (2 required)
2	W505003-S	Isolator-to-radiator grill reinforcement screw (3 required)
3	17E898	Isolator
4	N800488-S	Radiator grille reinforcement-to-radiator grille nut (12 required)
5	8A164	Radiator grill reinforcement
6	16A912	Radiator grille
7	-	Emblem retainer
8	8B262	Emblem
9	17D957	Bumper cover

REMOVAL AND INSTALLATION

1. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
2. Remove the isolator-to-bumper cover bolts.
3. Remove the isolator-to-radiator grille reinforcement screws.
4. Remove the isolator.
 - Release the isolator from the clips on the bumper cover.
5. Remove the radiator grille and radiator grille reinforcement.
 - Release the radiator grille reinforcement from the clips on the bumper cover.
6. Remove the radiator grille-to-radiator grille reinforcement nuts and remove the radiator grille.
7. To install, reverse the removal procedure.

ROOF RACK



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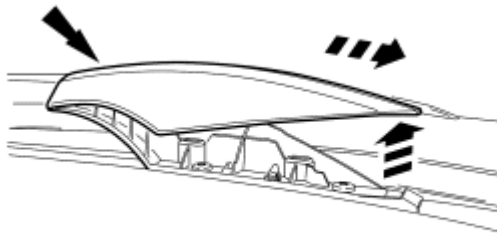
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Fig. 3: Exploded View Of Roof Rack With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	551A92-3	Roof rack bolt cover rear (RH)
2	551A90-1	Roof rack bolt cover front (RH)
3	551A92-4	Roof rack bolt cover rear (LH)
4	551A90-2	Roof rack bolt cover front (LH)
5	W712668-S	Roof rack bolts (8 required)
6	55128-8	Roof rack side rail (RH)
7	55128-9	Roof rack side rail (LH)
8	W712668-S	Roof rack retainer bolts (2 required)
9	02684	Roof rack retainer (RH)
10	02684	Roof rack retainer (LH)

NOTE: LH shown, RH similar.

1. Remove the rear roof rack bolt cover.
 - Lift upward on the rear of the bolt cover and then pull rearward.



N0060427

Fig. 4: Lifting Upward On Rear Of Bolt Cover & Pulling Rearward
Courtesy of FORD MOTOR CO.

2. Remove the front roof rack bolt cover.
 - Lift upward on the front of the bolt cover and then pull forward.

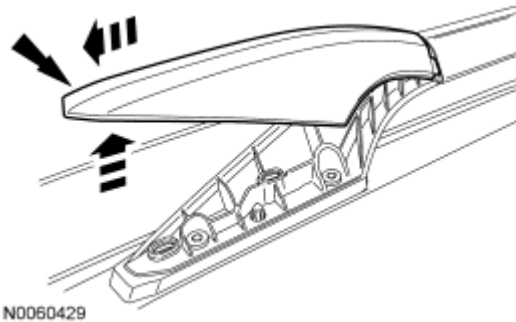


Fig. 5: Lifting Upward On Front Of Bolt Cover & Pulling Rearward
Courtesy of FORD MOTOR CO.

3. Remove the 2 front roof rack bolts.
 - To install, tighten to 6 N.m (53 lb-in).

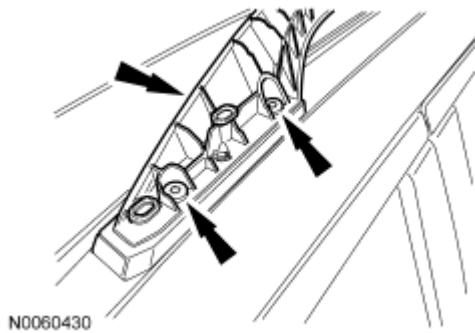


Fig. 6: Identifying Front Roof Rack Bolts
Courtesy of FORD MOTOR CO.

4. Remove the 2 rear roof rack bolts.
 - To install, tighten to 6 N.m (53 lb-in).

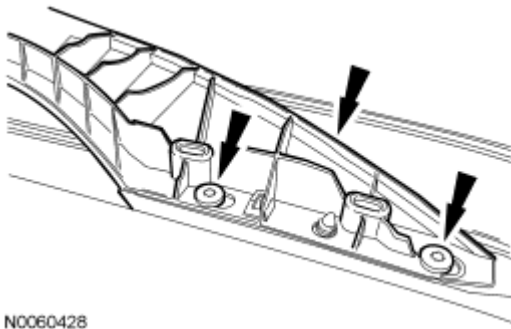


Fig. 7: Identifying Rear Roof Rack Bolts
Courtesy of FORD MOTOR CO.

NOTE: The roof rack side rail has a tab located at the front of the ditch moulding

and it is installed under the windshield moulding. When installing the roof rack side rail, start at the front of the vehicle and work toward the rear.

5. Remove the roof rack side rail.

- Start at the rear of the roof rack side rail working toward the front of the vehicle.

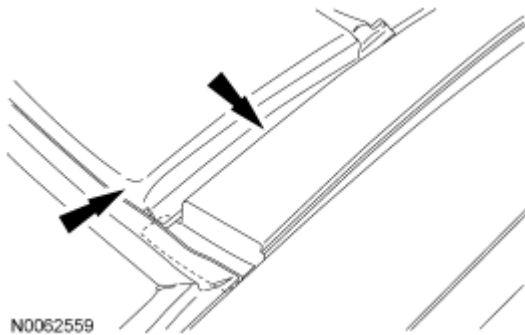
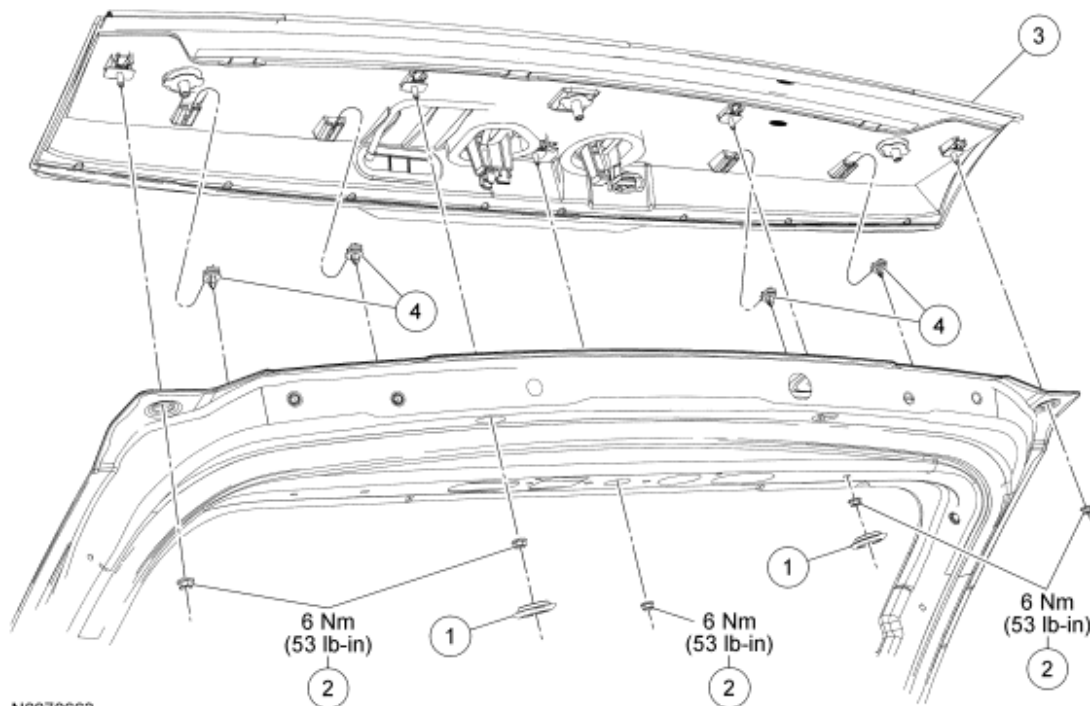


Fig. 8: Identifying Roof Rack Side Rail
Courtesy of FORD MOTOR CO.

6. To install, reverse the removal procedure.

- The 4 roof rack bolts are installed in a specific order starting with the front bolt and working toward the rear of the vehicle. The bolts must be installed in that order to make sure the roof rack side rail is correctly aligned.

SPOILER



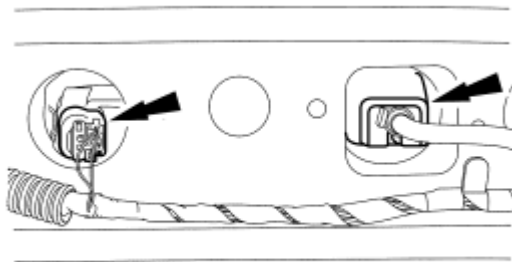
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Fig. 9: Exploded View Of Spoiler With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W651016-S	Spoiler nut covers (2 required)
2	W621926-S	Spoiler nuts (5 required)
3	010807	Spoiler
4	7841602	Spoiler retainers (4 required)

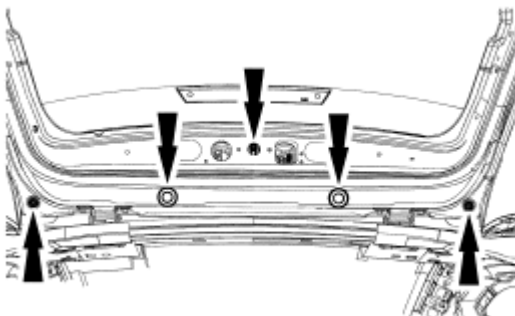
1. Remove the liftgate trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
2. Disconnect the electrical connector and the rear windshield washer hose from the rear spoiler.



N0054325

Fig. 10: Identifying Electrical Connector & Rear Windshield Washer Hose
Courtesy of FORD MOTOR CO.

3. Remove the 5 rear spoiler nuts.
 - Remove the 2 nut covers.
 - Remove the 5 nuts.
 - To install, tighten to 6 Nm (53 lb-in).



N0070662

Fig. 11: Locating Rear Spoiler Nuts
Courtesy of FORD MOTOR CO.

4. Place a piece of tape on the liftgate to protect the paint.

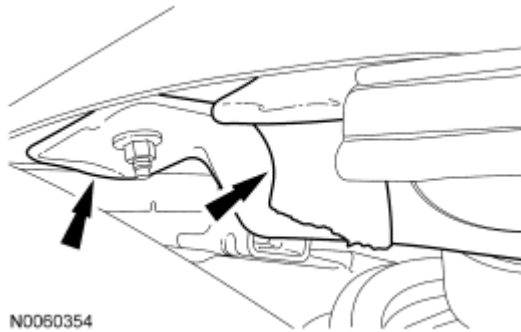


Fig. 12: Placing A Piece Of Tape On Liftgate To Protect Paint
Courtesy of FORD MOTOR CO.

NOTE: There are 4 pin-type retainers on the bottom of the spoiler. When removing the spoiler it is difficult to access the retainers to release them from the liftgate. It may be necessary to pry upward on the rear spoiler to release it from the retainers. Do not pry on the window.

5. Remove the rear spoiler.
 - Place a pry bar between the liftgate and the spoiler.

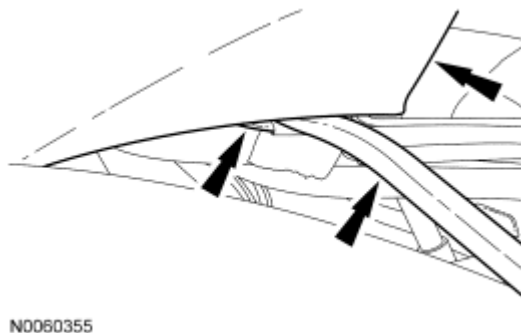


Fig. 13: Placing A Pry Bar Between Liftgate & Spoiler
Courtesy of FORD MOTOR CO.

6. To install, reverse the removal procedure.
 - Replace any retainers broken during removal.

ROCKER PANEL MOULDING

NOTE: RH shown, LH similar.

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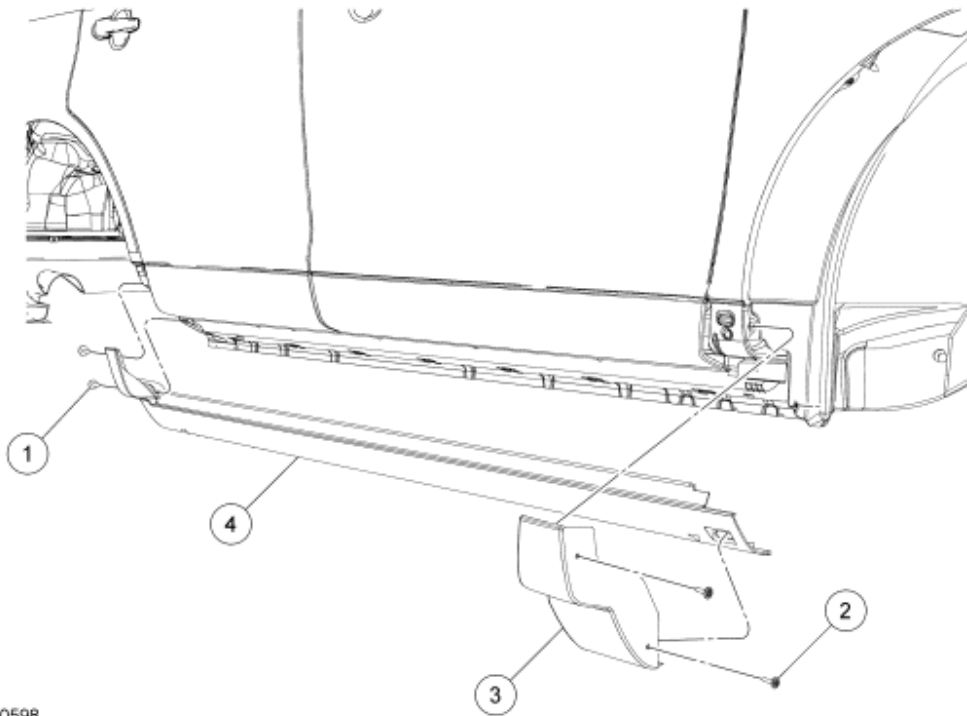


Fig. 14: Exploded View Of Rocker Panel Moulding
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W706805-S	Rocker panel moulding rear screw (2 required)
2	W706805-S	Rocker panel moulding cap front screw (2 required)
3	7810154 RH/ 7810155 LH	Rocker panel moulding cap
4	78101A04 RH/ 78101A05 LH	Rocker panel moulding

1. Remove the rocker panel moulding front cap.
 - Remove the 2 screws.
 - Remove the front cap.

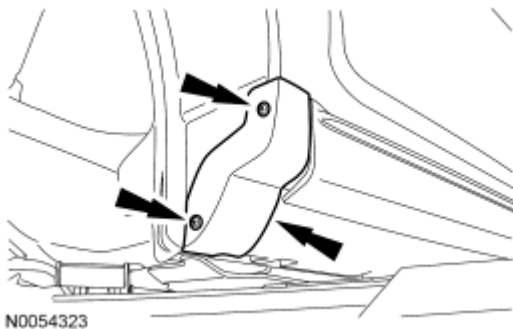


Fig. 15: Identifying Rocker Panel Molding End Cap & Screws
Courtesy of FORD MOTOR CO.

NOTE: RH shown, LH similar.

2. Remove the 2 screws from the rocker panel moulding inside the rear wheelhouse.

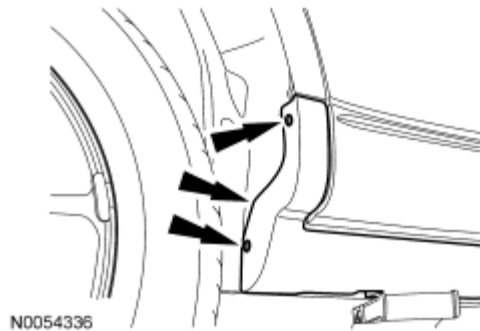


Fig. 16: Identifying Screws From Rocker Panel Moulding Inside Rear Wheelhouse
Courtesy of FORD MOTOR CO.

3. Open the front and rear doors.
4. Release the retainers on the bottom of the rocker panel moulding.

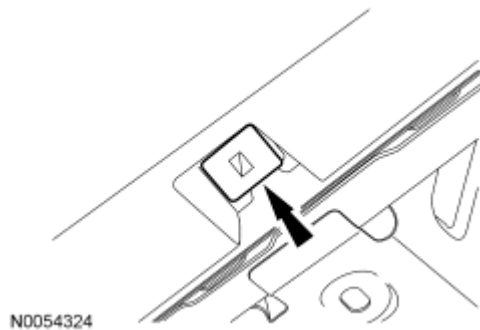


Fig. 17: Identifying Retainers On Bottom Of Rocker Panel Moulding
Courtesy of FORD MOTOR CO.

5. Pull outward to release the rocker panel moulding from the body, starting from the front of the vehicle moving rearward.
6. To install, reverse the removal procedure.